Navigation

Note on the solutions

These solutions have not been written to necessarily be the speediest, cleverest solution to each problem. Instead, I have tried to make the solutions clear and instructive to somebody at the level I expect a qual-studier will be. If you think to yourself "well, can't I just ...", possibly the answer is yes, but I don't care how short and elegant your solution is, that isn't the point.

When I personally have encountered great difficulty in a problem, which has happened frequently, I have tried to express in the solution both that the problem indeed is difficult (so don't be too hard on yourself if you are having trouble) as well as the way of thinking that led me eventually to a solution. I hope to give you the feeling that you too can solve the problem by learning enough, and having the patience and courage to follow your problem-solving instincts.

1 Mixes and Medleys

January 2026 Problem 1, edited On this problem, only your answer will be graded. No justification is required.

(a) How many ring homomorphisms
$$\mathbb{Z}[x, x^{-1}] \to \mathbb{F}_{27}$$
are there? Here, $\mathbb{Z}[x, x^{-1}]$ is the localization of $\mathbb{Z}[x]$ inverting x, and $\mathbb{F}_{27}$ is the finite field with 27 elements.

(b) Given $n \geq 2$, how many group homomorphisms
$$S_n \to \mathbb{Z}/2\mathbb{Z}$$
are there?

(c) V is an n-dimensional vector space over a field K. Let W be the vector space of alternating $(n+1)$-linear forms $V \times V \times \cdots \times V \to K$ (in $n+1$ vector arguments). What is $\dim_K(W)$?

(d) Suppose $K \supset E \supset F$ is a tower of field extensions such that both $K \supset E$ and $E \supset F$ are finite Galois extensions. Is it true that K must be a finite Galois extension of F?

(e) Find the cardinality of the tensor product
$$((\mathbb{Z}/20\mathbb{Z}) \oplus (\mathbb{Z}/26\mathbb{Z})) \otimes_{\mathbb{Z}} ((\mathbb{Z}/10\mathbb{Z})^2).$$

Solution

(a) A ring homomorphism must send 1 to 1, so a homomorphism $\mathbb{Z}[x, x^{-1}] \to \mathbb{F}_{27}$ is completely determined by the image of x. We can send x to any invertible element of $\mathbb{F}_{27}$, of which there are 26 choices, so there are 26 different homomorphisms. (The biggest danger here is inadvertantly mentally replacing $\mathbb{F}_{27}$ with $\mathbb{Z}/27\mathbb{Z}$, ask me how I know.)

(b) Two: the zero homomorphism and the sign homomorphism (the quotient $S_n \to S_n/A_n$). Indeed, suppose $\phi\colon S_n \to \mathbb{Z}/2\mathbb{Z}$ is nontrivial, hence surjective. For any $\sigma \in S_n$, we have $\phi(\pi\sigma\pi^{-1}) = \phi(\sigma)$ for any $\pi \in S_n$, because $\mathbb{Z}/2\mathbb{Z}$ is commutative. If $\tau_1, \tau_2 \in S_n$ are transpositions, then they are conjugate, and hence $\phi(\tau_1) = \phi(\tau_2)$. Since S_n is generated by transpositions, we have that $\phi(\tau)$ generates the $\operatorname{im}\phi$ for any transposition τ. We assumed ϕ is surjective, so $\phi(\tau) = 1$. Thus, all even permutations are in $\ker\phi$, i.e. $\ker\phi \supset A_n$. Since A_n is already index 2, we must have $\ker\phi = A_n$.

(c) This is the first time the language of alternating multilinear forms has appeared on the qual, so I'll include some definitions. An m-linear form is a function f which takes m vector inputs $v_1, \ldots, v_m \in V$ and outputs an element of K such that f is linear in each argument separately: $f(v_1, \ldots, av_i + bv_i', \ldots, v_m) = af(v_1, \ldots, v_i, \ldots, v_m) + bf(v_1, \ldots, v_i', \ldots, v_m)$ for each index $1 \leq i \leq m$. An m-linear form is called "alternating" if it evaluates to 0 whenever two or more of its arguments are the same, e.g. if $m = 2$ an alternating form satisfies $f(v, v) = 0$ for all vectors v. The determinant is an example of an alternating n-linear form. The set of all m-linear forms on V is a vector space over K, and the set of alternating forms is a subspace.

The key fact, which is an exercise if you don't already know it, is that an alternating form evaluates to 0 whenever its arguments are linearly dependent. Since V is dimension n, any set of $n + 1$ vectors will be dependent, and thus any alternating $(n + 1)$-linear form will evaluate to 0 on all inputs. Thus, $\dim_K(W) = 0$.

(d) K need not be a Galois extension of F (but will be finite). As written, the problem didn't say to give a counterexample, but you should anyway. The standard counterexample is $\mathbb{Q}(\sqrt[4]{2}) \supset \mathbb{Q}(\sqrt{2}) \supset \mathbb{Q}$.

(e) We use two tensor product facts: tensor products distribute over direct sums (justifying the notation), and $(\mathbb{Z}/a\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/b\mathbb{Z}) \cong \mathbb{Z}/\gcd(a, b)\mathbb{Z}$. Thus:

$$\begin{aligned}
((\mathbb{Z}/20\mathbb{Z}) \oplus (\mathbb{Z}/26\mathbb{Z})) \otimes_{\mathbb{Z}} ((\mathbb{Z}/10\mathbb{Z})^2) &\cong ((\mathbb{Z}/20\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/10\mathbb{Z})^2) \oplus ((\mathbb{Z}/26\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/10\mathbb{Z})^2) \\
&\cong ((\mathbb{Z}/20\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/10\mathbb{Z}))^2 \oplus ((\mathbb{Z}/26\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/10\mathbb{Z}))^2 \\
&\cong (\mathbb{Z}/10\mathbb{Z})^2 \oplus (\mathbb{Z}/2\mathbb{Z})^2,
\end{aligned}$$

which has order $10^2 \cdot 2^2 = 400$.

August 2025 Problem 1 On this problem, only your answer will be graded. No justification is required.

(a) Suppose $x_1, \ldots, x_n$ are n distinct complex numbers, and $a_1, \ldots, a_n \geq 0$, $b_1, \ldots, b_n \geq 0$ are integers. Put

$$f(x) = (x - x_1)^{a_1} \cdots (x - x_n)^{a_n} \qquad g(x) = (x - x_1)^{b_1} \cdots (x - x_n)^{b_n}.$$

(Some of the exponents a_i, b_i may be zero, so f and g need not share all roots.) Consider $\mathbb{C}[x]$-modules $M = \mathbb{C}[x]/(f(x))$ and $N = \mathbb{C}[x]/(g(x))$. Write a formula for

$$\dim_{\mathbb{C}}(M \otimes_{\mathbb{C}[x]} N)$$

in terms of $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$.

(b) In the situation of question (a), write a formula for $\dim_{\mathbb{C}} M_g$ in terms of $a_1, \ldots, a_n$ and $b_1, \ldots, b_n$. Here M_g is the localization of M with respect to $g \in \mathbb{C}[x]$.

(c) Let M be the $\mathbb{C}[x]$-module

$$\mathbb{C}[x]/(x^3 - x) \times \mathbb{C}[x]/(x^3 + x^2) \times \mathbb{C}[x]/(x^2 + 1).$$

Rewrite M in the form

$$M \simeq \mathbb{C}[x]/(p_1(x)) \times \mathbb{C}[x]/(p_2(x)) \times \cdots \times \mathbb{C}[x]/(p_k(x))$$

for non-constant polynomials $p_1(x), \ldots, p_k(x) \in \mathbb{C}[x]$ such that $p_k(x) \mid p_{k-1}(x) \mid \cdots \mid p_1(x)$ (that is, the $p_i(x)$ are the *invariant factors* of M).

(d) In the situation of question (c), consider M as a vector space over $\mathbb{C}$ via restriction of scalars to $\mathbb{C} \subset \mathbb{C}[x]$. Let the operator $T\colon M \to M$ be the action of $x \in \mathbb{C}[x]$ on M. Clearly, it is a $\mathbb{C}$-linear operator; what are its eigenvalues?

(e) Give an example of a ring R, an R-module N, and an exact sequence of R-modules

$$0 \to M_1 \to M_2 \to M_3 \to 0$$

such that the induced sequence

$$0 \to \operatorname{Hom}_R(N, M_1) \to \operatorname{Hom}_R(N, M_2) \to \operatorname{Hom}_R(N, M_3) \to 0$$

is <u>not</u> exact.

Solution

This is a reading-intensive problem 1.

(a) By standard tensor product properties, $R/I \otimes_R R/J \cong R/(I+J)$. In a PID like $\mathbb{C}[x]$, the sum of two ideals is just the ideal generated by the *gcd* of the two generators: $(f(x)) + (g(x)) = (\gcd(f(x), g(x))$. From the factorizations given, we can see that

$$\gcd(f(x), g(x)) = (x - x_1)^{\min(a_1,b_1)} \cdots (x - x_n)^{\min(a_n,b_n)}.$$

Finally, the dimension of $\mathbb{C}[x]/(\text{tomato}(x))$ is just $\deg(\text{tomato}(x))$. Thus, $\dim_{\mathbb{C}}(M \otimes_{\mathbb{C}[x]} N) = \sum_{i=1}^{n} \min(a_i, b_i)$.

(b) (This is a bit like January 2023 Problem 1(a).) Localizing with respect to g is the same as localizing with respect to every factor $(x - x_i)$ satisfying $b_i > 0$. Thus,

$$\begin{aligned} M_g &= (\mathbb{C}[x]/(f(x)))_g \\ &\cong \mathbb{C}[x]_g/(f(x))_g \\ &\cong \mathbb{C}[x]_g/(h(x))_g, \end{aligned}$$

where $h(x)$ has the factorization of $f(x)$ without any of the factors $(x - x_i)^{a_i}$ that have $b_i > 0$. So, $\dim_{\mathbb{C}}(M_g) = \displaystyle\sum_{i : b_i = 0} a_i$.

(c) We start by factoring these polynomials: $x^3 - x = x(x-1)(x+1)$, $x^3 + x^2 = x^2(x+1)$,

$x^2+1=(x-i)(x+i)$. For each irreducible polynomial appearing in these factorizations, record the powers to which it appears in descending order, as shown in the table below.

poly.	powers
x	2, 1
$x-1$	1
$x+1$	1, 1
$x-i$	1
$x+i$	1

To find the polynomial $p_i(x)$, we go through our table and include a factor of $(x-a)^{e_i}$, where e_i is the ith power appearing on the right side of the table (or 0 if the list is too short). So, we have $p_1(x)=x^2(x-1)(x+1)(x-i)(x+i)=x^6-x^2$ and $p_2(x)=x(x+1)=x^2+x$, and then we are done since none of the lists of exponents have more than 2 entries. We can verify that $p_2(x)\mid p_1(x)$, and use the Chinese Remainder Theorem to verify that $M=\mathbb{C}[x]/(x(x+1))\times\mathbb{C}[x]/(x^2(x-1)(x+1)(x-i)(x+i))$.

(d) The minimal polynomial of T is $p_1(x)=x^2(x-1)(x+1)(x-i)(x+i)$, so the eigenvalues of T are $0,\pm1,\pm i$.

(e) Take $N=\mathbb{Z}/2\mathbb{Z}$, and the short exact sequence

$$0\to\mathbb{Z}\xrightarrow{\times 2}\mathbb{Z}\to\mathbb{Z}/2\mathbb{Z}\to 0.$$

January 2025 Problem 1, edited On this problem, no justification is required; only the answers will be graded.

(a) Let R be a (not necessarily commutative) ring, and let M be a right R-module. Let A be the annihilator of M in R. Is A necessarily a two-sided ideal of R, necessarily a one-sided ideal of R, or not necessarily either of these things?

(b) Let H be the subgroup of S_5 generated by the element $(12)(345)$. Then H acts on S_5 by left multiplication. How many orbits are there of this action?

(c) Let $\mathbb{H}$ be the quaternion algebra $\mathbb{R}+\mathbb{R}i+\mathbb{R}j+\mathbb{R}k$, and let A be its subalgebra $\mathbb{R}+\mathbb{R}i$ (so A is isomorphic to the complex numbers). What is the dimension of $\mathbb{H}\otimes_A\mathbb{H}$ as a real vector space?

(d) True or false: if A is an $n\times n$ orthogonal matrix, then all n eigenvalues are either 1 or -1.

(e) Give an example of a prime ideal in the ring $\mathbb{C}[x,y,z]/(xy,xz,yz,x^2,z^2)$ which is not a maximal ideal.

Solution

(a) A is always a two-sided ideal of R. Indeed, for any $r\in R$, $a\in A$, and $m\in M$, $(ra)m=r(am)=r\cdot 0=0$, so $ra\in A$. Also, $(ar)m=a(rm)=0$ since $rm\in M$ and a annihilates M, so $ar\in A$. It is easy to see that A is closed under addition.

(b) Orbits are cosets of H, so there are $[S_5:H]=20$ orbits.

(c) 8

(d) False, e.g. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

(e) Prime ideals of this ring correspond to prime ideals of $\mathbb{C}[x, y, z]$ which contain (xy, xz, yz, x^2, z^2). Notice that every generator of that ideal is divisible by x or by z, so (x, z) is a prime ideal containing that ideal. On the other hand, (x, z) is not maximal because e.g. (x, y, z) is another prime ideal which is strictly bigger.

August 2024 Problem 1 On this problem, no justification is required; only the answers will be graded.

(a) The symmetric group on 5 letters S_5 acts on the set $\{1, \ldots, 5\} \times \{1, \ldots, 5\} \times \{1, \ldots, 5\}$ by the diagonal action $\sigma(a, b, c) = (\sigma(a), \sigma(b), \sigma(c))$. How many orbits are there of this action?

(b) The vector space $\mathbb{R}^3$ is a module for the ring of matrices $M_3(\mathbb{R})$. Let v be a nonzero vector in $\mathbb{R}^3$. What is the dimension of the annihilator of v in $M_3(\mathbb{R})$?

(c) Let $\mathbb{F}_5$ be the field with 5 elements. How many ideals does the ring $\mathbb{F}_5[T]/(T^2 + 1)$ have?

(d) Give an example of an abelian subgroup of $\mathrm{GL}_2(\mathbb{C})$ that is not contained in the group of diagonal matrices.

(e) Give an example of a simple ring which is not a division ring.

Solution

(a) There are 5 orbits: 1 orbit where none of the three coordinates are equal, 3 orbits where two of the three coordinates are equal and the third is distinct, and 1 orbit where all three coordinates are equal.

(b) If we extend the set $\{v\}$ to a basis $\{v, v_2, v_3\}$, then the annihilator of v is the same as the set of matrices whose first column consists entirely of 0s. This is a subspace of dimension 6 inside $M_3(\mathbb{R})$.

(c) By the lattice isomorphism theorem, ideals in $\mathbb{F}_5[T]/(T^2 + 1)$ correspond to ideals in $\mathbb{F}_5[T]$ that contain $(T^2 + 1)$. Since $\mathbb{F}_5[T]$ is a UFD, this corresponds to the number of divisors of the polynomial $T^2 + 1$ over $\mathbb{F}_5$. Over $\mathbb{F}_5$, the polynomial $T^2 + 1$ factors as $(T - 2)(T - 3)$, so it has four total divisors, giving us the four containing ideals: $(1), (T - 2), (T - 3), (T^2 + 1)$.

(d) The subgroup generated by

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

forms an abelian group isomorphic to $\mathbb{Z}$, but which is obviously not contained in the group of diagonal matrices.

(e) Any counterexample must be a noncommutative ring, because a simple commutative ring must be a field. The standard example is $M_2(\mathbb{R})$, which is simple because its only two-sided ideals are the zero ideal and the whole ring, but it is not a division ring because there are matrices that don't have inverses.

January 2024 Problem 1 On this problem, no justification is required; only the answers will be graded.

(a) Give an example of a simple ring which is not a division ring.

(b) Let $A\colon \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by $Ae_1 = e_1 + e_2$, $Ae_2 = e_2$, $Ae_3 = e_3$. Let V be the vector space of linear transformations from $\mathbb{R}^3$ to $\mathbb{R}^3$ which commute with A. What is $\dim_{\mathbb{R}}(V)$?

(c) Give an example of a group G and a subgroup H, and a subgroup K of H, such that K is normal in H, and H is normal in G, but K is not normal in G.

(d) Give an example of an inseparable field extension.

(e) Give an example of a pair (R, M), where R is a ring and M is an R-module that is not Noetherian.

Solution

(a) (deja vu) Any counterexample must be a noncommutative ring, because a simple commutative ring must be a field. The standard example is $M_2(\mathbb{R})$, which is simple because its only two-sided ideals are the zero ideal and the whole ring, but it is not a division ring because there are matrices that don't have inverses.

(b) $\dim_{\mathbb{R}}(V) = 5$. One way to do this problem is to write out the matrix of A, and an arbitrary matrix B, and compute the linear relations that must hold in order for them to commute. On the qual, if I saw a problem like this and didn't think of a clever workaround immediately, I would try this first.

Here is a trick that comes up often with commuting matrices which does not outright solve this problem, but does reduce the amount of computation required: if v is an eigenvector of A of eigenvalue λ, and B commutes with A, then Bv is *also* an eigenvalue of A of eigenvalue λ. Indeed, $ABv = BAv = \lambda Bv$. This immediately tells us that for $i = 2, 3$, $Be_i = b_{2i}e_2 + b_{3i}e_3$, i.e. that the latter two entries in the first row of B are 0. To finish this problem, I tried doing a similar computation for e_1: $ABe_1 = BAe_1 = Be_1 + Be_2$, which we can rearrange to get $(A - I)Be_1 = Be_2$. Here, writing out the matrix $A - I$ is very easy, and I quickly saw that $b_{11} = b_{22}$ and $b_{32} = 0$ were the last two linear conditions defining B.

(c) Any counterexample must be a nonabelian group. The two standard examples are $D_4 = \langle r, s : r^4 = s^2 = e, srs = r^{-1}\rangle$, with $H = \langle r^2, s\rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $K = \langle s\rangle$, and A_4, with $H = \langle (12)(34), (13)(24)\rangle \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and $K = \langle (12)(34)\rangle$. In fact, for any even number

$n = 2m$, D_n is a counterexample with $H = \langle r^m, s \rangle$ and $K = \langle s \rangle$. (If you're curious about cases where transitivity holds, look for "characteristic subgroup".)

(d) Any example must be over a non-perfect field, so in particular must be over an infinite field of characteristic $p \neq 0$. The standard example is to take the base field to be the field of rational function $\mathbb{F}_p(t)$, and the extension to be $\mathbb{F}_p(\sqrt[p]{t}) \cong \mathbb{F}_p(t)[x]/(x^p - t)$.

(e) Noetherian modules are less familiar than Noetherian rings are: a module M is Noetherian if its submodules satisfy the ACC, which is equivalent to saying every submodule is finitely generated. Maybe the easiest example of a non-Noetherian module would be to take R to be a non-Noetherian ring, and M to be a non-finitely-generated ideal in R. For example, if $R = \mathbb{C}[x_1, x_2, x_3, \dots]$ is the polynomial ring in countably many variables, and $M = (x_1, x_2, x_3, \dots)$ is the (maximal) ideal generated by the individual variables. If you don't like that, you can also take any infinitely-generated module over your favorite ring of choice, e.g. $R = \mathbb{Z}$, $M = \mathbb{Z}^{\oplus \mathbb{N}}$.

August 2023 Problem 1 On this problem, no justification is required; only the answers will be graded.

(a) Give an example of a field K, a finite algebraic field extension L/K, and a finite-dimensional K-algebra A such that A has no zero divisors but $A \otimes_K L$ does have zero divisors.

(b) Give an example of a zero divisor in the $A \otimes_K L$ you provided above.

(c) Give two elements x, y of the algebra of quaternions over $\mathbb{R}$ such that x and y both have norm 4 and x and y do not commute.

(d) Give an example of a ring with exactly three ideals (including the zero ideal and the ring itself).

(e) Give an example of a localization of $\mathbb{Z}$ which is not of the form $\mathbb{Z}[1/n]$ for any integer n.

Solution

(a) The example that came to mind for me is that if L/K is a finite algebraic field extension, say $L = K(\alpha)$, then L is moreover a finite-dimensional K-algebra. If we tensor $L \otimes_K L$ we will get $L \otimes_K K[x]/m_\alpha(x) \cong L[x]/m_\alpha(x)$, where $m_\alpha(x)$ is the minimal polynomial of α. As a concrete example, let $K = \mathbb{R}$, $L = \mathbb{C}$, then $\mathbb{C} \otimes_\mathbb{R} \mathbb{C} \cong \mathbb{C} \otimes_\mathbb{R} \mathbb{R}[x]/(x^2+1) \cong \mathbb{C}[x]/(x^2+1)$, which has zero-divisors because in this ring $(x-i)(x+i) = 0$.

(b) See above.

(c) We could take $x = 2i$ and $y = 2j$.

(d) You have lots of options for how to come up with examples here, see January 2020 Problem 3.

(e) I think this is a confusingly-phrased question, it took me a while to figure out what it was asking. What we want is a multiplicatively closed subset $S \subset \mathbb{Z}$ so that $S^{-1}\mathbb{Z}$ is not of the form $\mathbb{Z}[1/n]$ for some integer n. One idea might be to try throwing in two different denominators, say $\mathbb{Z}[1/2, 1/3]$, but this doesn't work because $\mathbb{Z}[1/2, 1/3] = \mathbb{Z}[1/6]$. For the same reason, throwing in only finitely many denominators doesn't work. What would work would be something like $\mathbb{Z}_{(2)}$, which as a reminder is where we throw in all denominators of numbers not in the ideal (2). Certainly, there is no way for us to get every odd denominator by only taking a ring of the form $\mathbb{Z}[1/n]$.

January 2023 Problem 1 On this problem, only your answer will be graded, no justification is necessary.

(a) Let R be a commutative ring and take $f, g \in R$. Consider the homomorphism $R \to R/(f) \otimes_R R_g$. Describe its kernel in terms of f and g.

(b) Consider the homomorphism

$$\phi\colon (\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z}) \to \mathbb{Z}/5\mathbb{Z}\colon (a, b) \mapsto a + b.$$

(Note that $a+b$ makes sense as an element of $\mathbb{Z}/5\mathbb{Z}$.) The kernel $\ker \phi$ is a finite abelian group. What is its isomorphism class. (That is, its canonical form, such as $(\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z})$.)

(c) True or False: if α and β are algebraic numbers such that $\mathbb{Q}(\alpha)$ is a Galois extension of $\mathbb{Q}$, and $\mathbb{Q}(\alpha, \beta)$ is a Galois extension of $\mathbb{Q}(\alpha)$, then $\mathbb{Q}(\alpha, \beta)$ is a Galois extension of $\mathbb{Q}$.

(d) True or False: For any non-zero $f \in \mathbb{C}[x]$, any finitely-generated torsion-free module over $\mathbb{C}[x, \frac{1}{f}]$ is free.

(e) True or False: For any n, any ideal $I \subset \mathbb{C}[x_1, \ldots, x_n]$ can be generated by (at most) n elements.

Solution

(a) It's important here that localization is exact, so $R/(f) \otimes_R R_g \cong R_g/(f)_g$, and the homomorphism we're looking at is just the composition of $R \to R_g \to R_g/(f)_g$. Certainly, we kill all the multiples of f, but because we're allowed to divide by g in R_g, the kernel will also contain any element that can be "multiplied into" the ideal (f) by some power of g. As a concrete example, suppose $R = \mathbb{Z}$, $f = 6$, $g = 9$, then 2 will be in the kernel of $\mathbb{Z} \to \mathbb{Z}[1/9]/6\mathbb{Z}[1/9]$ because $2 = 18/9 = 6(3/9) \in 6\mathbb{Z}[1/9]$.

(I think writing something like the above qualifies as a complete answer already, but I'll remark that the set in question $\{r \in R : \exists n,\ g^n r \in (f)\}$ has a name: it's the *ideal saturation* of the ideal (f) by the ideal (g), denoted $(f) : (g)^\infty$.)

(b) We can see that the group on the left has order 5^5, and the group on the right has order 5, and the map is certainly surjective, so the kernel must have order 5^4, and is a subgroup of $(\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z})$, so the possibilities are either $(\mathbb{Z}/125\mathbb{Z}) \times (\mathbb{Z}/5\mathbb{Z})$ or $(\mathbb{Z}/25\mathbb{Z}) \times (\mathbb{Z}/25\mathbb{Z})$.

We can check which it is by checking to see if the kernel has an element of order 125, which in fact it does, for example $(1,4)$.

(c) False. The standard counterexample is $\alpha = \sqrt{2}$, $\beta = \sqrt[4]{2}$.

(d) True. $\mathbb{C}[x]$ is a PID, hence all its localizations are also PIDs, and a finitely generated modules over a PID is torsion-free iff it is free.

(e) False. The standard example is $(x^2, xy, y^2) \in \mathbb{C}[x, y]$.

January 2023 Problem 5 Let p be a prime number and let G be the group of 2×2 matrices of the form

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix}, \qquad a \in \mathbb{F}_p,\ b \in \mathbb{F}_p^\times.$$

Fix a dimension n, and let $\rho\colon G \to \mathrm{GL}_n(\mathbb{C})$ be a homomorphism. Put

$$x = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix},\ y_b = \begin{pmatrix} 1 & 0 \\ 0 & b \end{pmatrix} \qquad (b \in \mathbb{F}_p^\times).$$

(a) Show that $\rho(x)$ is diagonalizable, and that its eigenvalues are pth roots of unity.

(b) Suppose $v \in \mathbb{C}^n$, $v \neq 0$ is an eigenvector of $\rho(x)$ of eigenvalue λ. Prove that for any $b \in \mathbb{F}_p^\times$, $\rho(y_b)v$ is also an eigenvector of $\rho(x)$, and find its eigenvalue.

(c) Let v be as in part (b). Show that $W = \operatorname{span}\langle \rho(y_b)v : b \in \mathbb{F}_p^\times \rangle$ is G-invariant (that is, $\rho(g)W \subseteq W$ for all $g \in G$), and that, if $\lambda \neq 1$, $\dim W = p - 1$.

Solution

(a) Note that in G, $x^p = I$, so $\rho(x)$ satisfies the polynomial $x^p - 1$. Thus, all the eigenvalues of $\rho(x)$ must be roots of $x^p - 1$, i.e. pth roots of unity. Furthermore, the minimal polynomial of $\rho(x)$ divides $x^p - 1$, which has all distinct roots, so the minimal polynomial must have all distinct roots, so $\rho(x)$ must be diagonalizable.

(b) In G, we have that $y_b x^b = x y_b$. Thus, $\rho(x)\rho(y_b)v = \rho(y_b)\rho(x)^b v = \lambda^b \rho(y_b)v$. Thus, $\rho(y_b)v$ is an eigenvector of eigenvalue λ^b.

(c) First note that x and the set of all y_b together generate G, since

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix} = y_b x^a.$$

Thus, since each of the generators of W is an eigenvector of $\rho(x)$, and these eigenvectors are permuted by each $\rho(y_b)$, the generating set of W is fixed by the action of G, and hence W is G-invariant. If $\lambda \neq 1$, λ is a primitive pth root of unity, and so each λ^b is distinct. Thus, the

$\rho(y_b)v$ are linearly independent, since they are eigenvectors of $\rho(x)$ of different eigenvalues. Since there are $p-1$ choices for b, this means $\dim W = p-1$.

August 2022 Problem 1 On this problem, only your answer will be graded, no justification is necessary.

(a) Let $K = \mathbb{Q}(2^{1/3})$. Is $K/\mathbb{Q}$ a Galois extension?

(b) What is the radical of the ideal $(x^3 - x^2) \subset \mathbb{R}[x]$?

(c) How many different (up to isomorphism) groups of order 49 are there?

(d) Give an example of two non-zero abelian groups M, N such that $M \otimes_{\mathbb{Z}} N = 0$.

(e) How many two-sided ideals are there in $M_2(\mathbb{C})$, the ring of 2×2 matrices with entries in $\mathbb{C}$?

Solution

(a) No, since $\zeta_3 2^{1/3}$ is a root of $x^3 - 2$ which does not lie in K.

(b) $(x^2 - x)$, in this case radical corresponds to "largest squarefree divisor".

(c) 2, $\mathbb{Z}/49\mathbb{Z}$ and $\mathbb{Z}/7\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z}$.

(d) $M = \mathbb{Z}/2\mathbb{Z}$, $N = \mathbb{Z}/3\mathbb{Z}$.

(e) 2, the zero ideal and the whole ring.

January 2022 Problem 2 For each of the following statements, either prove it or provide a counterexample.

(a) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{C})$ is diagonalizable.

(b) Every matrix of finite order in $\mathrm{GL}_n(\mathbb{Q})$ is diagonalizable.

(c) Every matrix in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ has finite order. (We write $\overline{\mathbb{F}}_p$ for the algebraic closure of the finite field $\mathbb{F}_p$.)

(d) Every matrix of finite order in $\mathrm{GL}_n(\overline{\mathbb{F}}_p)$ is diagonalizable.

Solution

(a) Suppose $A \in \mathrm{GL}_n(\mathbb{C})$ has finite order, so that $A^k = I$ for some k. Then we know the minimal polynomial of A divides $x^k - 1$. Now, over $\mathbb{C}$, $x^k - 1$ splits into k different linear factors, so the minimal polynomial also splits into different linear factors. Since the exponents on the linear factors of the minimal polynomial are the sizes of maximal Jordan blocks, this tells us that the size of a maximal Jordan block is 1, which is the same as saying A is diagonalizable.

(b) Everyone's favorite counterexample:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(c) Let $\mu_A(x)$ be the minimal polynomial of A. Let $\mathbb{F}_q \supset \mathbb{F}_p$ be the smallest extension containing all the roots of $\mu_A(x)$. Then all the roots of $\mu_A(x)$ are nonzero elements of $\mathbb{F}_q$, and so satisfy $\alpha^{q-1} = 1$. Since $\mu_A(x)$ factors over $\mathbb{F}_q$, write

$$\mu_A(x) = (x - \alpha_1)^{e_1} \cdots (x - \alpha_k)^{e_k}.$$

If all the e_i were 1 then we would just know that $\mu_A(x) \mid x^{q-1} - 1$, and we would be done, but if one of the $e_i > 1$ then we want to find another polynomial of the form $x^{\text{something}} - 1$ which μ_A divides. We know $x^{q-1} - 1$ has distinct roots, so if $e = \max_i(e_i)$, then $\mu_A(x) \mid (x^{q-1} - 1)^e$, but this isn't in the form we want yet. To get it in the desired form, we can instead choose the smallest power of p which is greater than e, since powers of p distribute over addition in characteristic p. Namely, set $m = \lceil \log_p(e) \rceil$, then we have $\mu_A(x) \mid (x^{q-1} - 1)^{p^m} = x^{p^m(q-1)} - 1$. Thus, $A^{p^m(q-1)} = I$. (I actually think this exponent might be optimal, but I'm not sure. Lemme know if you figure out one way or the other.)

(d) The problem with our proof from part (a) is that we said $x^k - 1$ splits into *distinct* linear terms over $\mathbb{C}$, but that need not be true in $\overline{\mathbb{F}}_p$. For example, $x^p - 1$ definitely doesn't. So, take the example of

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

over $\overline{\mathbb{F}}_2$.

August 2021 Problem 1 On this problem, only the answer will be graded.

(a) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization $R_f = 0$?

(b) Put $R = \mathbb{Z}/12\mathbb{Z}$. For which elements $f \in R$ is the localization R_f a field? (Recall that the zero ring is not a field.)

(c) The Jordan form of an operator $A : \mathbb{C}^6 \to \mathbb{C}^6$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

What is the Jordan form of the operator $A^3 - A$?

Solution

(a) $f = 0, 6$, since these are the only nilpotent elements of R.

(b) We can answer this question by thinking about the prime ideals of R_f. We know R has two prime ideals: (2) and (3). In order for R_f to be a field, it needs to have exactly one prime ideal, which must be the zero ideal. We know the primes in the localization will be the prime ideals of R which do not contain a power of f. A prime ideal contains a power of f iff it contains f, so in order to eliminate one of the prime ideals we need to choose either $f \in (3)$ or $f \in (2)$. We already determined in part (a) that localizing at $f = 6$ will give us the zero ring, which is not a field. Localizing at an element of (3) which is not in (6) (so $f = 3$ or $f = 9$) will leave us with a ring with three ideals ((1), (2), and (4) = (0)), which is not a field. So, our other option is localizing at any element of (2) which is not in (6) ($f = 2, 4, 8$, or 10), and all of these do work.

(c) It is possible to do this problem computationally, but working with 6×6 matrices seems like a lot of work, so let's try to do this using other properties of JNF. From the Jordan form of A, we can see that its minimal polynomial is $f(x) = x^4(x-1)^2$. Thus, the minimal polynomial of $A^3 - A$ is x^4, because in the ring $\mathbb{C}[x]/f(x)$, the polynomial $x^3 - x = x(x-1)(x+1)$ is nilpotent of nilpotence degree 4 (i.e. $(x^3 - x)^4 = 0$ but $(x^3 - x)^k \neq 0$ for $k < 4$). This tells us that the only eigenvalue of $A^3 - A$ is 0, and the size of the largest Jordan block is 4. This leaves us with only two possible Jordan forms for $A^3 - A$: it has Jordan blocks of sizes $(4, 2)$ or of sizes $(4, 1, 1)$.

We can tell these two possibilities apart by checking the dimension of the 0-eigenspace (i.e. the kernel) of $A^3 - A$: if it is 2-dimensional then we are in the former case, and if it is 3-dimensional we are in the latter case. The kernel of $A^3 - A$ will be the sum of the kernels of A, $A - I$, and $A + I$ (this is true in general for polynomials in terms of matrices). That is, $\ker(A^3 - A)$ will be the sum of the 0-, 1-, and -1-eigenspaces of A. We can tell from the Jordan form of A that the 0-eigenspace is 1-dimensional, as is the 1-eigenspace, and there are no eigenvectors of eigenvalue -1, so $\dim \ker(A^3 - A) = 2$, and we are in the case where we have Jordan blocks of size $(4, 2)$. Thus, the Jordan form of $A^3 - A$ is

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

January 2021 Problem 1 On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a finite extension of $\mathbb{Q}$ which is not Galois.

(b) Give an example of two-sided ideal of the ring of 2×2 matrices over $\mathbb{Z}$ which is neither 0 nor the whole ring.

(c) Give an example of an injective map $M \to N$ of $k[x]$-modules (k a field) and another $k[x]$-module R such that $M \otimes R \to N \otimes R$ is not injective.

(d) Give an example of a commutative ring R and prime ideals p, q, q' in R such that the extension $\bar{q}$ of q to the localization R_p is prime, while the extension $\overline{q'}$ of q' to R_p is not prime.

Solution

(a) $\mathbb{Q}(\sqrt[4]{2})$

(b) $M_2(2\mathbb{Z})$

(c) Consider $k[x] \xrightarrow{\cdot x} k[x]$ and $R = k[x]/(x)$. Then $k[x] \otimes_{k[x]} R \cong R \xrightarrow{\cdot x} k[x] \otimes_{k[x]} R \cong R$ is not injective since it is the zero map.

(d) Let $R = \mathbb{Z}$, $p = (2)$, $q = (0)$, and $q' = (3)$.

January 2021 Problem 4, edited Let $F/\mathbb{Q}$ be a quadratic extension, and let f and g be two maps of $\mathbb{Q}$-algebras from F to $M_2(\mathbb{Q})$. (**NB** in this problem we take algebra homomorphisms to be *unital*, meaning that $f(1) = g(1) = I$.)

(a) Prove that there exists a matrix $X \in \mathrm{GL}_2(\mathbb{Q})$ such that for all $\alpha \in F$, $Xf(\alpha)X^{-1} = g(\alpha)$.

(b) Prove that there does not exist a choice of f whose image is contained in the set of upper triangular matrices.

Solution

(a) For any $\mathbb{Q}$-algebra homomorphism $f\colon F \to M_2(\mathbb{Q})$, we must have that $f(q) = qI$ for all $q \in \mathbb{Q}$. So, if $F = \mathbb{Q}(\sqrt{n})$, we need only show that there is an invertible matrix X so that $Xf(\sqrt{n})X^{-1} = g(\sqrt{n})$. Let $A = f(\sqrt{n})$, then A must satisfy $A^2 - nI = 0$, so the minimal polynomial of A divides $x^2 - n$. Since $x^2 - n$ is irreducible over $\mathbb{Q}$, in fact it must be the minimal polynomial of A, so the eigenvalues of A are $\pm\sqrt{n}$. Thus, over $\mathbb{C}$ we can find an invertible matrix Y so that

$$YAY^{-1} = \begin{pmatrix} \sqrt{n} & 0 \\ 0 & -\sqrt{n} \end{pmatrix}.$$

The same is true for $B = g(\sqrt{n})$, so this shows that over $\mathbb{C}$ we can find an X so that $XAX^{-1} = B$. Then $XA - BX = 0$. Notice that $T \mapsto TA - BT$ is a linear transformation $M_2(\mathbb{Q}) \to M_2(\mathbb{Q})$, and we have just shown that it has nontrivial kernel as a linear transformation $M_2(\mathbb{C}) \to M_2(\mathbb{C})$. Therefore, it must have nontrivial kernel as a linear transformation $M_2(\mathbb{Q}) \to M_2(\mathbb{Q})$, and so we can take $X \in \mathrm{GL}_2(\mathbb{Q})$.

(b) As noted above, $A = f(\sqrt{n})$ has eigenvalues $\pm\sqrt{n}$. If A were upper triangular, its diagonal entries would have to be its eigenvalues, but $\pm\sqrt{n} \notin \mathbb{Q}$.

August 2020 Problem 1 On this problem, no justification is required; only the answer will be graded.

(a) Give an example of a non-diagonalizable matrix in $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(b) Give an example of a nontrivial proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_p)$ (for some p).

(c) Give an example of a one-sided ideal (either left or right) in $M_2(\mathbb{C})$ which is not a two-sided ideal.

(d) Give an example of a $\mathbb{Z}$-module which is flat but not free.

(e) Give an example of an exact sequence of abelian groups which is not split.

Solution

(a) Let $p = 2$ and consider $M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

(b) Let $p = 3$. Then $\mathrm{SL}_3(\mathbb{F}_3)$ is a proper normal subgroup of $\mathrm{GL}_3(\mathbb{F}_3)$.

(c) Let I be the ideal of matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$. Then I is a left ideal but not a right ideal.

(d) $\mathbb{Q}$

(e) $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$

August 2019 Problem 1 On this problem, only the answers will be graded.

(a) Let $R = M_2(\mathbb{Z})$ the ring of 2×2 matrices with entries in $\mathbb{Z}$. Give an example of a left ideal that is not a right ideal.

(b) Give an example in $R = M_2(\mathbb{Z})$ of a proper, nonzero, two-sided ideal.

(c) $\mathbb{Z}/100\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/35\mathbb{Z}$ is a finite abelian group. Compute its order.

Solution

(a) Let I be the ideal of matrices of the form $\begin{pmatrix} a & 0 \\ b & 0 \end{pmatrix}$. Then I is a left ideal but not a right ideal.

(b) The set of matrices with even entries is a proper, nonzero, two-sided ideal of $M_2(\mathbb{Z})$.

(c) We have that $\mathbb{Z}/100\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/35\mathbb{Z} \cong \mathbb{Z}/\gcd(100, 35)\mathbb{Z} = \mathbb{Z}/5\mathbb{Z}$, so the tensor product has order 5.

August 2019 Problem 3, edited Let L/K be a Galois extension of fields with Galois group G. Let S be the subset of roots of unity in L other than 1; that is, S is the set of elements $x \in L$ such that $x \neq 1$ but $x^k = 1$ for some positive integer k.

(a) Show that the action of G on L restricts to an action of G on S (i.e., the action of G preserves S.)

(b) Let $L = \mathbb{Q}(i)$ and $K = \mathbb{Q}$. Show in this case that G acts faithfully on S.

(c) Give an example of L/K such that the action of G on S is not faithful.

(d) Suppose that the action of G on S is transitive. Prove that $\text{char}(K) = 2$.

Solution

(a) The nth roots of unity in L are precisely the elements in L that are roots of the equations $t^n - 1$, and the roots of unity different from 1 are the roots of $(t^n - 1)/(t - 1)$. Since $(t^n - 1)/(t - 1) \in K[t]$, G must permute the roots of this polynomial. Thus, G also acts on S.

(b) In this case, $S = \{\text{-}1, \pm i\}$, and G is cyclic of order 2 generated by complex conjugation σ. So the action of G on S is given by id acting trivially and σ acting by $i \mapsto -i$. So this action is faithful since only the identity element in G acts trivially on S.

(c) Let $L = \mathbb{Q}(\sqrt{2})$ and $K = \mathbb{Q}$. Then $S = \{\text{-}1\}$ since ± 1 are the only roots of unity in $\mathbb{R}$, and $L \subseteq \mathbb{R}$. As in the previous example, G is a cyclic group of order 2. But both elements of G fix S since $S \subseteq \mathbb{Q}$.

(d) Imagine K has characteristic not equal to 2. Then -1 and 1 are distinct elements of $S \cap K$. But every element of G will fix -1 since it is in K, so the action could not be transitive.

January 2019 Problem 1 On this problem, only the answers will be graded.

(a) Give an example of a simple ring which is not a commutative ring.

(b) Give an example a ring R, R-modules M_1, M_2, M_3, and N and a short exact sequence $0 \to M_1 \to M_2 \to M_3 \to 0$ such that the corresponding sequence
$0 \leftarrow \text{Hom}(M_1, N) \leftarrow \text{Hom}(M_2, N) \leftarrow \text{Hom}(M_3, N) \leftarrow 0$ is not exact.

(c) Let Φ be the map $\mathbb{Z}^3 \to \mathbb{Z}^2$ given by the matrix $\begin{bmatrix} 2 & 3 & 5 \\ 12 & \text{-}4 & 8 \end{bmatrix}$. Let M be the cokernel of Φ, which we recall is $\mathbb{Z}^2/\operatorname{im}(\Phi)$. Compute the annihilator of M.

Solution

(a) Let R be the ring of 2×2 matrices over $\mathbb{F}_2$. Then R is not commutative, and R is simple since the only nonzero two-sided ideal is R.

(b) Let $R = \mathbb{Z}$ and consider $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$. Then applying $\operatorname{Hom}(-, \mathbb{Z}/2\mathbb{Z})$ gives

$$0 \to \operatorname{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \to \operatorname{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \to \operatorname{Hom}(\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z} \to 0$$

which is not exact on the right since the map is multiplication by $2 = 0$, so it is not surjective.

(c) I guess this space will be where I write about Smith Normal Form. Notice that computing the cokernel of Φ would be easy if we had a matrix of the form

$$\begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \end{bmatrix},$$

as then the cokernel would clearly be $\mathbb{Z}/a\mathbb{Z} \oplus \mathbb{Z}/b\mathbb{Z}$. The goal of Smith Normal Form is to rewrite a given matrix in a new pair of bases so that it looks like this.

Computing SNF is very similar to row-column reduction over a field, with two notable caveats:

(i) You cannot multiply or divide rows/columns by scalars (other than ± 1). Instead, your only available techniques are to swap rows/colums, and to add multiples of one row/column to another. You can use this to replace an element with the gcd of all the elements in its row/column, and then clear those other elements as usual.

(ii) Once you clear all the entries in the column below a pivot, you might have to "ruin" your work in order to clear the entries to the right in the row, and visa versa. However, by alternately clearing the entries below and to the right, eventually you will be able to clear both. (Luckily, this process doesn't actually come up in this particular problem.)

We begin with the matrix from the problem. I notice that in the first row, the elements have gcd 1, so I'm going to subtract the first column from the second column, then swap the first two columns:

$$\begin{bmatrix} 1 & 2 & 5 \\ \text{-}16 & 12 & 8 \end{bmatrix}.$$

Now I can use the 1 in the first row to clear the -16 below it, and the 2 and 5 to the right of it (the order that I do this does not matter, because it corresponds to multiplying by matrices on either side, and matrix multiplication is associative). Once I do, I arrive at this matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 44 & 88 \end{bmatrix}.$$

Luckily, the process ends right away, since I can use the 44 to clear the 88, so the SNF is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 44 & 0 \end{bmatrix}.$$

Thus, the cokernel of Φ is isomorphic to $\mathbb{Z}/44\mathbb{Z}$, and the annihilator is 44.

(Here's another true thing: the annihilator of the cokernel of Φ is equal to the gcd of the maximal subdeterminants of Φ. In this case, the subdeterminants are -44, -44 and 44, and the gcd of those numbers is easy to find. In general, this is a strictly more cumbersome algorithm than computing the Smith Normal Form directly, but it works nicely when you only have a small number of 2×2 subdeterminants.)

August 2018 Problem 5 Let A be a two-dimensional (unital) algebra over a field F. This means A is an associative, but not necessarily commutative, ring with a unit that contains F as a subring, such that the elements of F commute with the elements of A (that is, F is in the center) and A is two-dimensional as a vector space over F.

(a) Show that A must in fact be commutative.

(b) Show that if F is algebraically closed, then either $A \simeq F \times F$ or $A \simeq F[x]/(x^2)$.

(c) Suppose $F = \mathbb{R}$. List (with a proof) all the possibilities for A, up to isomorphism.

Solution

(a) Since A is 2-dimensional as an F-vector space, and contains F, we can choose a basis of the form $1, a$ for some $a \in A$. Then any element of A can be written as $x + ya$ with $x, y \in F$. Since 1 and a certainly commute with one another, in fact all the elements of A commute with one another.

(b) Since A is a commutative F-algebra generated as an algebra by the element a, there is a map $F[x] \to A$ sending $x \to a$, and the kernel of this map is an ideal of $F[x]$. Such an ideal is principal, since $F[x]$ is a PID, and its generator must be a degree 2 polynomial, since A is 2-dimensional. As F is algebraically closed, this polynomial factors into linear terms. If it factors into distinct linear terms, then we get $A \simeq F[x]/(x-a)(x-b) \simeq F[x]/(x-a) \times F[x]/(x-b)$ by the Chinese Remainder Theorem, so $A \simeq F \times F$. On the other hand, if it has a repeated root, then $A \simeq F[x]/((x-a)^2) \simeq F[x]/(x^2)$.

(c) In this case, in addition to the two possibilities in part (b), we could also have that the polynomial is irreducible over $\mathbb{R}$. In such a case, by the fundamental theorem of algebra we get that $A \simeq \mathbb{C}$.

August 2017 Problem 2 Let K be a field, and let A be an $n \times n$ matrix over K. Suppose that $f(x) \in K[x]$ is an irreducible polynomial such that $f(A) = 0$. Show that $\deg(f) \mid n$.

Solution

Let $m(x) \in K[x]$ be the minimal polynomial of A. Since $f(A) = 0$, $m(x)$ must divide $f(x)$, but $f(x)$ is irreducible so $m(x) = f(x)$. Let $c(x)$ be the characteristic polynomial of A, which has degree n. Since $c(x)$ and $f(x)$ have the same roots and $f(x)$ is irreducible, it must be the case that $c(x) = f(x)^\ell$ for some ℓ. Therefore, $\ell \cdot \deg(f) = \deg(c) = n$, so $\deg(f) \mid n$.

January 2016 Problem 5 Consider the group $\mathrm{GL}_n(\mathbb{Q})$ of invertible $n \times n$ matrices with rational coefficients. Suppose $G \subset \mathrm{GL}_n(\mathbb{Q})$ is a finite subgroup. Prove that every prime factor p of the order $|G|$ satisfies $p \leq n + 1$.

Solution

Suppose that $p \mid |G|$, then by Cauchy's theorem there is an element $A \in G$ of order p, $A^p = I$. Notice that the minimal polynomial of A divides $x^p - 1$, so the minimal polynomial of A is either $x^p - 1$ or $\Phi_p(x)$ (notice it cannot be $x - 1$). Since the minimal polynomial divides the characteristic polynomial, the degree of the minimal polynomial is at most n. Thus, if the minimal polynomial is $x^p - 1$ then we get $p \leq n$, and if the minimal polynomial is $\Phi_p(x)$ we get $p - 1 \leq n$. Thus, in all cases $p \leq n + 1$.

I was curious whether or not the bound on p can actually be achieved, and the answer is it can. For example, the 2×2 matrix

$$A = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}$$

has order 3.

2 Linear Algebra

January 2026 Problem 4 Let F be an algebraically closed field, and $a_1, \dots, a_n \in F - \{0\}$. Consider the matrix

$$A = \begin{pmatrix} 0 & 0 & \dots & 0 & a_n \\ a_1 & 0 & \dots & 0 & 0 \\ 0 & a_2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & a_{n-1} & 0 \end{pmatrix}$$

over F.

(a) Suppose F has characteristic 0. Find the eigenvalues and eigenspaces of A. (Of course, the answer would involve the a_i's.)

(b) Suppose F has characteristic 0. Find the Jordan form of A.

(c) Suppose $n = p$ is prime and F has characteristic p. Find the Jordan form of A.

Solution

(a) To find the eigenvalues, let us first find the characteristic polynomial. Expanding $\det(\lambda I - A)$ along the first row, we get

$$\det \begin{pmatrix} \lambda & 0 & \dots & 0 & -a_n \\ -a_1 & \lambda & \dots & 0 & 0 \\ 0 & -a_2 & \ddots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & -a_{n-1} & \lambda \end{pmatrix} = \lambda \det \begin{pmatrix} \lambda & 0 & \dots & 0 \\ -a_2 & \lambda & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & -a_{n-1} & \lambda \end{pmatrix} +$$

$$(-1)^n a_n \det \begin{pmatrix} -a_1 & \lambda & \dots & 0 \\ 0 & -a_2 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \lambda \\ 0 & 0 & \dots & -a_{n-1} \end{pmatrix}.$$

The first matrix is lower triangular and the second is upper triangular, so we easily compute that the characteristic polynomial of A is $\lambda^n + (-1)^{2n-1} \prod_j a_j = \lambda^n - \prod_j a_j$. This makes sense since the matrix A cycles the standard basis vectors and multiplies by the constants a_i as it does so: $Ae_i = a_i e_{i+1}$. Thus, we can see that $A^n e_i = \left(\prod_j a_j\right) e_i$ for all i, so A^n is the scalar matrix with $\prod_j a_j$ along the diagonal.

Set $P = \prod_j a_j$, then the eigenvalues of A are $\zeta^k \sqrt[n]{P}$ for $0 \leq k \leq n-1$, where ζ is a primitive nth root of unity. Since the characteristic polynomial has distinct roots, the eigenspaces of A are all 1-dimensional.

By "find the eigenspaces", I assume they mean write down eigenvectors corresponding to each eigenvalue. That seems a bit mean, because the eigenvectors don't look particularly nice. Here's the cleanest way I found to write them. Reckon all indices cyclically, so $n+1$ means the same thing as 1. Let v be the vector whose ith component is

$$v_i = \zeta^{-ik} \sqrt[n]{\prod_{j=1}^{n-1} a_{i+j}^j} = \zeta^{-ik} \sqrt[n]{a_{i+1}a_{i+2}^2 \cdots a_{i-1}^{n-1}}.$$

Then the ith component of Av is

$$\begin{aligned} a_{i-1}v_{i-1} &= a_{i-1} \cdot \zeta^{-(i-1)k} \sqrt[n]{a_i a_{i+1}^2 \cdots a_{i-2}^{n-1}} \\ &= \zeta^{-(i-1)k} \sqrt[n]{a_i a_{i+1}^2 \cdots a_{i-2}^{n-1} a_{i-1}^n} \\ &= \zeta^k \sqrt[n]{a_i a_{i+1} \cdots a_{i-1}} \left(\zeta^{-ik} \sqrt[n]{a_{i+1} a_{i+2}^2 \cdots a_{i-1}^{n-1}} \right) \\ &= \zeta^k \sqrt[n]{P} v_i, \end{aligned}$$

so v is an eigenvector of eigenvalue $\zeta^k \sqrt[n]{P}$.

(b) The Jordan form of A in this case is a diagonal matrix with $\zeta^k \sqrt[n]{P}$ down the diagonal.

(c) In this case, now $\lambda^n - P = (\lambda - \sqrt[n]{P})^n$ no longer has distinct roots, but the $\sqrt[n]{P}$-eigenspace will still be 1-dimensional. Indeed, $\operatorname{rank}(\sqrt[n]{P}I - A) = n-1$ since the first $n-1$ rows are linearly independent. Thus, in this case the Jordan form of A will consist of a single Jordan block of eigenvalue $\sqrt[n]{P}$.

August 2025 Problem 4 Fix a field F.

(a) Let V be a finite dimensional vector space over F, and let $U \subset V$ be a subspace. Put $n = \dim(V)$, $k = \dim(U)$. Denote by V^* the dual space of V, so its elements are functionals (linear transformations) $\phi\colon V \to F$. Find the dimension of the subspace

$$V_U^* = \{\phi : \phi|_U = 0\} \subset V^*.$$

(b) Denote by $Gr(n,k)$ the set of all k-dimensional subspaces of the space F^n. Construct a bijection between $Gr(n,k)$ and $Gr(n,n-k)$.

Solution

(a) The elementary but tedious way: Let $v_1, \ldots, v_k$ be a basis for U, and complete this to a basis $v_1, \ldots, v_n$ for V. Let ϕ_i be the functional that takes a vector v to the coefficient of v_i when v is written in this basis (also known as the dual basis). The ϕ_i are linearly independent since for any linear combination $a_1\phi_1 + \cdots + a_n\phi_n$ with, say, $a_i \neq 0$, when we evaluate at v_i we get $a_1\phi_1(v_i) + \cdots + a_n\phi_n(v_i) = a_i$, so this linear combination is clearly not the 0 homomorphism. We claim that $\phi_{k+1}, \ldots, \phi_n$ span V_U^*. Indeed, let $\phi \in V_U^*$, and let $b_i = \phi(v_i)$. Then the linear combination $b_{k+1}\phi_{k+1} + \cdots + b_n\phi_n$ agrees with ϕ at every basis vector v_i, and so $\phi = b_{k+1}\phi_{k+1} + \cdots + b_n\phi_n$. Thus, $\dim V_U^* = n - k$.

The fast way: a functional vanishing on U is the same thing as a functional on V/U. Thus, $V_U^* \cong (V/U)^*$. Since the dual of a finite dimensional vector space has the same dimension as the original vector space (by taking the dual basis), $\dim(V/U)^* = \dim(V/U) = n - k$.

(b) Notice that $(F^n)^* \cong F^n$. Define a function $Gr(n,k) \to Gr(n, n-k)$ by sending a k-dimensional subspace U to the $n-k$-dimensional subspace $(F^n)_U^*$ inside $(F^n)^* \cong F^n$.

January 2025 Problem 4 Let V be a dimension 4 vector space over $\mathbb{C}$ and $\phi\colon V \to V$ an operator. Suppose that a dimension 2 subspace W of V is invariant in the sense that $\phi(W) \subset W$; then ϕ induces operators $W \to W$ and $V/W \to V/W$.

Suppose that the Jordan form of ϕ on both W and V/W is a single 2×2 Jordan block with eigenvalue 0. Find all possible Jordan forms of ϕ on V.

(For full credit, you need to provide examples that show your answers indeed can happen and prove that there are no other possibilities.)

Solution

We know the only possible eigenvalue of ϕ is 0, and that ϕ must have a Jordan block of size at least 2. There are four possible Jordan forms satisfying those conditions, and we will show that only three can arise. Namely, the Jordan form of ϕ can be: a single 4×4 block; a 1×1 block and a 3×3 block; or two 2×2 blocks. That is, the Jordan matrix of ϕ can be any of

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \text{or} \quad \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Here are two helpful things for approaching this problem. First, saying we know how ϕ acts on W and on V/W is basically the same as saying we know 3 out of 4 blocks of a block matrix representing ϕ. Specifically, if we choose a basis v_1, v_2, v_3, v_4 so that v_1, v_2 are a Jordan basis for W, and $v_3 + W, v_3 + W$ are a Jordan basis for V/W, then we know the matrix of ϕ acting on v_1, v_2 is a 2×2 Jordan block. We also know that ϕ acting on v_3, v_4 is a 2×2 Jordan block modulo W, which is the same as saying it's a 2×2 Jordan block plus some linear combination of v_1 and v_2. This means that the matrix of ϕ with respect to this basis look like

$$\begin{pmatrix} 0 & 1 & * & * \\ 0 & 0 & * & * \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

where the $*$ represent unknown entries. So, our task is essentially to understand all the possible

Jordan forms that can come about for different values of the $*$ entries.

The second helpful thing is what exactly you're looking for to determine the Jordan form of ϕ. To get a matrix in Jordan form, you need a basis of Jordan chains, and a Jordan chain of eigenvalue λ is a sequence of vectors $v_k, v_{k-1}, \ldots, v_1$ such that $\phi(v_k) = \lambda v_k + v_{k-1}$. (This is where the staircase shape of Jordan matrices comes from.) The general way to build a Jordan chain is to start with v_k which you know is a generalized eigenvector of eigenvalue λ, and recursively define $v_{i-1} = \phi(v_i) - \lambda v_i = (\phi - \lambda I)v_i$.

So, in this case, to build our Jordan chain we should start with v_4 and recursively work our way up. We know $\phi(v_4) = v_3 + w$ for some $w \in W$. We can change basis by declaring $v_3' = v_3 + w$, so without loss of generality we may assume $w = 0$. Now, there are several possibilities for $\phi(v_3)$:

(a) We could have $\phi(v_3) = 0$. In this case, v_1, v_2, v_3, v_4 is already a Jordan basis for ϕ, and it has two Jordan blocks of size 2.

(b) Perhaps $\phi(v_3) \neq 0$ but $\phi(\phi(v_3)) = 0$, that is $\phi(v_3) \in \ker \phi$. In this case, we know $\phi(v_3) = cv_1$ for some nonzero scalar c. In this case, notice that $\phi(cv_2 - v_3) = cv_1 - cv_1 = 0$, so v_1 and $cv_2 - v_3$ are two linearly independent elements of $\ker \phi$. Thus, $cv_2 - v_3, cv_1, v_3, v_4$ is a Jordan basis for ϕ, having one Jordan block of size 1 and one of size 3.

(c) Perhaps $\phi(v_3) \notin \ker \phi$. In this case, we can change basis by $v_2' = \phi(v_3)$, $v_1' = \phi(v_2')$, and the basis v_1', v_2', v_3, v_4 is a Jordan basis for ϕ, having one Jordan block of size 4.

All that is left to do is to translate this proof that these are the only possibilities into examples of all three happening. For the Jordan block of size 4, we can take $W = \mathrm{span}(e_1, e_2)$, for the Jordan blocks of sizes (1,3) we can take $W = \mathrm{span}(e_1 + e_3, e_2)$, and for the two Jordan blocks of size 2 we can again take $W = \mathrm{span}(e_1, e_2)$.

(There is a way to approach this question as a question about $\mathbb{C}[x]$-modules, where subtly what this question is asking is almost but not quite to compute an Ext group.)

August 2024 Problem 3 For an integer $n \geq 1$ consider the following $n \times n$ matrix:

$$A = \begin{pmatrix} 0 & & & & 1 \\ & & & 1 & \\ & & \iddots & & \\ & 1 & & & \\ 1 & & & & 0 \end{pmatrix}.$$

Let V denote the set of all $n \times n$ matrices over $\mathbb{C}$ that are upper triangular and commute with A.

(a) Show that V is a vector space over $\mathbb{C}$.

(b) Find a basis for V.

(c) Find the dimension of V.

Solution

(a) You know how to do this.

(b) I recommend starting by computing this in the cases $n = 2, 3$. I think it's worth noting that $A = A^{-1}$, so saying that $MA = AM$ is the same as saying that $AMA = M$. However, if M is upper triangular then AMA is lower triangular, so in fact every matrix in V is diagonal. Moreover, AMA reverses the order of the diagonal elements, so we must also have that $m_{ii} = m_{n-i,n-i}$ for all i. Thus, we have a basis for V composed of matrices M_i which has $m_{ii} = m_{n-i,n-i} = 1$ and all other entries equal to 0.

(c) $\lceil \frac{n}{2} \rceil$.

August 2023 Problem 3 Let V be a four-dimensional real vector space with a positive definite inner product, and let $R\colon V \to V$ be an orientation-preserving automorphism that respects the inner product (that is, $\langle Rv, Rw\rangle = \langle v, w\rangle$ for all vectors $v, w \in V$).

(a) Show that V decomposes as the direct sum of two subspaces V_1 and V_2, each of dimension 2, each of which is preserved by R.

(b) Show furthermore that V_1 and V_2 can be chosen to be orthogonal (that is, $\langle v_1, v_2\rangle = 0$ for any $v_1 \in V_1$, $v_2 \in V_2$).

(**NB** in this context, "orientation-preserving" means $\det R > 0$.)

Solution

Below, I'm writing the solution in the level of abstraction given in the problem. The main idea is that, if we pretend the inner product is the usual dot product, then the condition that R preserves the inner product means that R is an orthogonal matrix, and the condition of being orientation-preserving means $\det R = 1$, so in fact R lies in the special orthogonal group $\mathrm{SO}_4(\mathbb{R})$. An orthogonal transformation is a composition of rotations and reflections, and since R is orientation-preserving we can decompose it in a way that has either 0, 2, or 4 reflections. If there are 0 reflections, we want to find 2 orthogonal subspaces in which R acts by rotations. If there are 2 reflections, we take the subspace of reflected vectors as one of our subspaces, and want to find an orthogonal subspace in which R acts by rotations. If the whole space is reflected, then any pair of orthogonal subspaces will do.

To read a bit more about this from a concrete perspective, here's a short document I found which I think is pretty good.

(a) If we're looking for a decomposition of a vector space, the most natural place to look is at eigenspaces. Unfortunately, the map R need not have any eigenvectors, since its eigenvalues might be complex. To remedy this, we can take $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ to be the complexification of V. That is to say, $V_{\mathbb{C}}$ is a 4-dimensional complex vector space, and we can think of any basis for V as being a basis for $V_{\mathbb{C}}$ (once we allow for complex scalars). We can extend the inner product on V to a Hermitian inner product on $V_{\mathbb{C}}$ via $\langle \alpha v, \beta w\rangle = \alpha\overline{\beta}\langle v, w\rangle$ (and extending bilinearly to sums). Note that R can now be interpreted as a map $V_{\mathbb{C}} \to V_{\mathbb{C}}$, and it preserves this Hermitian inner product. We can also view $V_{\mathbb{C}}$ as an 8-dimensional **real** vector space which contains V (as the real subspace where all the complex parts are equal to 0). When we view $V_{\mathbb{C}}$ in this way, we will denote the subspace corresponding to our original V as $V_{\mathbb{R}}$.

Let's study the eigenspaces of R acting on $V_{\mathbb{C}}$. Since $V_{\mathbb{C}}$ is a complex vector space, for every root λ of the characteristic polynomial of R, there is an eigenvector $v \in V_{\mathbb{C}}$ of eigenvalue λ. If v is such an eigenvector, then $\langle v, v\rangle = \langle Rv, Rv\rangle = |\lambda|^2\langle v, v\rangle$, so since $v \neq 0$, we must have $|\lambda| = 1$ for all eigenvalues of R. Since the characteristic polynomial of R has real coefficients, if λ is an eigenvalue of R, so too is $\bar{\lambda}$, and indeed $R\bar{v} = \overline{Rv} = \bar{\lambda}\bar{v}$.

If λ is real, then we already have that $v \in V_{\mathbb{R}}$. If $\lambda \in \mathbb{C} \setminus \mathbb{R}$, write $\lambda = a + bi$. Note that $u_1 = v + \bar{v}$ and $u_2 = (v - \bar{v})/i$ are linearly independent vectors in $V_{\mathbb{R}}$, and moreover $Ru_1 = au_1 - bu_2$, $Ru_2 = bu_1 + au_2$. So, each complex eigenvalue gives us a 2-dimensional subspace of $V = V_{\mathbb{R}}$ preserved by R, which is what we are looking for. Each real eigenvalue gives us a 1-dimensional subspace, but since complex eigenvalues occur in pairs, if we have 1 real eigenvalue, we are guaranteed to have 2, and so we can pair up those eigenspaces as well. Furthermore, since we know that $\det R > 0$, we can always find pairs of real eigenvalues of the form $(1, 1)$ or (-1, -1). The last thing we need to check is that if we have an eigenvalue which is a root of the characteristic polynomial of multiplicity greater than 1, then we actually can find the right number of independent eigenvectors. That is, we need R to be diagonalizable.

This argument is one instance of Schur's Lemma, which has a few different possible forms it can take. Let v be an eigenvector of eigenvalue λ, and let $W = \langle v\rangle$ be the subspace generated by v. Consider the action of R on the subspace $W^\perp$, the orthogonal complement of W. Since R preserves the Hermitian inner product, for any $w \in W^\perp$, $\langle v, Rw\rangle = (1/\lambda)\langle Rv, Rw\rangle = (1/\lambda)\langle v, w\rangle = 0$, so R indeed maps $W^\perp$ to itself. But then, R must have an eigenvector inside $W^\perp$. Continuing this process, we find that R has 4 eigenvectors, and moreover these eigenvectors can be chosen to be orthogonal.

(b) This follows immediately from the argument that the eigenvectors can be taken to be orthogonal.

You might wonder (as I did) why we extend the inner product by declaring $\langle \alpha v, \beta w\rangle = \alpha\bar{\beta}\langle v, w\rangle$ rather than $\alpha\beta\langle v, w\rangle$, which seems more straightforward and natural. The reason comes up when we discussed the eigenvalues of R: if we don't impose conjugate-linearity, we will get a nondegenerate, symmetric bilinear form, but we might have $\langle v, v\rangle = 0$ for nonzero vectors v. For example, choose an orthonormal basis $v_1, \ldots, v_4$ for V, then if we extended the inner product bilinearly, we would get $\langle v_1 + iv_2, v_1 + iv_2\rangle = 1 - 1 = 0$.

January 2022 Problem 4 Let V and W be vector spaces over a field k, not necessarily finite dimensional. As usual, we denote by $\mathrm{Hom}_k(V, W)$ the space of linear maps from V to W, and by $V^\vee$ the dual of V, so that $V^\vee = \mathrm{Hom}_k(V, k)$.

(a) Construct a natural injective linear operator

$$\Phi \colon V^\vee \otimes_k W \to \mathrm{Hom}(V, W),$$

and prove that it is indeed injective. ("Natural" means it should be independent of any arbitrary choices.)

(b) Prove that the operator Φ is an isomorphism if V and W are finite-dimensional.

(c) Prove a necessary and sufficient condition that a linear map $A \in \mathrm{Hom}(V, W)$ is in the image of Φ. (The assumption that V and W are finite-dimensional does not apply to this part.)

Solution

(a) Let $f \colon V \to k$ be an arbitrary element of $V^\vee$, and $w \in W$. We define $\Phi(f \otimes w)$ to be the linear map that sends $v \in V$ to $f(v)w \in W$. Then, for an arbitrary linear combination of tensors in $V^\vee \otimes_k W$, we extend the map linearly. (You are free to decide for yourself how "natural" you find this map to be.)

Showing this map is injective is a bit annoying. Take any nonzero tensor $t = \sum_i f_i \otimes w_i$ in $V^\vee \otimes_k W$, and assume the sum for t above has the fewest summands among all possible representations, and in particular that all the f_i and w_i are linearly independent. (As an example, suppose $t = f_1 \otimes w_1 + f_2 \otimes w_2 + f_3 \otimes w_3$, and that $w_3 = a_1 w_1 + a_2 w_2$. Then we can rewrite t as $(f_1 + a_1 f_3) \otimes w_1 + (f_2 + a_2 f_3) \otimes w_2$, which has fewer terms.) Then $\Phi(t)(v) = \sum_i f_i(v) w_i$, which is not equal to zero unless $f_i(v) = 0$ for all i by the assumption that the w_i are independent. But we can always find a $v \in V$ so that some $f_i(v) \neq 0$, so $\Phi(t)$ is not the zero map $V \to W$. Thus, Φ is injective.

(b) When V and W are finite-dimensional, we have

$$\dim(V^\vee \otimes_k W) = \dim(V \otimes_k W) = \dim(V)\dim(W) = \dim(\mathrm{Hom}(V, W)).$$

Therefore, any injective function $V^\vee \otimes_k W \to \mathrm{Hom}(V, W)$ is an isomorphism.

(c) A linear map $A \in \mathrm{Hom}(V, W)$ is in the image of Φ if and only if $\dim \operatorname{im} A < \infty$. Certainly, any $A \in \operatorname{im} \Phi$ has finite-dimensional image, because if we write $A = \Phi\left(\sum_i f_i \otimes w_i\right)$, then $\operatorname{im} A$ is spanned by the finitely many w_i. On the other hand, given any $A \colon V \to W$ with finite-dimensional image, choose any basis $v_1, v_2, \ldots$ for V, and $w_1, \ldots, w_n$ a basis for $\operatorname{im} A$. Then for $i = 1, \ldots, n$, the function

$$f_i(v_j) = \text{coefficient of } w_i \text{ in } A(v_j)$$

is a linear map $V \to k$, and clearly $A = \Phi\left(\sum_i f_i \otimes w_i\right)$. (Precisely, if we extend the w_i to a basis for W, f_i is the composition of A followed by the map $W \to k$ that sends w_i to 1 and all $w_{i'}$ to 0 for $i' \neq i$.)

August 2020 Problem 2 Let M be an $n \times n$ matrix with real entries. Let M^T be its transpose. Each of these two matrices defines a linear transformation $\mathbb{R}^n \to \mathbb{R}^n$; let A denote the transformation corresponding to M, and let B denote that corresponding to M^T.

(a) Prove that any vector in $\ker(A)$ is orthogonal to any vector in $\operatorname{im}(B)$ with respect to the standard inner product (dot product).

(b) Prove that $\mathbb{R}^n$ is the direct sum of $\ker(A)$ and $\operatorname{im}(B)$.

Solution

(a) The key thing here is that $v \cdot u = v^T u$. Let $v \in \ker(A)$ and $u' \in \operatorname{im}(B)$. Then there is some $u \in \mathbb{R}^n$ so that $u' = M^T u$. Then we have $v \cdot u' = v \cdot (M^T u) = v^T (M^T u) = (Mv)^T u = 0 \cdot u = 0$.

(b) This is the slick proof, using the internal direct product test. We need to check that $\ker(A)$ and $\operatorname{im}(B)$ (i) have dimensions that add up to n and (ii) have trivial intersection. By the rank-nullity theorem,

$$n = \operatorname{rank}(M) + \operatorname{null}(M) = \operatorname{rank}(M^T) + \operatorname{null}(M) = \dim \operatorname{im}(B) + \dim \ker(A).$$

Also, since any vector in $\ker(A)$ is orthogonal to any vector in $\operatorname{im}(B)$, $\ker(A) \cap \operatorname{im}(B) = \{0\}$. Thus, $\mathbb{R}^n = \ker(A) + \operatorname{im}(B)$.

January 2020 Problem 4 Let V be a finite-dimensional vector space over a field. Let $A \colon V \to V$ be a linear transformation.

(a) Show that there exists $B \colon V \to V$ such that $ABA = A$.

(b) Show that there exists $C \colon V \to V$ such that $ACA = A$ and $CAC = C$.

(**NB** a matrix C like in part (b) is known as a “pseudo-inverse” to A.)

Solution

(a) By row and column reducing, we can find bases $\mathcal{V} = \{v_1, \ldots, v_n\}$ and $\mathcal{W} = \{w_1, \ldots, w_n\}$ for $\mathbb{R}^n$ so that, with respect to the pair of bases $(\mathcal{V}, \mathcal{W})$, the matrix of A is block-diagonal

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

where $r = \operatorname{rank}(A)$. Then we can choose B to be the linear transformation whose matrix with respect to the bases $(\mathcal{W}, \mathcal{V})$ is

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $ABA = A$.

(b) The B from part (a) already does this.

August 2019 Problem 4 This problem involves 4×4 matrices with entries in $\mathbb{R}$. We will say that a matrix M is an **E-matrix** if M has 1's along the diagonal and a single nonzero entry off the diagonal. For instance:

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 31 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

is an E-matrix. Express the matrix A below as a product of E-matrices.

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$$

Solution

This is a straight up row reducing problem. Let $E_{i,j}$ be the matrix with 1s on the diagonal, a 1 in position (i, j), and 0s elsewhere. Then

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix} = E_{2,1}E_{3,1}E_{4,1}E_{1,2}E_{1,3}E_{1,4}$$

January 2019 Problem 5 Let F be a field and n be a positive integer. Fix an $n \times n$ matrix S over F that is invertible and symmetric. Writing A^t for the transpose of a matrix, we let

$$V := \{n \times n \text{ matrices } A \text{ over } F : A^t = SAS^{-1}\}.$$

Note that V is a vector space (you do not need to prove this). Find the dimension of V in terms of n.

Solution

Since S is an arbitrary invertible symmetric matrix, let's first think about V when $S = \mathrm{Id}_n$. In this case, V is the space of matrices that are equal to their own transpose, i.e. those that are symmetric. Thus, $\dim(V) = n + (n-1) + \cdots + 1 = \frac{1}{2}(n^2 + n)$. Now the question is, why is this the dimension of V for any invertible symmetric S?

Observe that $A^t = SAS^{-1}$ if and only if $A^tS = SA$, and since $S^t = S$, this is equivalent to $(SA)^t = SA$. So, we want to find the dimension of the space of matrices A such that SA is symmetric. Consider the map $M : M_n(F) \to M_n(F)$ defined by $A \mapsto SA$. Since S is invertible, this map is an isomorphism. Therefore, $\dim(V)$ is the dimension of the space of symmetric matrices, which is exactly $\frac{1}{2}(n^2 + n)$ as before.

August 2018 Problem 3 Denote by $M_n(\mathbb{R})$ the ring of all $n \times n$ matrices over $\mathbb{R}$.

(a) Show that for any $A \in M_n(\mathbb{R})$, there exists a $B \in M_n(\mathbb{R})$ such that $AB = 0$ and $\operatorname{rank}(A) + \operatorname{rank}(B) = n$.

(b) Prove or disprove that for any $A \in M_n(\mathbb{R})$, there exists $B \in M_n(\mathbb{R})$ such that $BA = AB = 0$ and $\operatorname{rank}(A) + \operatorname{rank}(B) = n$.

Solution

(a) By row and column reducing, we can find bases $\mathcal{V} = \{v_1, \ldots, v_n\}$ and $\mathcal{W} = \{w_1, \ldots, w_n\}$ for $\mathbb{R}^n$ so that, with respect to the pair of bases $(\mathcal{V}, \mathcal{W})$, the matrix of A is block-diagonal

$$\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

where $r = \operatorname{rank}(A)$. Then we can choose B to be the linear transformation whose matrix with respect to the pair of bases $(\mathcal{W}, \mathcal{V})$ is

$$\begin{pmatrix} 0 & 0 \\ 0 & I_{n-r} \end{pmatrix}.$$

Then $AB = 0$ and $\operatorname{rank}(A) + \operatorname{rank}(B) = n$.

(b) The B from part (a) does this already.

January 2017 Problem 2 Let $n > 0$ be an integer. Let F be a field of characteristic zero, V a vector space over F of dimension n, and $T \colon V \to V$ an invertible F-linear map such that $T^{-1} = T$. Denote by W the vector space of linear transformations from V to V that commute with T. Find a formula for $\dim(W)$ in terms of n and the trace of T.

Solution

Since $T^{-1} = T$, we see that $T^2 - \mathrm{Id}_n = 0$, so T satisfies the polynomial $x^2 - 1 \in F[x]$. Therefore, the minimal polynomial of T must divide $x^2 - 1 = (x-1)(x+1)$. So the only possible eigenvalues of T are 1 and -1. Since the minimal polynomial of T has distinct roots, we can choose a basis so that T is a diagonal matrix with only 1's and -1's on the diagonal. Furthermore, we can rearrange the basis so that T has the block matrix form

$$T = \begin{pmatrix} \mathrm{Id}_a & 0 \\ 0 & -\mathrm{Id}_b \end{pmatrix}$$

where $a + b = n$. So the trace of T is $\operatorname{Tr}(T) = a - b$.

We want to understand matrices that commute with T. Given an $n \times n$ matrix S, we consider it as a block matrix of the form $S = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ where A has size $a \times a$ and D has size $b \times b$. Then multiplying by T on either side gives

$$TS = \begin{pmatrix} A & B \\ -C & -D \end{pmatrix} \qquad \text{and} \qquad ST = \begin{pmatrix} A & -B \\ C & -D \end{pmatrix}.$$

Therefore, $S \in W$ if and only if $B = -B = 0$ and $C = -C = 0$. Since there are no constraints on A and D, we see that $\dim(W) = a^2 + b^2 = \frac{1}{2}((a+b)^2 + (a-b)^2) = \frac{1}{2}(n^2 + \mathrm{Tr}(T)^2)$.

August 2016 Problem 2 Let V be a non-zero finite dimensional vector space over $\mathbb{C}$. Given a linear transformation $A \colon V \to V$, show that the following are equivalent.

(i) There exists a linear transformation $P \colon V \to V$ such that $P^2 = I$ and $AP = -PA$.

(ii) There exists an invertible linear transformation $P \colon V \to V$ such that $AP = -PA$.

(iii) There exists a direct sum decomposition $V = V_1 \oplus V_2$ such that $AV_1 \subset V_2$ and $AV_2 \subset V_1$.

(Hint: Thinking about (iii) $\Rightarrow$ (i) helped me to think about (ii) $\Rightarrow$ (iii).)

Solution

The implication (i) $\Rightarrow$ (ii) is immediate.

For (ii) $\Rightarrow$ (iii), we first begin by showing that for every eigenvalue λ of AP, $-\lambda$ is also an eigenvalue. Suppose $\lambda \neq 0$ is an eigenvalue of AP, and v is an eigenvector for λ. Then notice that $APAv = -A(AP)v = -\lambda Av$. Furthermore, we have that $Av = -P^{-1}APv = -\lambda P^{-1}v \neq 0$, so Av is an eigenvector for AP of eigenvalue $-\lambda$.

Now, divide the eigenvalues of AP into two sets, S_1 and S_2, where if a nonzero $\lambda \in S_1$, then $-\lambda \in S_2$, and visa versa. Let V_1 be the union of the λ-generalized eigenspaces for all $\lambda \in S_1$, and V_2 that for those $\lambda \in S_2$. Then our calculation in the previous paragraph shows that $AV_1 \subset V_2$ and $AV_2 \subset V_1$, as desired.

Finally, for (iii) $\Rightarrow$ (i), let P be given by the block diagonal matrix $\begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$. It is easy to check this matrix satisfies the properties we desire.

January 2015 Problem 4 Let F be a field, and let V be a finite-dimensional vector space over F that has dimension $n \geq 1$. Let q be a nonzero element of F such that $q^i \neq 1$ for $1 \leq i \leq n$. Let $X : V \to V$ and $Y : V \to V$ be two F-linear transformations such that $XY = qYX$. Show that XY is nilpotent.

Solution

Suppose $\lambda \neq 0$ is an eigenvalue of XY. Then $XYv = \lambda v$ for some nonzero $v \in V$, and since $XY = qYX$, we also have that $XY(Yv) = qYXYv = qY(XYv) = q\lambda Yv$. So, $q\lambda$ is also an eigenvalue for XY with eigenvector Yv. (Note that $Yv \neq 0$ since $XYv \neq 0$.) Iterating this gives that $q^i\lambda$ is an eigenvalue of XY for each $0 \leq i \leq n$. Since $q^i \neq 1$ for $1 \leq i \leq n$, this means that XY has at least $n+1$ distinct eigenvalues, which is impossible since V is n-dimensional. Therefore, $\lambda = 0$ is the only eigenvalue of XY, and hence XY is nilpotent.

August 2014 Problem 3

(a) Let V be a finite-dimensional vector space over $\mathbb{C}$, and let $T\colon V \to V$ be a linear transformation. Show that T has an eigenvector.

(b) Give an example of a nonzero finite-dimensional vector space V over $\mathbb{R}$ and a linear transformation $T\colon V \to V$ such that T does not have an eigenvector.

(c) Does a linear transformation of an infinite-dimensional vector space over $\mathbb{C}$ have to have an eigenvector? Either prove this is the case, or give an example of a linear transformation of an infinite-dimensional vector space with no eigenvector.

(d) Suppose that T and U are two linear transformations on a finite-dimensional vector space V over $\mathbb{C}$ which satisfy $TU = UT$. Prove that there is some $v \in V$ which is an eigenvector for both T and U.

Solution

(a) Let $f_T(x)$ be the characteristic polynomial of T. Then by the fundamental theorem of algebra, f_T has a root λ. This means that $\det(T - \lambda\,\mathrm{Id}) = 0$, so $T - \lambda\,\mathrm{Id}$ has nontrivial kernel. For any $v \in \ker(T - \lambda\,\mathrm{Id})$, we have $Tv = \lambda v$, so v is an eigenvector for T.

(b) The classic example is a rotation by $\frac{\pi}{2}$ radians:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

(c) This does not have to be the case. Consider the vector space V whose elements are sequences of complex numbers. So a vector looks like $v = (z_1, z_2, \dots)$. Let T be the "right shift" operator, so $Tv = (0, z_1, z_2, \dots)$. If we try setting up $Tv = \lambda v$, then looking at the first entry tells us the only possible value of λ is 0. But, you can check that T has trivial kernel, so it can't have any eigenvectors.

(d) Let λ be an eigenvalue for T, and let $W \subset V$ be the subspace consisting of all eigenvectors for λ (and the zero vector). (We know we can find such things by part (a).) Here's the standard trick: for any $w \in W$, we have that $TUw = UTw = \lambda Uw$, so $Uw \in W$. Thus, we can think about U as a linear transformation on the smaller vector space W, and by part (a), U must have an eigenvector w'. But then w' is an eigenvector for both U and T, as desired.

January 1994 Problem 4 Let X be a subspace of $M_n(\mathbb{C})$, the $\mathbb{C}$-vector space of all $n \times n$ complex matrices. Assume that every nonzero matrix in X is invertible. Prove that $\dim_{\mathbb{C}}(X) \leq 1$.

Solution
Let A and B be arbitrary matrices in X, and suppose $B \neq 0$. Then B must be invertible, so consider the matrix AB^{-1}. Since we are over $\mathbb{C}$ this matrix must have an eigenvalue λ. So, $AB^{-1} - \lambda\,\mathrm{Id}$ is not invertible, which means that $\det(AB^{-1} - \lambda\,\mathrm{Id}) = 0$. Then we have

$$\det(A - \lambda B) = \det(AB^{-1} - \lambda\,\mathrm{Id})\det(B) = 0$$

which implies that $A - \lambda B = 0$ since it is an element of X that is not invertible. Thus, $A = \lambda B$. Since all of the matrices in X are scalar multiples of each other, we have that $\dim_{\mathbb{C}}(X) \leq 1$.

January 1983 Problem 1 Suppose V is a finite dimensional vector space over $\mathbb{C}$, and $T\colon V \to V$ is a linear operator. Suppose the subspace of V spanned by all eigenvectors of T is 1-dimensional. Prove that the minimal polynomial of T is $(x-\alpha)^n$, where $n = \dim V$ and α is some complex number. (**NB** As an extra challenge, try doing this without JNF.)

Solution
not using Jordan normal form: the characteristic polynomial of T is a degree n polynomial whose only root is α, so it must be $(x-\alpha)^n$. By the Cayley-Hamilton theorem, T satisfies its characteristic polynomial, so the minimal polynomial of T divides $(x-\alpha)^n$. We just need to check that $(T - \alpha\,\mathrm{Id})^k \neq 0$ for any $k < n$. Notice that $\dim\ker(T - \alpha\,\mathrm{Id}) = 1$, and for any linear transformations A, B, $\dim\ker(AB) \leq \dim\ker(A) + \dim\ker(B)$. Thus, $\dim\ker((T - \alpha\,\mathrm{Id})^k) \leq k < n$, so $(T - \alpha\,\mathrm{Id})^k \neq 0$.

Using Jordan normal form: if we take the Jordan normal form of T, all the Jordan blocks have α down the diagonal. However, if there is more than one block, then T will have two linearly independent eigenvectors, so actually the Jordan form of T is

$$\begin{pmatrix} \alpha & 1 & 0 & \dots & 0 \\ 0 & \alpha & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \alpha & 1 \\ 0 & \dots & \dots & 0 & \alpha \end{pmatrix}.$$

From here, one can compute that $(T - \alpha\,\mathrm{Id})^n = 0$, but $(T - \alpha\,\mathrm{Id})^k \neq 0$ for any $k < n$. (Indeed, for any upper triangular $n \times n$ matrix W with 0 on the diagonal, $W^n = 0$.)

3 Group Theory

January 2026 Problem 5 Let G be a nontrivial finite group. A normal subgroup $N \trianglelefteq G$ is *minimal* if $N \neq \{e\}$ and it contains no proper subgroups normal in G. For example, G itself is a minimal normal subgroup if and only if it is simple.

(a) Let $N_1, N_2 \trianglelefteq G$ be distinct minimal normal subgroups of G. Show that the subgroup $\langle N_1, N_2 \rangle$ generated by N_1 and N_2 is isomorphic to $N_1 \times N_2$.

(b) Let $N_1, \ldots, N_r \trianglelefteq G$ be minimal normal subgroups, $r \geq 1$. Put $H = \langle N_1, \ldots, N_r \rangle$. Prove that

$$H \cong \prod_{i \in S} N_i$$

for some subset $S \subset \{1, \ldots, r\}$.

(c) Let $N \trianglelefteq G$ be a minimal normal subgroup, and let $H \trianglelefteq N$ be any normal subgroup, $H \neq \{e\}$. Show that N is generated by the conjugates of H:

$$N = \left\langle \bigcup_{g \in G} gHg^{-1} \right\rangle.$$

(d) Show that for any minimal normal subgroup $N \trianglelefteq G$, there exists a simple group S and $k > 0$ such that $N \cong S^k$.

Solution

(a) Since the intersection of two normal subgroups is again normal, $N_1 \cap N_2$ is a normal subgroup of G properly contained in both, hence $N_1 \cap N_2 = \{e\}$ by the minimality assumption. Thus, $\langle N_1, N_2 \rangle \cong N_1 \times N_2$ by the internal direct product test.

(b) We proceed inductively from the $r = 2$ case above. Note that a subgroup generated by normal subgroups is again normal. So, if we have that

$$H' \coloneqq \langle N_1, \ldots, N_{r-1} \rangle = \prod_{i \in S'} N_i$$

for $S' \subset \{1, \ldots, r-1\}$, then $H' \cap N_r$ is either trivial or all of N_r. In the former case, $\langle H', N_r \rangle = H' \times N_r$, and in the latter case $\langle H', N_r \rangle = H'$.

(c) Note that the subgroup generated by the conjugates of H is preserved by conjugation in G, by construction, hence is a nontrivial normal subgroup of G. On the other hand, since $gHg^{-1} \subset gNg^{-1} = N$, the subgroup generated by the conjugates of H is contained in N. Since N is minimal, we must have equality. (We did not actually use that H was normal in N, but the hypothesis was probably included to be suggestive for the next part.)

(d) Let S be a minimal normal subgroup of N. By part (c), the G-conjugates of S generate N. Since each conjugate of S is again a minimal normal subgroup of N, by part (b) we have that N is the internal direct product of conjugates of S. Since each conjugate of S is isomorphic to S, this tells us that $N \cong S^k$ for some k.

Finally, we will show that S is simple. Suppose that $H \triangleleft S$, then H is normal in a direct factor of N, hence $H \triangleleft N$. But we supposed that S was a minimal normal of N, so we must have either $H = S$ or $H = \{e\}$. Thus, S is simple. (I think it's somewhat funny that we show S is simple by first showing that $N \cong S^k$. I tried to think how to show S is simple directly, and it doesn't seem easy.)

August 2025 Problem 3 Let G be a finite group.

(a) Suppose $H \subset G$ is a normal subgroup such that the quotient group G/H is cyclic of order n. Show that for every divisor d of n, there exists a normal subgroup $K \subset G$ such that the quotient group G/K is cyclic of order d.

(b) Suppose $H_1, H_2 \subset G$ are two normal subgroups such that the quotients are cyclic:

$$G/H_1 \simeq \mathbb{Z}/n_1\mathbb{Z} \qquad G/H_2 \simeq \mathbb{Z}/n_2\mathbb{Z}.$$

Assume further that n_1 and n_2 are coprime. Show that there exists a normal subgroup $H \subset G$ such that

$$G/H \simeq \mathbb{Z}/(n_1 n_2 \mathbb{Z}).$$

Solution

(a) By the lattice isomorphism theorem, normal subgroups of G/H correspond to normal subgroups of G which contain H. Since G/H is cyclic, for every d dividing n, it contains a normal subgroup K' of index d, and the quotient will be cyclic. This normal subgroup corresponds to a normal subgroup K of G, and by the third isomorphism theorem(!) $G/K \cong (G/H)/K'$ is cyclic of order d.

(b) (Solution courtesy of Caroline Nunn.) Let $\pi_1 \colon G \to G/H_1$ and $\pi_2 \colon G \to G/H_2$ be the quotient maps. We have a map $D \colon G \to G/H_1 \times G/H_2$ given by $D(g) = (\pi_1(g), \pi_2(g))$. We have $G/H_1 \times G/H_2 \cong (\mathbb{Z}/n_1\mathbb{Z}) \times (\mathbb{Z}/n_2\mathbb{Z}) \cong \mathbb{Z}/(n_1 n_2 \mathbb{Z})$ by the Chinese Remainder Theorem. So, if we can show that D is surjective, then we will be done, since $\ker D$ will be our desired subgroup.

We will show that this map is surjective by showing that we can find elements $g_1, g_2 \in G$ so that $D(g_1) = (\text{generator}, e)$ and $D(g_2) = (e, \text{generator})$. Let $x \in G$ be any element so that $\pi_1(x)$ generates G/H_1. Since n_1 and n_2 are coprime, $\pi_1(x^{n_2})$ is also a generator of G/H_1, but now $\pi_2(x^{n_2}) = \pi(x)^{n_2} = e$. Similarly, if $\pi_2(y)$ is a generator for G/H_2, so is $\pi_2(y^{n_1})$, but $\pi_1(y^{n_1}) = e$. Thus, the choice $g_1 = x^{n_2}$, $g_2 = y^{n_1}$ does what we wanted.

January 2025 Problem 2 Let $n \geq 2$ be an integer, and $p \geq 3$ be a prime. Denote by B the group of invertible $n \times n$ upper triangular matrices with entries in $\mathbb{Z}/p\mathbb{Z}$, and let U be the subgroup of B consisting of upper triangular matrices whose diagonal entries are all 1.

(a) What is the order of the quotient group B/U as a function of p and n?

(b) Give an example of a nontrivial element in the center of U.

(c) Prove that every nontrivial subgroup of U has nontrivial center.

(d) Compute the center of B.

Solution

(a) Any upper triangular matrix t can be written as $t = du$ for a diagonal matrix d and $u \in U$. Indeed, let d be the diagonal matrix whose entries are the same as the diagonal entries of t, then $d^{-1}t \in U$. So, the elements of the quotient B/U are represented by invertible diagonal matrices, of which there are $(p-1)^n$.

(b) Let z be the matrix with 1s on the diagonal, a 1 in the top right corner, and 0s elsewhere. Then for any $u \in U$, zu adds 1 to the top right entry of u, and so does uz, so $zu = uz$, so $z \in Z(U)$.

(c) The group U has order a power of p, so every subgroup of U also has order a power of p. Every p-group has nontrivial center, so every subgroup of U must have nontrivial center.

(d) The center of B consists of all scalar matrices. Clearly, the scalar matrices lie in $Z(B)$, so we need only show that any element of $Z(B)$ is a scalar matrix. First, any central element must be a diagonal matrix, for if b is not a diagonal matrix, say $b_{ij} \neq 0$ for $i < j$. We can take the diagonal matrix $d = \underbrace{\operatorname{diag}(1, \dots, x, \dots 1)}_{x \text{ in position } i}$ for some $x \neq 1$ (which we can find since $p \geq 3$), and then the (i, j) entry of dbd^{-1} is $xb_{ij} \neq b_{ij}$, so $b \notin Z(B)$. Further, all the diagonal entries must be equal, for say that $b_{ii} \neq b_{jj}$. Let e_{ij} be the matrix with 1s down the diagonal, a 1 in position (i, j), and 0s elsewhere. Then $e_{ij}b \neq be_{ij}$, so $b \notin Z(B)$. Hence, an element of $Z(B)$ must be a scalar matrix.

August 2024 Problem 4, August 2016 Problem 1 Let G be a group. By a maximal subgroup of G, we mean a subgroup $M \neq G$ such that the only subgroups containing M are M and G.

(a) Describe all the maximal subgroups of the dihedral group of order $2p$ where p is an odd prime. How many are there?

(b) Show that if a finite group G has only one maximal subgroup, then G is cyclic.

(c) Show that if a maximal subgroup $M \triangleleft G$ is normal, then the index of M in G is finite and prime.

Solution

(a) As in the previous problem, we have $D_p = \langle r, f : r^p = f^2 = rfrf = 1 \rangle$ and every element of D_p can be written as $r^i f^j$ where $0 \leq i \leq p-1$ and $0 \leq j \leq 1$. Since $\varphi(p) = p-1$, every rotation element r^i with $1 \leq i \leq p-1$ has order p, and each of these generates the same subgroup $\langle r \rangle \cong \mathbb{Z}/p\mathbb{Z}$. Every element of the form $r^i f$ with $0 \leq i \leq p-1$ has order 2, and each generates its own subgroup $\langle r^i f \rangle \cong \mathbb{Z}/2\mathbb{Z}$. Since $|D_p| = 2p$, we know that any maximal subgroup must have order 2 or p. Therefore, the maximal subgroups are exactly the single subgroup $\langle r \rangle$ of order p and the p subgroups $\langle r^i f \rangle$ of order 2.

(b) Suppose $M < G$ is the only maximal subgroup of G. Then $M \neq G$, so there exists some $g \in G \setminus M$. Then $\langle g \rangle$ is a nontrivial subgroup of G, so it is either contained in a maximal subgroup or all of G (since every proper subgroup must be contained in a maximal subgroup). Since, $g \notin M$, we must have that $\langle g \rangle = G$. Thus, G is cyclic.

(c) Suppose $M \triangleleft G$ is maximal. Then the lattice theorem gives that the only subgroups of G/M are the trivial group and the whole group. So, in particular G/M has only one maximal subgroup, the trivial group, so G/M is cyclic by the previous part. Let $n = \#G/M$, and notice that if n were not prime then G/M would have a subgroup of order $d \neq 1, n$ for some $d \mid n$. So, we conclude that in fact n is prime, and since $\#G/M = [G : M]$ we conclude that the index of M is also prime.

January 2024 Problem 3 Let G be a group acting transitively on a set X. Let $H \subset G$ be a normal subgroup. Then H naturally acts on X, but the action need not be transitive. Prove that all orbits of H on X have the same cardinality. (You may assume G and X are finite if you like, but this is not necessary!)

Solution

Let's find a bijection between any two orbits. Let $x, y \in X$ be arbitrary, and denote their orbits under the action of H by Hx, Hy, respectively. Since G acts transitively, we know that there is some $g \in G$ so that $gx = y$, and hence $Hy = H(gx) = (Hg)x$, where by $(Hg)x$ I mean the set $\{x' \in X : x' = hgx \text{ for some } h \in H\}$. (The set Hg is only a right coset, and yet it still makes sense for it to act on X.) Since H is normal in G, the right coset Hg is equal to the left coset gH, and thus $Hy = gHx$.

This gives us an easy bijection between the two: we get a map $Hx \to Hy$ given by $x' \mapsto gx'$, and a map $Hy \to Hx$ given by $x' \mapsto g^{-1}x'$. Our earlier argumentation shows that these maps indeed have the codomains we claim they have, and the two maps are clearly inverses of one another, which is enough to get what we need.

August 2023 Problem 5, edited By S_n we mean the symmetric group on n elements.

(a) Let G be a subgroup of S_n. Suppose the action of G on $\{1, \ldots, n\}$ is transitive. (**NB** such a subgroup is called a "transitive" subgroup.) Prove that for any $x, y \in \{1, \ldots, n\}$, the stabilizer of x in G and the stabilizer of y in G are conjugate subgroups of G.

(b) Prove that if G is a transitive abelian subgroup of S_n, then $|G| = n$.

(c) Give an example of an integer n and a subgroup G of S_n such that G is transitive and abelian but not cyclic.

(d) For which integers n does S_n contain a transitive abelian subgroup G that is not cyclic? (Justify your answer.)

Solution

(a) Since the action of G is transitive, in particular there is some $g \in G$ so that $g \cdot x = y$. Then the conjugation functions

$$\begin{aligned} \phi\colon \mathrm{Stab}(x) &\to \mathrm{Stab}(y) \\ h &\mapsto ghg^{-1} \end{aligned} \qquad\qquad \begin{aligned} \psi\colon \mathrm{Stab}(y) &\to \mathrm{Stab}(x) \\ f &\mapsto g^{-1}fg \end{aligned}$$

are in fact group homomorphisms, and they are clearly mutually inverse, so the two subgroups are conjugate.

(b) By part (a) we know that for any $x, y \in \{1, \ldots, n\}$, $\mathrm{Stab}(x)$ is conjugate to $\mathrm{Stab}(y)$. If G is abelian, then then any subgroup of G is fixed by conjugation, so in fact if some $g \in G$ fixes some $x \in \{1, \ldots, n\}$, it fixes every x, and therefore is the identity element. Therefore, we have

$$\begin{aligned} n &= |G \cdot 1| && \text{(since } G \text{ is transitive)} \\ &= [G : \mathrm{Stab}(1)] && \text{(by Orbit-Stabilizer)} \\ &= |G|. && \text{(since } \mathrm{Stab}(1) = \{e\}) \end{aligned}$$

(c) Inside S_4, we have the abelian subgroup $\{e, (12)(34), (13)(24), (14)(23)\} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(d) This occurs iff n has a square factor. If n has no square factor, then by the classification theorem there is only one abelian group of order n, and it is cyclic. On the other hand, if n has a square factor, then there is some non-cyclic abelian group of order n, and by Cayley's theorem this group embeds as a transitive subgroup of S_n. (It acts transitively on itself by left multiplication.)

January 2023 Problem 2 We say a group G has *big finite quotients* if for every $k > 0$, there is a normal subgroup N of G such that G/N is finite and has order at least k.

(a) Give (with proof) an example of an infinite group that does *not* have big finite quotients.

(b) Show that $\mathrm{SL}_2(\mathbb{Z})$ (the group of 2×2 integer matrices with determinant 1) has big finite quotients.

Solution

(a) The first infinite groups I think of are the additive groups of common rings. The group $\mathbb{Z}$ actually does have big finite quotients, so let's try $\mathbb{Q}$. The group $\mathbb{Q}$ is divisible, so all of its quotients are divisible, so all of its quotients are either trivial or infinite. Hence, $\mathbb{Q}$ does not have big finite quotients.

(b) For every $p \in \mathbb{Z}$, there is a quotient map $\mathrm{SL}_2(\mathbb{Z}) \to \mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ given by reducing each entry in the matrix mod p. One can compute that $|\mathrm{SL}_2(\mathbb{F}_p)| = (p^2-1)(p^2-p)/(p-1) = p(p^2-1)$, so for any k we can certainly just choose a $p > k$ and get a big quotient.

August 2022 Problem 2 Let $\mathbb{F}_p$ be the finite field with p elements; here p is a prime number. Let $V = \mathbb{F}_p^2$, and recall that $G = \mathrm{GL}_2(\mathbb{F}_p)$ is the group of invertible linear transformations on V. G acts on V in the usual way (by multiplication).

(a) Describe the orbits of this action.

(b) Describe the stabilizer in G of the vector

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in V.$$

(c) Consider now the action of G on $V \times V$ (acting independently on each of the two vectors). How many orbits does this action have?

(d) What is the cardinality of G? Remember to justify your answer.

Solution

(a) There are two orbits, the zero vector, and everything else. For any two nonzero vectors u_1, v_1, there is some invertible matrix A such that $Au_1 = v_1$; for example, we can choose complementary vectors u_2, v_2 so that $\{u_1, u_2\}$ and $\{v_1, v_2\}$ are both bases, and define an invertible linear map by $u_1 \mapsto v_1, u_2 \mapsto v_2$.

(b) The stabilizer of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ consists of all matrices of the form

$$\begin{pmatrix} 1 & a \\ 0 & b \end{pmatrix},$$

where $a \in \mathbb{F}_p, b \in \mathbb{F}_p^\times$. Certainly, every matrix in the stabilizer must have $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ as the first column, and it is easy to verify that every such matrix is in the stabilizer.

(c) There's definitely an orbit corresponding to when $v_1 = 0$ and $v_2 = 0$. By our answer to part (a), there's also an orbit for when $v_1 \neq 0, v_2 = 0$, and another one for $v_1 = 0, v_2 \neq 0$.

Indeed, somewhat more generally, if v_1 and v_2 are linearly dependent, say they are nonzero and $v_1 = cv_2$ with $c \in \mathbb{F}_p^\times$, then A must take them to a new pair of vectors satisfying the same linear relation: $Av_1 = cAv_2$. This gives us $p-1$ more orbits. Finally, we have an orbit corresponding to when v_1 and v_2 are linearly independent vectors in V, because for any two such pairs there is a change-of-basis matrix taking one to the other. In total this gives us $p+3$ orbits.

(d) $(p^2-1)(p^2-p)$. One way to calculate this would be to use the calculation from part (b), together with the orbit-stabilizer theorem:

$$\begin{aligned}|G| &= \left|\operatorname{Stab}\begin{pmatrix}1\\0\end{pmatrix}\right| \cdot \left|G \cdot \begin{pmatrix}1\\0\end{pmatrix}\right| \\ &= p(p-1)\cdot(p^2-1).\end{aligned}$$

January 2022 Problem 1 On this problem, only the answer will be graded. Consider the symmetric group S_4.

(a) What is the order of S_4?

(b) What is the order of its center $Z(S_4)$?

(c) What is the order of its commutator subgroup $[S_4, S_4]$?

(d) Is S_4 a simple group?

(e) Is S_4 a solvable group?

(f) How many conjugacy classes does S_4 have?

Solution

(a) 24

(b) 1

(c) 12 ($[S_4, S_4] = A_4$; this is always the case)

(d) No, A_4 is a nontrivial normal subgroup.

(e) Yes, since A_4 contains a normal subgroup isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, and the quotient is isomorphic to $\mathbb{Z}/3\mathbb{Z}$.

(f) 5, one for each partition of 4.

August 2021 Problem 3 Let G be a finite group. For given $n > 1$, we say that G *admits roots of degree* n if, for every $x \in G$, there exists $y \in G$ such that $y^n = x$.

(a) Show that if n is coprime to $|G|$, G admits roots of degree n.

(b) Conversely, prove that if G admits roots of degree n, then n is coprime to $|G|$.

Solution

(a) We will prove the proposition in the case where G is cyclic, and the result will follow because each $x \in G$ lives in the cyclic subgroup $\langle x \rangle$. So, let G be cyclic and generated by x, and consider the group homomorphism $\phi_n \colon G \to G$ given by $x \mapsto x^n$. We want to show that this map is surjective. Since G is finite, it is enough to show that this map is injective, and hence it is enough to show that $\ker \phi_n = \{e\}$. Suppose $x \in \ker \phi_n$, then $x^n = e$, so the order x, $\operatorname{ord} x$, divides n. But also $\operatorname{ord} x$ divides $|G|$. Since n is coprime to $|G|$ we must have that $\operatorname{ord} x = 1$, i.e. $x = e$. Thus, $\ker \phi_n$ is trivial.

(b) We will prove this by contradiction: suppose $\gcd(n, |G|) > 1$ and G admits roots of degree n. If G admits roots of degree n, it also admits roots of degree p for every $p \mid n$, so it suffices to consider the case where $n = p$ for some prime p dividing $|G|$. Let $p \mid |G|$, and let $g \in G$ generate a cyclic subgroup of order p^k which is not contained in any cyclic subgroup of order p^{k+1}. By Cauchy's theorem, there a cyclic subgroup of order p, and we are just taking a maximal cyclic p-group containing such a subgroup. Since G admits roots of degree p, we can find $h \in G$ with $h^p = g$. First, notice that for any k, $(g^k)^p$ lies in a subgroup of index $\geq p$ inside $\langle g \rangle$, and therefore $h \notin \langle g \rangle$. But, on the other hand, $\operatorname{ord} h = p^{k+1}$, and by construction g is not contained in any cyclic group of order p^{k+1}, a contradiction. Therefore, G cannot admit roots of degree p.

January 2020 Problem 5

(a) Find a nontrivial finite subgroup of $\mathrm{GL}_2(\mathbb{R})$.

(b) Find an infinite, abelian, but non-cyclic subgroup of $\mathrm{GL}_2(\mathbb{R})$.

(c) Find a proper, normal subgroup of $\mathrm{GL}_2(\mathbb{Z})$ of finite index.

Solution

(a) Consider the subgroup of order 2 $\{I, -I\}$.

(b) Consider the subgroup consisting of diagonal matrices.

(c) For each prime p there is a map $\mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{F}_p)$ given by reducing each entry mod p. The kernel of such a map must be proper and normal, and since the target is a finite group it must be of finite index.

August 2019 Problem 5 Let G be the group of 2×2 invertible upper triangular matrices over the field $\mathbb{F}_p$.

(a) Prove that G has only one subgroup of order p, and that this subgroup is normal. (Hint: first show that an element of order p must have 1's on the diagonal.)

(b) Prove that G is solvable by exhibiting a homomorphism f from G to an abelian group A such that the kernel of f is also abelian.

(c) Prove that if $p \neq 2$, G is not nilpotent.

Solution

(a) Given a matrix

$$A = \begin{pmatrix} a & b \\ 0 & c \end{pmatrix},$$

we have that the diagonal entries of A^n are a^n and c^n. Thus, if A has order p, then $a^p = c^p = 1$, but by Fermat's Little Theorem, $a^p = a$, and so $a = c = 1$. Thus, the p matrices of the form

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

form the only subgroup of order p. Since conjugating this subgroup would take it to another subgroup of order p, and we have determined it is the unique subgroup of order p, we conclude that this subgroup must be fixed by conjugation, hence normal.

(b) Let the subgroup from part (a) be U, and let T be the subgroup of diagonal matrices. Then $U \cap T = \{I\}$, and we can write any matrix

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & c \end{pmatrix} \begin{pmatrix} 1 & a^{-1}b \\ 0 & 1 \end{pmatrix}$$

where the first factor is in T and the second is in U. Thus, $G = U \rtimes T$, and so G/U is isomorphic to T, which is abelian. Thus, $f \colon G \to G/U$ is the homomorphism we want.

(c) Assume $p \neq 2$. This is probably easiest to show using the lower central series. Notice that the diagonal entries of any commutator $ABA^{-1}B^{-1}$ are going to be 1, so actually every commutator falls inside U. So, $[G, G]$ is a subgroup of U, and it isn't the trivial subgroup since G is not abelian, so the only other option is $[G, G] = U$. But now, the second term of the lower central series $G_2 = [G, U]$ is still equal to U, since for example the matrices

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

do not commute (recall that we've assumed $p \neq 2$). Thus, $ABA^{-1}B^{-1}$ is a nontrivial commutator in $[G, U]$, so $[G, U]$ is a nontrivial subgroup of U, hence is all of U.

January 2019 Problem 2 If G is a group and $x, y \in G$, then we let $\langle x, y\rangle$ denote the subgroup of G generated by x and y. For $x \in G$, define $\mathrm{Cyc}_G(x) := \{y \in G \ : \ \langle x, y\rangle \text{ is cyclic}\}$.

(a) Show by example that $\mathrm{Cyc}_G(x)$ need not be a subgroup of G.

(b) Let $\mathrm{Cyc}(G) = \bigcap_{x\in G} \mathrm{Cyc}_G(x)$. Show that $\mathrm{Cyc}(G)$ is a subgroup of G.

(c) Show that $\mathrm{Cyc}(G)$ lies in the center of G.

(d) Assume now that G is finite. Show that $\mathrm{Cyc}(G)$ is cyclic.

Solution

(a) When thinking of a counterexample, it's helpful to rephrase the definition of $\mathrm{Cyc}_g(x)$. For any cyclic subgroup $\langle g\rangle$ which contains x, we will have that $\langle x, y\rangle \le \langle g\rangle$ for all $y \in \langle g\rangle$, so $\langle x, y\rangle$ will also be cyclic. Thus, every $y \in \langle g\rangle$ is in $\mathrm{Cyc}_g(x)$, and this holds for all $\langle g\rangle$, so $\mathrm{Cyc}_g(x)$ is the union of all the cyclic subgroups containing x:

$$\mathrm{Cyc}_g(x) = \bigcup_{\substack{g\in G\\ x=g^k}} \langle g\rangle.$$

However, a union of subgroups need not be a subgroup, so we should try to think of examples where that fails.

As a specific example, $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$ and $x = (0, 2)$. Then $(1, 1)$ and $(0, 1)$ are in $\mathrm{Cyc}_G(x)$ since $\langle (0, 2), (1, 1)\rangle = \langle (1, 1)\rangle)$ and $\langle (0, 2), (0, 1)\rangle = \langle (0, 1)\rangle$, but $(1, 0) = (1, 1) - (0, 1)$ is not in $\mathrm{Cyc}_G(x)$ since $\langle (0, 2), (1, 0)\rangle = \mathbb{Z}/2\mathbb{Z} \times 2\mathbb{Z}$.

(b) It's helpful to think about what it means to say "y is an element of $\mathrm{Cyc}(G)$". If $y \in \mathrm{Cyc}(G)$, then $y \in \mathrm{Cyc}_G(x)$ for every $x \in G$, so $\langle x, y\rangle$ is cyclic for every $x \in G$. (And the implications go the other way as well.)

We first show that $\mathrm{Cyc}(G)$ is closed under taking inverses. Let $y \in \mathrm{Cyc}(G)$. We need to show that for each $x \in G$, $\left\langle x, y^{-1}\right\rangle$ is cyclic. But $\left\langle x, y^{-1}\right\rangle = \langle x, y\rangle$, which is cyclic by assumption, so $y^{-1} \in \mathrm{Cyc}(G)$.

Now we show that $\mathrm{Cyc}(G)$ is closed under multiplication. It will be convenient to define a function (not a homomorphism) $g\colon G \times \mathrm{Cyc}(G) \to G$ by choosing $g_y(x)$ to be any generator for the cyclic group $\langle x, y\rangle$. (I put the y in the subscript to make the following easier to read.) Now, let $y, z \in \mathrm{Cyc}(G)$, and let $x \in G$ be arbitrary. Observe that $\langle x, y, z\rangle = \langle g_y(x), z\rangle = \langle g_z(g_y(x))\rangle$ is cyclic. Since $y, z \in \langle x, y, z\rangle$, $yz \in \langle x, y, z\rangle$, so $\langle x, yz\rangle \le \langle x, y, z\rangle$ is a subgroup of a cyclic group, hence is itself cyclic. Thus, $yz \in \mathrm{Cyc}(G)$.

(c) Let $y \in \mathrm{Cyc}(G)$. We want to show that $y \in Z(G)$, that is, that y commutes with every element of G. We know that for any $x \in G$, $\langle x, y\rangle$ is cyclic, and hence abelian. Therefore, x and y must commute.

(d) We will reuse our function $g\colon G \times \mathrm{Cyc}(G) \to G$ from part (b). Since G is finite, we can find finitely many $y_1, y_2 \ldots, y_n$ that generate $\mathrm{Cyc}(G)$. But then, similar to part (b),

$$\mathrm{Cyc}(G) = \langle y_1, \ldots, y_n \rangle = \langle g_{y_2}(y_1), y_3, \ldots, y_n \rangle = \cdots = \big\langle g_{y_n} \circ g_{y_{n-1}} \circ \ldots g_{y_2}(y_1) \big\rangle.$$

Hence, $\mathrm{Cyc}(G)$ is cyclic.

August 2018 Problem 2 For a finite group G, denote by $s(G)$ the number of subgroups of G.

(a) Show that $s(G)$ is finite.

(b) Show that if H is a nontrivial normal subgroup of G, then $s(G/H) < s(G)$.

(c) Show that $s(G) = 2$ if and only if G is cyclic of prime order.

(d) Show that $s(G) = 3$ if and only if G is cyclic of order p^2 for a prime p.

Solution

(a) Certainly every subgroup is also a subset, and there are at most $2^{|G|}$ subsets of G, so $s(G) \leq 2^{|G|} < \infty$.

(b) By the lattice isomorphism theorem, the subgroups of G/H are in bijection with subgroups of G containing H. Certainly the number of subgroups of G containing H is at most the number of subgroups of G, and since H is nontrivial we know the trivial subgroup does not contain H, which makes the inequality strict.

(c) Certainly, if G is cyclic of order p, then its only subgroups are $G, \{e\}$. On the other hand, suppose $s(G) = 2$. Since $s(G) > 1$, G is not trivial. Let p be a prime dividing $|G|$, then by Cauchy's theorem there is a $g \in G$ of exact order p. But then $\langle g \rangle$ and $\{e\}$ are 2 subgroups of G, so we have already found all the subgroups of G.

(d) Certainly, if G is cyclic of order p^2 generated by g, then its only subgroups are $G, \langle g \rangle, \{e\}$. On the other hand, suppose $s(G) = 3$, so that G has exactly 1 nontrivial proper subgroup. As before, we can find an element g of exact order p. Furthermore, p is the only prime which divides $|G|$, as if another prime q divided $|G|$ we would have another nontrivial proper subgroup, but we know G has a unique such subgroup. Thus, $|G| = p^n$ for some n. Now, let $h \in G \setminus \langle g \rangle$, then h generates a nontrivial subgroup, so that subgroup cannot be proper, so $\langle h \rangle = G$, and G is cyclic. Since G is cyclic of order p^n, it has $n + 1$ subgroups, generated by h^{p^k} for $k \leq n$. Thus, $n + 1 = 3$, so $n = 2$.

January 2018 Problem 4, modified Let G be a finite group. Denote by $\operatorname{Aut}(G)$ the group of automorphisms of G, and by $Z(G) \subset G$ the center of G.

(a) Let $\operatorname{Inn}(G) \subset \operatorname{Aut}(G)$ be the subgroup consisting of automorphisms coming from conjugation, i.e. automorphisms of the form $x \mapsto gxg^{-1}$ for some $g \in G$. Prove that $\operatorname{Inn}(G)$ is isomorphic to $G/Z(G)$.

(b) Show that if $G/Z(G)$ is cyclic, then G is abelian.

(c) Show that if $\operatorname{Aut}(G)$ is cyclic, then G is abelian.

(d) Show that if G is abelian, then $x \mapsto x^{-1}$ is an automorphism of G. (**NB** in fact this is an if and only if.)

(e) Deduce that there is no group G such that $\operatorname{Aut}(G)$ is a nontrivial cyclic group of odd order.

Solution

(a) Denote by conj_g the "conjugation by g" automorphism, $\operatorname{conj}_g \colon G \to G$, $\operatorname{conj}_g(x) = gxg^{-1}$. Define a map $G \to \operatorname{Inn}(G)$ by $g \mapsto \operatorname{conj}_g$. We will show that the kernel of this map is equal to $Z(G)$. Certainly, if $z \in Z(G)$, then $zxz^{-1} = x$ for all x, so $\operatorname{conj}_z = \mathrm{id}$. On the other hand, suppose $\operatorname{conj}_g = \mathrm{id}$, that is, that $gxg^{-1} = x$ for all x. Then multiplying on the right by g gives us $gx = xg$ for all x, so $g \in Z(G)$.

(b) Suppose $G/Z(G)$ is cyclic with generator $gZ(G)$. Then any $x, y \in G$ can be written uniquely as $x = g^{n_1}z_1$, $y = g^{n_2}z_2$, for $z_1, z_2 \in Z(G)$. But then

$$xy = g^{n_1}z_1g^{n_2}z_2 = g^{n_1}g^{n_2}z_1z_2 = g^{n_2}g^{n_1}z_2z_1 = g^{n_2}z_2g^{n_1}z_1 = yx,$$

and G is abelian.

(c) If $\operatorname{Aut}(G)$ is cyclic, then its subgroup $\operatorname{Inn}(G) \cong G/Z(G)$ is also cyclic.

(d) Notice that $(xy)^{-1} = y^{-1}x^{-1} = x^{-1}y^{-1}$ since G is abelian.

(e) By the proceeding, if $\operatorname{Aut}(G)$ is nontrivial and cyclic, then G is abelian. Then $x \to x^{-1}$ is an element of $\operatorname{Aut}(G)$ of order dividing 2, so either $2 \mid |\operatorname{Aut}(G)|$ or every element of G has order exactly 2. In the other case, we will have $G \cong (\mathbb{Z}/2\mathbb{Z})^n$ for some $n > 1$ by the classification of finite abelian groups. But then there is an automorphism defined by swapping e_1 and e_2 and leaving the other basis elements fixed, and this is an automorphism of order 2, so even in this case $2 \mid |\operatorname{Aut}(G)|$. Thus, it is impossible for $\operatorname{Aut}(G)$ to be a nontrivial cyclic group of odd order.

August 2017 Problem 3 What is the smallest n such that the 3-Sylow subgroup of S_n is non-abelian? (You may use the Sylow theorem that all Sylow subgroups are conjugate, so that one 3-Sylow subgroup is non-abelian if and only if they all are.)

Solution

Any group of order p or p^2 is abelian, so the smallest n such that the 3-Sylow subgroup of S_n could be nonabelian is $n = 9$. We have to be a little careful showing that in this case the 3-Sylow subgroup is nonabelian. We can certainly find two elements whose orders are powers of 3 and don't commute, but there's always the possibility that they live in different Sylow subgroups, in which case we will not have demonstrated that the 3-Sylow subgroup is nonabelian. (As an aside: by other considerations, there are at least 4 3-Sylow subgroups, so if we were to choose our elements randomly we would get two elements in the same Sylow subgroup at best 1/4 of the time. Those aren't great odds.)

However, we can say that a 3-Sylow subgroup must contain a 3-cycle (abc), because certainly every 3-cycle lives in *some* 3-Sylow subgroup, but all such subgroups are conjugate, and conjugation does not change cycle type, so each 3-Sylow subgroup gets at least one of them. For the same reason, we can see that every 3-Sylow subgroup has a 9-cycle. However, a 3-cycle never commutes with a 9-cycle. Indeed, given a 3-cycle τ and a 9-cycle ν, there is always some $n \in \{1, \ldots, 9\}$ so that $\tau(n) \neq n$ and $\tau \circ \sigma(n) = \sigma(n)$. (E.g. if $\tau = (123)$ and $\nu = (123456789)$, then $n = 3$ is such a number, and is the only such number.) But then clearly $\sigma \circ \tau(n) \neq \tau \circ \sigma(n)$, so these two do not commute.

January 2017 Problem 1 Give an example of each of the following.

(a) A group G with a normal subgroup N such that G is not a semidirect product $N \rtimes G/N$.

(b) A finite group G that is nilpotent but not abelian.

(c) A group G whose commutator subgroup $[G, G]$ is equal to G.

(d) A non-cyclic group G such that every Sylow subgroup of G is cyclic.

(e) A transitive action of S_3 on a set X of cardinality greater than 3.

Solution

(a) The smallest such example is $G = \mathbb{Z}/4\mathbb{Z}$, $N = 2\mathbb{Z}/4\mathbb{Z}$. Then $G/N \cong \mathbb{Z}/2\mathbb{Z}$, but there is only one nontrivial map $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/4\mathbb{Z}$, and it is not a splitting for the quotient map. Therefore, $\mathbb{Z}/4\mathbb{Z}$ is not a semidirect product of $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z}$. A non-abelian example would be the quaternion group Q with subgroup $N = \{\pm 1\}$. An infinite example would be $\mathbb{Z}$ with subgroup $N = 2\mathbb{Z}$.

(b) The smallest examples are D_4 and Q, the quaternion group.

(c) The smallest such example is $G = A_5$, which is simple and therefore its commutator subgroup must be equal to the whole group.

(d) Take $G = D_n$ for n odd.

(e) Consider the action of S_3 on itself by left multiplication.

January 2016 Problem 4 Let D_k be the dihedral group of order $2k$ where $k \geq 3$.

(a) Show that the number of automorphisms of D_k is equal to $k \cdot \varphi(k)$, where φ is Euler's φ-function.

(b) Let $\mathrm{Aut}(D_k)$ denote the group of automorphisms of D_k. What is the structure of $\mathrm{Aut}(D_k)$? Describe the group as explicitly as you can.

Solution

(a) Recall that D_k has presentation given by

$$\left\langle r, f : r^k = f^2 = rfrf = 1 \right\rangle$$

where r is a rotation of the k-gon that sends each vertex to an adjacent one and f is a reflection of the k-gon through a line of symmetry. This presentation shows that every element of D_k can be written in the form $r^i f^j$, where $0 \leq i \leq k-1$ and $0 \leq j \leq 1$. Notice that the only elements of D_k of order k are those of the form r^i where i is relatively prime to k, and the only elements of order 2 are those of the form $r^i f$.

Any element of $\mathrm{Aut}(D_k)$ is determined by where it sends r and f, and r must be sent to another element of order k and f must be sent to another element of order 2.Therefore, there are $\varphi(k)$ choices for the image of r and k choices for the image of f. This shows that $|\mathrm{Aut}(D_k)| \leq k \cdot \varphi(k)$. It remains to check that each of these choices gives an automorphism of D_k. It is routine to check that each map is a homomorphism (just need to check the relations still hold), and given the map φ where $\varphi(r) = r^i$ and $\varphi(f) = r^j f$, we can construct an inverse. Since $\gcd(i, k) = 1$, there exist $a, b \in \mathbb{Z}$ such that $ai + bk = 1$, and then φ^{-1} is given by $r \mapsto r^a$ and $f \mapsto r^{-ja} f$. Thus, $|\mathrm{Aut}(D_k)| = k \cdot \varphi(k)$.

(b) By part (a), we know that any automorphism of D_k is determined by a pair of integers (j, i) where $0 \leq j \leq k-1$ and $\gcd(i, k) = 1$. Let's next understand the multiplication in this group. Given two automorphism determined by (j, i) and (n, m), we see that their composition maps $r \mapsto r^m \mapsto r^{im}$ and $f \mapsto r^n f \mapsto r^{in+j} f$. Therefore, we have $(j, i) \circ (n, m) = (in + j, im)$. This looks like it might indicate a semi-direct product. Indeed, if we consider the map $\alpha : (\mathbb{Z}/k\mathbb{Z})^* \to \mathrm{Aut}(\mathbb{Z}/k\mathbb{Z})$ which sends i to the multiplication by i map, we see that $\mathrm{Aut}(D_k) \cong \mathbb{Z}/k\mathbb{Z} \rtimes (\mathbb{Z}/k\mathbb{Z})^*$ with respect to α.

4 Field and Galois Theory

January 2026 Problem 3 Consider the field $K = \mathbb{Q}(i)$, and put $L = K[t]/(t^4 - 5)$. Clearly, L is a ring containing the field K (so it is a K-algebra).

(a) Show that L/K is a Galois field extension.

(b) List the roots of the polynomial $x^4 - 5$ in L.

(c) Determine the action of the Galois group $\mathrm{Gal}(L/K)$ on the roots of the polynomial $x^4 - 5$ in L.

(d) Consider L as an extension of $\mathbb{Q} \subset K$. Show that this is a Galois extension and determine the action of the Galois group $\mathrm{Gal}(L/\mathbb{Q})$ on the roots of the polynomial $x^4 - 5$ in L.

Solution

(a) First, let us show that L is a field. This requires us to show that $t^4 - 5$ is irreducible over K. Below, I'll give three arguments, the first elementary and computational, the second more conceptual and clean but a tad tricky, the third short and easy but requiring some insight into the ring $\mathbb{Q}(i)$.

   i. We can see that $t^4 - 5$ has no roots in K, so it certainly has no linear factors. Suppose it factors as $(t^2 + at + b)(t^2 + ct + d)$. Multiplying out and comparing coefficients, this means that: $a + c = 0$, $b + d + ac = 0$, $a(d - b) = 0$, and $bd = 5$. Since $\sqrt{5} \notin K$, the last equation implies $b \neq d$. But then the third equation implies $a = 0$, and the second implies that $b = -d$, but $-\sqrt{5} \notin K$, so this is a contradiction. Thus, no such factorization exists.

   ii. Note that $t^4 - 5$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion for $p = 5$. Thus, if it factors over K, it must factor as $g(t)\bar{g}(t)$ for some $g \in K[t]$, where $\bar{g}$ is the polynomial whose coefficients are the complex conjugates of the coefficients of g. The roots of $\bar{g}$ are the complex conjugates of the roots of g. Since $\sqrt[4]{5}$ is a root of $t^4 - 5$, it must be a root of either g or $\bar{g}$, but then since $\sqrt[4]{5}$ is real it must be a root of both of them, so it must be a double root of $t^4 - 5$. However, $t^4 - 5$ is separable, so this is a contradiction.

   iii. The polynomial $t^4 - 5$ is Eisenstein over the ring $\mathbb{Z}[i]$ for the prime $2 + i$, hence is irreducible.

   To show that L is Galois, probably the easiest way is to do part (b), and observe that L is the splitting field of $t^4 - 5$ over K, hence is Galois.

(b) The roots are $\pm\sqrt[4]{5}$, $\pm i\sqrt[4]{5}$.

(c) Since the ratio of any two roots is in K, any element of $\mathrm{Gal}(L/K)$ must fix the ratios between the roots. Thus, the action of any $\sigma \in \mathrm{Gal}(L/K)$ is determined completely by where it sends $\sqrt[4]{5}$, for which there are 4 options. The Galois group is thus cyclic of order 4, and it acts by cyclically permuting the roots of $x^4 - 5$.

(d) L is still the splitting field of $x^4 - 5$ over $\mathbb{Q}$, so still Galois. In addition to the four automorphisms found above, $\mathrm{Gal}(L/\mathbb{Q})$ contains complex conjugation and its product with the other three nontrivial automorphisms of L/K. (This is another way to argue that $L/\mathbb{Q}$ is Galois: by finding $8 = [L : \mathbb{Q}]$ automorphisms.) Thus, $\mathrm{Gal}(L/\mathbb{Q})$ is the dihedral group D_4. Picturing the roots of $x^4 - 5$ as the vertices of a diamond in the complex plane, $\mathrm{Gal}(L/\mathbb{Q})$ acts as the full group of symmetries of that diamond.

August 2025 Problem 2 Let E/F be a Galois field extension whose Galois group $G = \mathrm{Gal}(E/F)$ is isomorphic to $\mathbb{Z}/8\mathbb{Z}$.

(a) Give an example of such a Galois extension. No proof is required.

(b) Let $\sigma \in G$ be a generator. Suppose $a \in E$ is such that the elements

$$\{a, \sigma(a), \sigma^2(a), \ldots, \sigma^7(a)\}$$

form a basis of E over F. Let $b \in E$ be a non-zero linear combination

$$b = c_0 a + c_1 \sigma(a) + c_2 \sigma^2(a) + c_3 \sigma^3(a),$$

where $c_0, c_1, c_2, c_3 \in F$ are not all zero. Show that b is a primitive element: $E = F(b)$.

Solution

(a) The easiest way to do this would be to find a prime p which is 1 mod 8, and look for a subfield of $\mathbb{Q}(\zeta_p)$. Since $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \cong (\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic, it has a normal subgroup H with $[G : H] = 8$, and then the fixed field $\mathbb{Q}(\zeta_p)^H$ has Galois group cyclic of order 8. So, in particular, $\mathbb{Q}(\zeta_{17})$ has Galois group $(\mathbb{Z}/17\mathbb{Z})^\times \cong \mathbb{Z}/16\mathbb{Z}$, and complex conjugation forms the unique order 2 subgroup. The fixed field of complex conjugation is $\mathbb{Q}(\zeta_{17}) \cap \mathbb{R} = \mathbb{Q}(\cos(2\pi/17))$, so this is an example of a field with Galois group $\mathbb{Z}/8\mathbb{Z}$.

(b) It is easy to see that no power of σ fixes the element b, so the orbit of b under the action of the Galois group has size 8. Since the orbit of b is the same thing as the roots of the minimal polynomial of b, the minimal polynomial of b must have degree 8. Thus, $F(b)$ is a degree 8 extension of F contained in E, so must in fact be equal to E.

January 2025 Problem 3 Let F be a field, $K \supset F$ be a finite extension of F, and let a be an element of K.

(a) Suppose the degree $[K : F]$ is relatively prime to 6. Show that $F(a) = F(a^3)$.

(b) Suppose the degree $[K : F]$ is not divisible by 3. Either prove that we still have $F(a) = F(a^3)$, or show by example that this is false.

Solution

(a) We have that $F(a^3) \subset F(a)$, so we have a tower of field extensions:

$$\begin{array}{c} K \\ | \\ F(a) \\ | \\ F(a^3) \\ | \\ F \end{array}$$

Also, a is a root of the polynomial $x^3 - a^3$ over $F(a^3)$, so the degree $[F(a) : F(a^3)]$ is at most 3. On the other hand, the degree of $[F(a) : F(a^3)]$ divides $[K : F]$, which we are told is coprime to 6, so we must have $[F(a) : F(a^3)] = 1$, i.e. $F(a) = F(a^3)$.

(b) This can be false. For example, let ω be a primitive cube root of unity, and consider the extension $F = \mathbb{Q}$, $K = \mathbb{Q}(\omega)$. Then $[K : F] = 2$, which is coprime to 3, but $\mathbb{Q}(\omega^3) = \mathbb{Q}(1) = \mathbb{Q}$.

August 2024 Problem 5 Denote by ζ_m a primitive mth root of unity, i.e. a complex number such that $\zeta_m^m = 1$ but $\zeta_m^k \neq 1$ for any $k < m$.

(a) Prove that, for any m, the field extension $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is Galois.

(b) What is the Galois group of $\mathbb{Q}(\zeta_6)/\mathbb{Q}$?

(c) Give an example of an m such that the Galois group of $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ is not cyclic.

Solution

(a) There are several equivalent definitions of what it means for a field extension to be Galois, and for this question it's important to choose one that makes your life easiest. The easiest way I know to do this is to note that ζ_m is a root of the mth cyclotomic polynomial $\Phi_m(x)$, and the degree of the mth cyclotomic polynomial is $\phi(m)$, so we have $|\mathrm{Aut}(\mathbb{Q}(\zeta_m)/\mathbb{Q})| \leq [\mathbb{Q}(\zeta_m) : \mathbb{Q}] \leq \phi(m)$. Now, we construct $\phi(m)$ automorphisms of $\mathbb{Q}(\zeta_m)$ as follows: for each $1 \leq a \leq m$ such that $\gcd(a, m) = 1$, we have an automorphism given by $\zeta_m \mapsto \zeta_m^a$. Thus, in fact we must have $|\mathrm{Aut}(\mathbb{Q}(\zeta_m)/\mathbb{Q})| = [\mathbb{Q}(\zeta_m) : \mathbb{Q}] = \phi(m)$, and the size of the automorphism group being equal to the degree of the extension is one of the equivalent definitions of a Galois extension, so $\mathbb{Q}(\zeta_m)$ is Galois. (Incidentally, along the way this also proves that $\Phi_m(x)$ is irreducible, which we did not need to assume for any other part of the proof.)

(b) $(\mathbb{Z}/6\mathbb{Z})^\times \cong \mathbb{Z}/2\mathbb{Z}$

(c) By the Chinese Remainder Theorem, if we factor $m = p_1^{e_1} \dots p_n^{e_n}$, then

$$(\mathbb{Z}/m\mathbb{Z})^\times \cong \prod_{i=1}^{n} (\mathbb{Z}/p_i^{e_i}\mathbb{Z})^\times.$$

Thus, the Galois group will be non-cyclic as soon as we either have a non-cyclic $(\mathbb{Z}/p_i^{e_i}\mathbb{Z})^\times$ or if we have more than one nontrivial factor. The latter is easier to manufacture, as any number divisible by more than one odd prime works, for example $m = 15$. Another thing that would work is any number divisible by 4 and an odd prime, so $m = 12$ also works. Something that you don't need to know, but which is neat, is that the only time $(\mathbb{Z}/p_i^{e_i}\mathbb{Z})^\times$ is *not* cyclic is if $p_i = 2$ and $e_i \geq 3$, so any m divisible by 8 also works, and $m = 8$ is in fact the minimal such example. (You could also easily find $m = 8$ by just checking small values of m, since $m = 2, \dots, 7$ don't work by basic group theory knowledge.)

January 2024 Problem 2 The irreducible factors of $x^{11} - 1$ over $\mathbb{Q}$ are $f_1 = x - 1$ and $f_2 = x^{10} + x^9 + x^8 + \cdots + 1$. (You do not need to prove this.) Let K be the splitting field of f_2.

(a) What is the Galois group of K over $\mathbb{Q}$?

(b) Let $a = \cos(2\pi/11) = (\exp(2\pi i/11) + \exp(\text{-}2\pi i/11))/2$. Show that $K/\mathbb{Q}(a)$ is a Galois extension and find its Galois group.

(c) Show that $\mathbb{Q}(a)/\mathbb{Q}$ is a Galois extension and find its Galois group.

Solution

(a) $(\mathbb{Z}/11\mathbb{Z})^\times \cong \mathbb{Z}/10\mathbb{Z}$

(b) Since K is Galois over $\mathbb{Q}$, K is Galois over any intermediate field extension, and in particular is Galois over $\mathbb{Q}(a)$. Notice that $\mathbb{Q}(a)$ is the fixed field of complex conjugation, so $[K : \mathbb{Q}(a)] = 2$, so $\mathrm{Gal}(K/\mathbb{Q}(a)) = \mathbb{Z}/2\mathbb{Z}$ generated by complex conjugation.

(c) Since $\mathrm{Gal}(K/\mathbb{Q})$ is abelian, all its subgroups are normal, and hence every intermediate field is Galois over $\mathbb{Q}$. We have that $\mathrm{Gal}(\mathbb{Q}(a)/\mathbb{Q}) \cong (\mathbb{Z}/10\mathbb{Z})/(5\mathbb{Z}/10\mathbb{Z}) \cong \mathbb{Z}/5\mathbb{Z}$. (Here, I have modded out by the unique subgroup of $\mathbb{Z}/10\mathbb{Z}$ of order 2, which we determined is the Galois group of $K/\mathbb{Q}(a)$.)

August 2023 Problem 2 Let p be an odd prime number. Prove that any two fields of order p^2 are isomorphic. (Do not just quote the theorem that there's a unique isomorphism class of fields of any prime power order; your job is to prove a case of that theorem!)

Solution

Any field of order p^2 must be a degree 2 extension of $\mathbb{F}_p$, and hence is of the form $\mathbb{F}_p[\sqrt{a}]$ for some non-square $a \in \mathbb{F}_p$, since $p \neq 2$. The multiplicative group $\mathbb{F}_p^\times$ is cyclic, so let $g \in \mathbb{F}_p^\times$ be some generator. Suppose $a \in \mathbb{F}_p^\times$ is nonsquare, and write $a = g^k$ for some odd k. Then we have that $(\sqrt{g}^k)^2 = a$, and thus $\sqrt{g}^k$ is a square root of a, so $\sqrt{a} \in \mathbb{F}_p[\sqrt{g}]$. Thus, $\mathbb{F}_p[\sqrt{a}] \subset \mathbb{F}_p[\sqrt{g}]$, but both are degree 2 extensions of $\mathbb{F}_p$, so they must be equal. Hence, every field of order p^2 is $\mathbb{F}_p[\sqrt{g}]$.

(Of course, this result is true also if $p = 2$, and for higher powers than just p^2. However, imitating the above argument is not a good way to try to prove the general case, a more clever argument is required. You can think for a while about how you might do this, then see the argument in Dummit and Foote.)

January 2023 Problem 4 Consider the finite field $\mathbb{F}_p$ where p is a prime number.

(a) How many monic irreducible polynomials of degree 5 are there over $\mathbb{F}_p$? (Your answer should be a function of p.)

(b) Let $P(x), Q(x) \in \mathbb{F}_p[x]$ be irreducible polynomials. Give a necessary and sufficient condition for there to exist a third polynomial $R(x) \in \mathbb{F}_p[x]$ such that $Q(x)$ divides $P(R(x))$. (The polynomial $R(x)$ need not be irreducible.)

Solution

(a) If f is an irreducible polynomial of degree 5, then it has 5 distinct roots in $\mathbb{F}_{p^5}$, which is the unique degree 5 extension of $\mathbb{F}_p$. Distinct irreducible polynomials will have distinct roots. There are p^5 total elements of $\mathbb{F}_{p^5}$, but p of those elements lie in the base field $\mathbb{F}_p$. Thus, the number of degree 5 elements of $\mathbb{F}_{p^5}$ is $p^5 - p$, and since each irreducible polynomial corresponds to 5 distinct roots, we have that the number of irreducible polynomials is $\frac{p^5-p}{5}$. (Can you independently confirm that for any prime number p, $p^5 - p \equiv 0 \pmod 5$?)

(b) Saying that $Q(x) \mid P(R(x))$ is the same as saying $P(R(x)) = 0 \mod Q(x)$. But $\mathbb{F}_p[x]/Q(x) \cong \mathbb{F}_{p^d}$ for $d = \deg Q(x)$, where the isomorphism is given by $x \mapsto \alpha$ for some α a root of $Q(x)$ in $\mathbb{F}_{p^d}$. Then saying $P(R(x)) = 0 \mod Q(x)$ is the same as saying $P(R(\alpha)) = 0$ in $\mathbb{F}_{p^d}$. But then $R(\alpha)$ is simply some element in $\mathbb{F}_{p^d}$, and every element of $\mathbb{F}_{p^d}$ can be written as $R(\alpha)$ for some R, so this is the same as saying P has a root in $\mathbb{F}_{p^d}$, which happens if and only if $\deg P \mid \deg Q$.

August 2022 Problem 4 Let p be a prime number, $\mathbb{F}_p$ the finite field with p elements, and $K = \mathbb{F}_p(t)$ the field of rational functions in one variable t over $\mathbb{F}_p$. (So that K is the fraction field of the polynomial ring $\mathbb{F}_p[t]$).

(a) Show that the splitting field of the polynomial $x^p - t \in K[x]$ over K is inseparable.

(b) Show that the splitting field of $x^p - t - 1 \in K[x]$ over K is the same as the splitting field of $x^p - t$.

(c) Show that K has a single degree p inseparable extension.

Solution

(a) A field extension L/K is said to be *inseparable* if there exists some $\alpha \in L$ whose minimal polynomial is inseparable. In this case, let L be the splitting field of $x^p - t$, and let $\alpha = \sqrt[p]{t}$ be a root of $x^p - t$. The polynomial $x^p - t$ is irreducible over K by Eisenstein's criterion, so this is the minimal polynomial of $\sqrt[p]{t}$. However, $x^p - t$ factors as $(x - \sqrt[p]{t})^p$, so this polynomial is not separable.

(b) Note that $(\sqrt[p]{t} - 1)^p = (\sqrt[p]{t})^p - 1^p = t - 1$, so $\sqrt[p]{t-1} \in K(\sqrt[p]{t})$ and visa versa, so these are the same extension.

(c) Let L/K be a degree p inseparable field extension. First, we want to show that $L = K(\alpha)$ for some inseparable element α. Certainly, L has an element α whose minimal polynomial is inseparable. Furthermore, since field degrees are multiplicative, we have $p = [L : K] = [L : K(\alpha)][K(\alpha) : K]$, and moreover $[K(\alpha) : K] \neq 1$ (or else $\alpha \in K$ is separable), so $[K(\alpha) : K] = p$ and thus $K(\alpha) = L$.

Now, it is a fact that any inseparable irreducible polynomial is of the form $g(x^p)$ for some irreducible polynomial $g(x)$. Thus, the only inseparable irreducible polynomials of degree p are of the form $x^p - f$ for some $f \in K$. But notice that $f(\sqrt[p]{t})^p = f(\sqrt[p]{t}^p) = f(t)$, so that $\sqrt[p]{f} \in K(\sqrt[p]{t})$. Thus, the only degree p inseparable extension of K is $K(\sqrt[p]{t})$.

January 2022 Problem 5 Let F be a field of prime characteristic p. Suppose $E = F(\alpha)$ is a simple extension such that $\alpha \notin F$ but $\alpha^p - \alpha \in F$.

(a) Find $[E : F]$.

(b) Prove that E/F is a Galois extension.

(c) Find the Galois group $\mathrm{Gal}(E/F)$.

Hint: What is $(x + 1)^p - (x + 1)$ in characteristic p?

Solution

(a) Let $\alpha^p - \alpha = b \in F$, so that α is a root of the polynomial $f(x) = x^p - x - b$. Thus, $[E : F] \leq p$. We will show that $[E : F] = p$, but we will come back to it, because it's easier to first show that E is Galois.

(b) Notice that $(x + 1)^p - (x + 1) = x^p - x$, so $\alpha, \alpha + 1, \ldots, \alpha + p - 1$ are the p roots of f. Thus, f splits in E. Furthermore, f clearly has no repeated roots, so f is separable. Thus, E is the splitting field of a separable polynomial, and thus is Galois.

(c) The Galois group $\mathrm{Gal}(E/F)$ has size at most p, since we have already seen that $[E : F] \leq p$. But it is easy to check that $\alpha \mapsto \alpha + i$ gives p different automorphisms of $\mathrm{Gal}(E/F)$, so this must be all of them. Thus, $\mathrm{Gal}(E/F) \cong \mathbb{Z}/p\mathbb{Z}$, generated by $\alpha \mapsto \alpha + 1$.

(a) (again) Since E/F is a Galois extension and $|\mathrm{Gal}(E/F)| = p$, we must have $[E : F] = p$.

August 2021 Problem 2 Let $n > 1$ be a square-free integer.

(a) Let $\alpha \in \mathbb{C}$ be a fourth root of n, i.e. $\alpha^4 = n$. Prove that $K = \mathbb{Q}(\alpha)$ is a degree 4 extension of $\mathbb{Q}$.

(b) Find (with proof) the Galois group of the normal closure of K (in $\mathbb{C}$) over $\mathbb{Q}$.

Solution

(a) The polynomial $x^4 - n$ is p-Eisenstein for any $p \mid n$, so it is irreducible, and thus is the minimal polynomial of α. Hence, $[K : \mathbb{Q}] = \deg \alpha = 4$.

(b) One can see that $x^4 - n$ does not split over K, but does split over $K(i)$, and $[K(i) : K] = 2$, so $\mathbb{Q}(\alpha, i)$ is the normal closure of K. The Galois group is D_4, which can be shown by demonstrating two generating automorphisms which satisfy the relationships $\sigma^2 = \mathrm{id}$, $\tau^4 = \mathrm{id}$, $\sigma\tau\sigma = \tau^{-1}$.

January 2021 Problem 5 Let $\mathbb{F}_p$ denote the field with p elements.

(a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?

(b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a.

(c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

Solution

(a) Observe that $x^4 + 1 = \Phi_8(x)$, so the roots of this polynomial are primitive 8th roots of unity. There are no primitive 8th roots of unity in $\mathbb{F}_3$, but in $\mathbb{F}_9$ every element is a root of $x^9 - x$, so every nonzero element is a root of $x^8 - 1$, and so there are primitive 8th roots of unity. In particular, if $i^2 = -1$ in $\mathbb{F}_9$, then $\pm 1 \pm i$ are primitive 8th roots of unity. Pairing these up, we can find that $x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1)$.

(b) Consider $a = 2$. Then $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$. Indeed, it has no linear factors since $\alpha^4 \in \{0, 1\}$ for all $\alpha \in \mathbb{F}_5$. To show it has no quadratic factors, we need to show that $x^4 - 2$ does not split over $\mathbb{F}_{25}$. For every $x \in \mathbb{F}_{25}$, we have that $x^5 \in \mathbb{F}_5$, so if $x \in \mathbb{F}_{25} \setminus \mathbb{F}_5$ and $x^5 = n \in \mathbb{F}_5$, then $x^4 = n/x \notin \mathbb{F}_5$. Thus, $x^4 - 2$ has no roots in $\mathbb{F}_{25}$, and hence has no quadratic factors over $\mathbb{F}_5$.

(c) First, let's rephrase the question in terms of field theory facts. Saying $x^4 - a$ factors that way is the same as saying that it has an irreducible quadratic factor. On the other hand, every irreducible quadratic over $\mathbb{F}_{103}$ splits in $\mathbb{F}_{103^2}$ (because finite fields are neat), so saying $x^4 - a$ has an irreducible quadratic factor is the same as saying (i) $x^4 - a$ does not split in $\mathbb{F}_{103}$, but (ii) $x^4 - a$ *does* split in $\mathbb{F}_{103^2}$. We'll prove those two statements, in that order.

First, note that the map $\mathrm{pow}_4 : \mathbb{F}_{103}^\times \to \mathbb{F}_{103}^\times$ given by $x \mapsto x^4$ is a group homomorphism. The kernel of this map is an abelian 2-group (since every element satisfies $x^4 = 1$) which contains ± 1, and has order dividing $103 - 1 = 102$. Since $4 \nmid 102$, we must have $\ker \mathrm{pow}_4 = \{\pm 1\}$. Since for any $a \in \mathbb{F}_{103}$, the solutions to $x^4 - a$ are the preimages of a under pow_4, there are either 0 or 2 such solutions, and $x^4 - a$ so cannot split over $\mathbb{F}_{103}$.

On the other hand, notice that $x^2 + 1$ has a root $i = \sqrt{-1}$ in $\mathbb{F}_{103^2}$, so if $x^4 - a$ has a root α in $\mathbb{F}_{103^2}$, then $\pm\alpha, \pm i\alpha$ are 4 distinct roots lying in $\mathbb{F}_{103^2}$. Thus, to show that $x^4 - a$ splits in $\mathbb{F}_{103^2}$, we need only show that $x^4 - a$ has a root in $\mathbb{F}_{103^2}$.

The multiplicative group $\mathbb{F}_{103^2}^\times$ is cyclic, so let it be generated by g, so g has exact order $103^2 - 1 = 102 \cdot 104$. Recall that the Galois group of $\mathbb{F}_{103^2}$ is cyclic of order 2, generated by the Frobenius automorphism $x \mapsto x^{103}$. Then notice that for any k, $g^{104k} \in \mathbb{F}_{103}$, since $(g^{104k})^{103} = (g^{102\cdot 104})^k g^{104k} = g^{104k}$, so g^{104k} is fixed by the Galois group. Moreover, there are 102 choices for k, which give the 102 distinct elements of $\mathbb{F}_{103}^\times$. Since any $a \in \mathbb{F}_{103}^\times$ can be written as g^{104k} for some k, this means that g^{26k} is a solution to $x^4 - a = 0$, and we're done.

(In fact, if a is not a 4th power in $\mathbb{F}_{103}$, then both a and -1 represent the nontrivial element in $\mathbb{F}_{103}^\times / \operatorname{im} \mathrm{pow}_4 \cong \mathbb{Z}/2\mathbb{Z}$. Thus -$a$ is a 4th power, so say $d^4 = -a$. Since $103 \equiv 7 \mod 8$, number theory tells us that $\sqrt{2} \in \mathbb{F}_{103}$, and so we can factor $x^4 - a$ as
$(x^2 + \sqrt{2}d + d^2)(x^2 - \sqrt{2}d + d^2)$.)

Below are alternative, more explicit computational solutions for (a) and (b):

(a) Observe that $x^4 + 1$ has no roots in $\mathbb{F}_3$, so it has no linear factors over $\mathbb{F}_3[x]$. Therefore, if it factors it must be as $(x^2 + ax + b)(x^2 + cx + d)$ for some $a, b, c, d \in \mathbb{F}_3$. Multiplying out, we see that we must have $a + c = 0, ac + b + d = 0$, and $bd = 1$. The last equation only has solutions $b = d = 1$ and $b = d = 2$. The former gives no solutions to the remaining equations, and the latter gives the solutions $a = 1, c = 2$ and $a = 2, c = 1$. Thus, $x^4 + 1 = (x^2 + x + 2)(x^2 + 2x + 2)$ over $\mathbb{F}_3[x]$.

(b) Consider $a = 2$. Then $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$. Indeed, it has no linear factors since $\alpha^4 \in \{0, 1\}$ for all $\alpha \in \mathbb{F}_5$. So, we just need to show that it has no quadratic factors $x^2 + bx + c$ with $b, c \in \mathbb{F}_5$. Observe that if $x^2 = -bx - c$, then $x^4 = (-b^3 + 2bc)x - b^2c + c^2$. So, we need to show that if $-b^3 + 2bc = 0$, then $-b^2c + c^2 \neq 2$. If $b = 0$, then it is clear that $c^2 \neq 2$ for any choice of c. If $b \neq 0$, then the only possible solutions for $-b^3 + 2bc = 0$ are $c = 3, b \in \{1, 4\}$ and $c = 2, b \in 2, 3$. In the first case, $-b^2c + c^2 = 0$, and in the second case, $-b^2c + c^2 = 3$. Therefore, $x^4 - 2$ is irreducible in $\mathbb{F}_5[x]$.

August 2020 Problem 3 Let L/K be a degree p Galois extension, where p is prime, and let σ be a nontrivial element of $\mathrm{Gal}(L/K)$. Then σ is a linear transformation of the K-vector space L.

(a) What is the characteristic polynomial of σ?

(b) Write L^0 for the subspace of L consisting of elements with trace 0. Show that σ sends L^0 to L^0.

(c) What is the characteristic polynomial of σ acting on L^0?

Solution

(a) Since $|\mathrm{Gal}(L/K)| = [L : K] = p$, $\mathrm{Gal}(L/K)$ must be the cyclic group of order p. So, in particular, σ satisfies $x^p - 1 = 0$. Furthermore, σ in fact generates $\mathrm{Gal}(L/K)$, so $\mathrm{id}, \sigma, \sigma^2, \dots, \sigma^{p-1}$ are all the automorphisms of L/K. By linear independence of characters, these automorphisms are all linearly independent over K, so σ does not satisfy any polynomial of degree $< p$ over K. Hence, $x^p - 1$ is both the minimal and characteristic polynomial of σ.

(b) Recall that the trace of an element is the sum of its Galois conjugates. So, if $\mathrm{Tr}(\alpha) = 0$, then we have $\mathrm{Tr}(\sigma(\alpha)) = \sum_{g\in\mathrm{Gal}} g(\sigma(\alpha)) = \sum_{g'\in\mathrm{Gal}} g'(\alpha) = 0$. Here we have used that the action of a group on itself by right multiplication is free.

(c) Since $\mathrm{Tr}(\alpha) = 0$ is a single K-linear condition, we know that L^0 is a dimension $p - 1$ vector space over K. So, we're looking for a degree $p - 1$ polynomial. We recall that the roots of the characteristic polynomial are the eigenvalues of σ. Since L^0 is a subspace of L, every eigenvector of σ acting on L^0 is also an eigenvector of σ acting on L, thus the characteristic polynomial of σ acting on L^0 divides $x^p - 1$.

Now we divide into cases based on the characteristic. If $\mathrm{char}(K) = p$, then $x^p - 1 = (x - 1)^p$, so the only degree $p-1$ divisor of $x^p - 1$ is $(x - 1)^{p-1}$. If $\mathrm{char}(K) \neq p$, then all but one of the distinct roots of $x^p - 1$ is an eigenvalue of σ acting on L^0. And let's be honest, it'd be pretty weird if the missing eigenvalue wasn't 1. If $\alpha \in L^0$ is an eigenvector for σ of eigenvalue 1, that means $\sigma(\alpha) = \alpha$, which means that α is fixed by $\mathrm{Gal}(L/K)$ since σ generates the Galois group, which means that $\alpha \in K$. But then $\mathrm{Tr}(\alpha) = p\alpha$, and since $\mathrm{char}(K) \neq p$, if $p\alpha = 0$ then $\alpha = 0$, so σ has no eigenvectors of eigenvalue 1 in L^0. Thus, the characteristic polynomial of σ acting on L^0 is $\frac{x^p-1}{x-1}$.

January 2020 Problem 2

(a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.

(b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F.

(c) Compute the Galois group $\mathrm{Gal}(F(b)/F)$.

Solution

(a) The roots of $x^4 - 7$ are $\pm\sqrt[4]{7}$ and $\pm i\sqrt[4]{7}$. If $a = \pm\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $i\sqrt[4]{7}$ so it is not a splitting field, and thus not Galois. If $a = \pm i\sqrt[4]{7}$, then $\mathbb{Q}(a)$ does not contain $\sqrt[4]{7}$, so again is not a splitting field so not Galois over $\mathbb{Q}$.

(b) Since $x^4 - 7 \equiv x^4 - 2 \mod 5$, we get that the roots of this polynomial over F are $\pm b$ and $\pm ib$, where $i = \sqrt{-1}$. Again, since $\text{char}(F) = 5$, 2 is a square root of $-1 = 4$, so the roots are $\pm b$ and $\pm 2b$, or, more simply, $b, 2b, 3b$, and $4b$. Since each of these is in $F(b)$, it is the splitting field of $x^4 - 7$, and since this polynomial is separable, $F(b)$ is Galois over F.

(c) An automorphism $\sigma \in \text{Gal}(F(b)/F)$ is determined by which root it sends b to. Thus, the elements in $\text{Gal}(F(b)/F)$ are $\sigma_1 = Id : b \mapsto b$, $\sigma_2 : b \mapsto 2b$, $\sigma_3 : b \mapsto 3b$, and $\sigma_4 : b \mapsto 4b$. Observe that $\sigma_2^2 = \sigma_3^2 = \sigma_4$ and $\sigma_4^2 = Id$. Therefore, $\text{Gal}(F(b)/F) \cong (\mathbb{Z}/5\mathbb{Z})^\times$ where the isomorphism is given by $\sigma_n \mapsto n$.

January 2019 Problem 4

(a) Let E be the splitting field over $\mathbb{Q}$ of $e(x) = x^3 + 2x^2 + 3x + 4$, which has discriminant -200. Find $\text{Gal}(E/\mathbb{Q})$.

(b) Let F be the splitting field over $\mathbb{Q}$ of $f(x) = x^3 + 4x^2 + 7x + 6$, which has discriminant -72. Find $\text{Gal}(F/\mathbb{Q})$.

(c) Find $\text{Gal}(EF/\mathbb{Q})$.

Solution

(a) First, check to see if $e(x)$ has any rational roots. The possible rational roots are $\pm1, \pm2, \pm4$, none of work actually work. Since this is a degree 3 polynomial with no roots in $\mathbb{Q}$, this means that it is irreducible over $\mathbb{Q}$, so $\text{Gal}(E/\mathbb{Q})$ is a transitive subgroup of S_3. So, it is either A_3 or S_3. Also, since $e'(x) = 3x^2 + 4x + 3$ has no real roots, $e(x)$ has no relative extrema, so $e(x)$ must just have one real root. This means that the other two roots are complex conjugates, so complex conjugation is an order 2 element of $\text{Gal}(E/\mathbb{Q})$. Thus, $\text{Gal}(E/\mathbb{Q}) = S_3$.

Another way to get that $\text{Gal}(E/\mathbb{Q}) \neq A_3$ is because $\text{Gal}(E/\mathbb{Q}) \subseteq A_3$ iff the discriminant of $e(x)$ is a square in $\mathbb{Q}$. Since -200 is not a square, we get that $\text{Gal}(E/\mathbb{Q}) \not\subseteq A_3$.

(b) Using the rational root theorem again, we find that -2 is a root of $f(x)$, and it factors as $f(x) = (x+2)(x^2 + 2x + 3)$. Therefore, the roots of $f(x)$ are $-2, -1+\sqrt{-2}, -1-\sqrt{-2}$. This gives that $F = \mathbb{Q}(\sqrt{-2})$, so $\text{Gal}(F/\mathbb{Q}) = \mathbb{Z}/2\mathbb{Z}$.

(c) By definition, the discriminant of $e(x)$ is $[(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3)]^2$ where $\alpha_1, \alpha_2, \alpha_3$ are the roots of $e(x)$. Therefore, the square root of the discriminant is an element in E. This gives that $\sqrt{-200} = 10\sqrt{-2} \in E$, so in particular, $F = \mathbb{Q}(\sqrt{-2}) \subseteq E$. Therefore, $EF = E$ and $\text{Gal}(EF/\mathbb{Q}) = S_3$.

August 2018 Problem 4

(a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.

(b) Let $K = \mathbb{F}_q(t)$, and let $L = \mathbb{F}_q(t^{1/p})$ where q is a power of p. Prove that there are no automorphisms of L fixing K.

Solution

(a) Consider $K = \mathbb{Q}(\zeta_5)$ where ζ_5 is a primitive 5th root of unity. We have that $\mathrm{Gal}(K/\mathbb{Q}) = \mathbb{Z}/5\mathbb{Z}^\times = \mathbb{Z}/4\mathbb{Z}$.

(b) Observe that $t^{1/p}$ satisfies $x^p - t \in K[x]$, and because K is characteristic p, this polynomial factors as $x^p - t = x^p - (t^{1/p})^p = (x - t^{1/p})^p$. Therefore, $t^{1/p}$ is its only root. Since L is the splitting field of $x^p - t$ over K, any automorphism of L over K is determined by how it permutes the roots of $x^p - t$. But since there is only one root, the only automorphism is the identity on L sending $t^{1/p} \mapsto t^{1/p}$.

January 2018 Problem 5 Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\mathrm{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)

Solution

Since $f(x)$ has degree 4, we know that $|\mathrm{Gal}(K/\mathbb{Q})| \leq 4!$ and that $\mathrm{Gal}(K/\mathbb{Q})$ is a transitive subgroup of S_4. So, it could be $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, D_4, A_4, or S_4. To find the roots of $f(x)$, notice that it is a quadratic in disguise. Letting $t = x^2$, we have that $f(t) = t^2 - t - 1$, which has roots $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$. Therefore, the roots of $f(x)$ are given by $x = \pm\sqrt{\alpha}, \pm\sqrt{\beta}$, and thus, $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$.

Observe that K contains the quadratic subfield $\mathbb{Q}(\alpha) = \mathbb{Q}(\beta) = \mathbb{Q}(\sqrt{5})$, so the Galois group must contain an index 2 subgroup. Therefore, it cannot be A_4. Also, since $K = \mathbb{Q}(\sqrt{\alpha}, \sqrt{\beta})$ has degree at most 4 over $\mathbb{Q}(\sqrt{5})$, the Galois group of $K/\mathbb{Q}$ can have order at most 8. So, it cannot be S_4.

We will now show that $\mathrm{Gal}(K/\mathbb{Q}) = D_4$, a group of order 8, by showing that $[K : \mathbb{Q}] > 4$, which will eliminate $\mathbb{Z}/4\mathbb{Z}$ and the Klein 4 group as options. Consider the subfield $L = \mathbb{Q}(\sqrt{\alpha}, \beta) \subseteq K$, which is not all of K since $L \subseteq \mathbb{R}$ and $K \not\subseteq \mathbb{R}$. We will show that $[L : \mathbb{Q}] = 4$, which implies that $[K : \mathbb{Q}] > 4$. Since $\mathbb{Q}(\sqrt{5}) \subseteq L$, it is enough to show that L is strictly larger than $\mathbb{Q}(\sqrt{2})$. Suppose instead that $\sqrt{\alpha} = a + b\sqrt{5}$ with $a, b \in \mathbb{Q}$. Then squaring both sides gives

$$\frac{1}{2} + \frac{1}{2}\sqrt{5} = a^2 + 5b^2 + 2ab\sqrt{5}$$

which implies that $\frac{1}{2} = a^2 + 5b^2$ and $\frac{1}{2} = 2ab$. Substituting $b = \frac{1}{4a}$ into the first equation and simplifying (since $a \neq 0$) gives $16a^4 - 8a^2 + 5 = 0$. You can then use the rational root theorem to show that none of the possible roots in $\mathbb{Q}$ are actually roots. Therefore, $\sqrt{\alpha} \notin \mathbb{Q}(\sqrt{5})$, so $[L : \mathbb{Q}] = 4$, and we are done.

August 2017 Problem 4 Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K : \mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m. Show that $\mathrm{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Solution
Since $K/\mathbb{Q}$ is Galois and degree 4, its Galois group is either $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. We can show that it is the latter by showing that K has two quadratic subfields, which means that $\mathrm{Gal}(K/\mathbb{Q})$ has two index 2 subgroups. We know that $\mathbb{Q}(\sqrt{-m}) \subseteq K$ is one quadratic subfield since $\sqrt{-m} \in K$ and the minimal polynomial of this element is $x^2 + m$. Another quadratic subfield is given by the fixed field of complex conjugation. Let $c : \mathbb{C} \to \mathbb{C}$ be complex conjugation. Then c is a nontrivial automorphism of K since c is not the identity on the subfield $\mathbb{Q}(\sqrt{-m})$. Since c has order 2, the fixed field of K by c is a quadratic subfield of K different from $\mathbb{Q}(\sqrt{-m})$.

January 2017 Problem 4 This is a question about "biquadratic extensions" in two parts.

(a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\mathrm{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E.

(b) Give an example of a field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$.

Solution

(a) Suppose $F = E(\sqrt{x}, \sqrt{y})$, where none of x, y, and xy are squares in E. This is a degree 4 extension, since it has basis $1, \sqrt{x}, \sqrt{y}, \sqrt{xy}$. It has four easy-to-think-of automorphisms: the identity, σ which flips the sign of $\sqrt{x}$ and $\sqrt{xy}$, τ which flips the sign of $\sqrt{y}$ and $\sqrt{xy}$, and $\sigma\tau$ which flips the sign of $\sqrt{x}$ and $\sqrt{y}$. Since the automorphism group can have size at most 4, we know we've found all the automorphisms, so this is in fact a Galois extension.

On the other hand, suppose F/E is a degree 4 Galois extension with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$. Then, by the Fundamental Theorem of Galois Theory, from the subgroup structure of $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$, we know that F has three intermediate extension, which are degree 2 over E. Since E is not characteristic 2, the quadratic formula tells us that any degree 2 extension is of the form $E(\sqrt{d})$ for some nonsquare d. So, we can say two of the intermediate extensions are $E(\sqrt{x})$ and $E(\sqrt{y})$. Since these are distinct extensions of E, $\sqrt{x} \notin E(\sqrt{y})$ and visa versa, so $E(\sqrt{x}, \sqrt{y})$ is a degree 4 extension of E sitting inside F, and thus in fact we have $F = E(\sqrt{x}, \sqrt{y})$.

(b) $E = \mathbb{F}_2$ works, because all the extensions of $\mathbb{F}_2$ have cyclic Galois group.

August 2016 Problem 5, modified Consider the field $K = \mathbb{Q}(\zeta_7)$. Find an element $x \in K$ which generates a quadratic subextension, that is, so that $[\mathbb{Q}(x) : \mathbb{Q}] = 2$. (Proving that such an x exists will earn you partial credit. For full credit, express x explicitly as a polynomial in ζ_7, such as $42\alpha^3 - 1337\alpha^2$.)

We know that $\mathrm{Gal}(K/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times$ is cyclic of order 6, so has a unique subgroup H of index 2. Then we know that the fixed field K^H is the field we're looking for. Explicitly, H is the subgroup consisting of all the squares mod 7, $H = \sqrt{1,2,4}$. To find a specific element fixed by this subgroup, we can take an orbit under this subgroup and add all the elements in that orbit together. So, one possible solution would be $x = \zeta_7 + \zeta_7^2 + \zeta_7^4$. This x is clearly fixed by H, but not fixed by all of $\mathrm{Gal}(K/\mathbb{Q})$, so it generates a degree 2 extension of $\mathbb{Q}$.

January 2016 Problem 1 Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f.

(a) Show that if $\mathrm{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.

(b) Is the converse statement true? That is, suppose $L = K(\alpha)$; must $\mathrm{Gal}(L/K)$ be abelian?

Solution

(a) Assume that $G = \mathrm{Gal}(L/K)$ is abelian. We know that $K(\alpha) \subseteq L$, so the Galois correspondence implies that there is a subgroup $H \leq G$ such that $H = \mathrm{Gal}(L/K(\alpha))$. Since G is abelian, H is a normal subgroup, so $K(\alpha)$ is a Galois extension of K. Since $f(x)$ is irreducible, it is the minimal polynomial of α over K, so f splits in $K(\alpha)$ since $K(\alpha)/K$ is Galois and contains a root of f so must contain all of the roots of f. Thus, $K(\alpha) = L$.

(b) This is not true. Let L be any finite Galois extension of $\mathbb{Q}$ with non-abelian Galois group. Then the primitive element theorem gives that $L = \mathbb{Q}(\alpha)$ for some $\alpha \in L$.

August 2015 Problem 5 Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x. Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

Solution

Let ζ be a primitive nth root of unity, let $\sigma_i : \mathbb{C}(x) \to \mathbb{C}(x)$ be the automorphism induced by $x \mapsto \zeta^i x$ for $0 \leq i \leq n-1$, and let τ be the automorphism defined by $x \mapsto \frac{1}{x}$. Notice that all of the σ_i and τ fix $\mathbb{C}(y)$ since $\zeta^n = 1$, so their products $\sigma_i\tau$ also fix $\mathbb{C}(y)$. Therefore, $Id, \sigma_1, \ldots, \sigma_{n-1}, \tau, \sigma_1\tau, \ldots, \sigma_{n-1}\tau$ are all elements of $\mathrm{Aut}(\mathbb{C}(x)/\mathbb{C}(y))$, so $|\mathrm{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \geq 2n$. On the other hand, we have that x is a root of $t^{2n} + 1 - (x^n + x^{-n})t^n \in \mathbb{C}(y)[t]$, which is a degree $2n$ polynomial, so $[\mathbb{C}(x) : \mathbb{C}(y)] \leq 2n$. Since $|\mathrm{Aut}(\mathbb{C}(x)/\mathbb{C}(y))| \leq [\mathbb{C}(x) : \mathbb{C}(y)]$, they must both be equal to $2n$. Thus, $\mathbb{C}(x)/\mathbb{C}(y)$ is a Galois extension of degree $2n$.

August 2011 Problem 3 Let $K \subseteq F \subseteq E$ be fields with $E = F[\alpha]$ and with $\alpha^n \in F$ for some positive integer n. Suppose K contains a primitive nth root of unity, and let L be a field with $K \subseteq L \subseteq E$ and $L \cap F = K$.

(a) If L is Galois over K, show that $L = K[\beta]$ for some element β with $\beta^n \in K$.

(b) Show by example that L need not be Galois over K. (Hint. Take $n = 2$.)

Solution

(a) Let $\zeta \in K$ be a primitive nth root of unity. If $\alpha^d \in F$ for some $d < n$, then $d \mid n$. Since K certainly contains a primitive dth root of unity, namely $\zeta^{n/d}$, so without loss of generality we may assume that n is the least integer such that $\alpha^n \in F$. Then $[E : F] \leq n$. Notice that for each i, $\alpha \mapsto \zeta^i \alpha$ gives a distinct automorphism of E fixing F, so E/F is Galois with Galois group C_n. For each $d \mid n$, $F(\alpha^{n/d})$ is a degree d extension of F contained in E. But we know that such extensions are in bijection with subgroups of C_n, and C_n has a unique subgroup of each order dividing n, so in fact every intermediate extension is of the form $F(\alpha^{n/d})$.

Now, consider the particular intermediate extension FL, the composite of F and L. This is equal to $F(\alpha^{n/d})$ for some d, so is the splitting field of $x^d - \alpha^n$.

(b) Let $K = \mathbb{Q}$, F and L be different fields containing cube roots of 2, say $F = \mathbb{Q}(\sqrt[3]{2})$ and $L = \mathbb{Q}(\zeta_3 \sqrt[3]{2})$, and $E = FL = \mathbb{Q}(\sqrt[3]{2}, \zeta_3)$. Then E is a quadratic extension of F, so it can be written as $F(\sqrt{\alpha})$ (in fact $F(\sqrt{-3})$). However, L cannot be written as $\mathbb{Q}(\sqrt{\beta})$ because it is a degree 3 extension.

5 Rings and Modules

Commutative rings

January 2026 Problem 2 Let M be a free R-module of rank n, where $n > 0$ is an integer and R an integral domain.

(a) Given an example of n elements of M which are linearly independent but do not form a basis (for some M, R, and n). No justification is required.

(b) Show that any $n + 1$ elements of M are linearly dependent over R.

Solution

(a) Let $R = M = \mathbb{Z}$ ($n = 1$), then the set $\{2\}$ is linearly independent but does not form a basis for M.

(b) Let $K = \mathrm{Frac}(R)$, and let $V = M \otimes_R K$. Then V is a vector space over K of dimension n, and we have an injective map of R-modules $M \to V$. Given $n + 1$ elements of M, they are certainly linearly dependent in V. By clearing denominators, we obtain a linear dependence over R as well.

January 2025 Problem 5 Let A be a commutative Noetherian ring, and M an A-module. We say that M has finite length if there exists a positive integer n such that every chain

$$M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_k$$

satisfies $k \leq n$. The smallest such n is then called the length of M.

(a) Prove that if M has finite length, then so does every localization $M_{\mathfrak{p}}$ at a prime ideal $\mathfrak{p}$ of A.

(b) Now assume that A is local with maximal ideal $\mathfrak{m}$. Assume furthermore that the field $\kappa = A/\mathfrak{m}$ embeds into A as a subring. Prove that the length of M equals the dimension of M over κ via restriction of scalars. (**NB** if the length and the dimension are both infinite, we consider them to be equal, we're not going to be particular about different ordinals/cardinals.)

(c) Prove that the following converse of (a) holds if A has finitely many maximal ideals: under this assumption, M has finite length if $M_{\mathfrak{m}}$ has finite length for every maximal ideal $\mathfrak{m}$ of A. (You can use without proof the fact that an A-module M is zero if and only if $M_{\mathfrak{m}}$ is zero for every maximal ideal $\mathfrak{m}$ of A.)

(d) Prove that the assumption in part (c) on the finiteness of the number of maximal ideals is necessary by providing an example of a ring A and an A-module M which is not of finite length, even though $M_{\mathfrak{m}}$ is of finite length for every maximal ideal $\mathfrak{m}$ of A. (Hint: you can find such an example where A is the ring of integers.)

Solution

(a) Given a prime $\mathfrak{p} \subset A$, we will show that $\operatorname{len}(M_{\mathfrak{p}}) \leq \operatorname{len}(M)$. Let $\ell\colon M \to M_{\mathfrak{p}}$ be the localization map. Given any chain contained in $M_{\mathfrak{p}}$, $\widetilde{M_0} \subsetneq \cdots \subsetneq \widetilde{M_k}$, we want to show that M contains a chain at least as long. Define $M_i = \ell^{-1}(\widetilde{M_i}) \subset M$. Clearly $M_i \subseteq M_{i+1}$ for all i, so if we show that the inclusions are strict we will be done.

Since $\widetilde{M_i} \subsetneq \widetilde{M_{i+1}}$, we can find some $m/s \in \widetilde{M_{i+1}} \setminus \widetilde{M_i}$ where $s \notin \mathfrak{p}$. If we look in the quotient $\widetilde{M_{i+1}}/\widetilde{M_i}$, the annihilator of the class $[m/s]$ is an ideal of $A_{\mathfrak{p}}$. Since $A_{\mathfrak{p}}$ is a local ring, and since the class $[m/s]$ was chosen not to be the zero class, the annihilator must be contained in the maximal ideal $\mathfrak{p}$. Since $s \notin \mathfrak{p}$, s does not annihilate $[m/s]$, so $s[m/s] \neq [0]$ in the quotient, i.e. $s(m/s) = m/1 \notin \widetilde{M_i}$. Hence, $m/1 \in \widetilde{M_{i+1}} \setminus \widetilde{M_i}$, and so $m \in M_{i+1} \setminus M_i$.

(b) First, we will show that $\operatorname{len}(M) \leq \dim_\kappa(M)$. Suppose $M_0 \subsetneq \cdots \subsetneq M_k$ is any chain in M. We can find elements $m_i \in M_i \setminus M_{i-1}$ for $i = 1, \ldots, k$. We will be done if we can show that the m_i are κ-linearly independent. Suppose $c_1 m_1 + \cdots + c_k m_k = 0$ for some $c_i \in \kappa$, if we look at this identity in the quotient M_k/M_{k-1}, it becomes $c_k[m_k] = 0$. If c_k were not 0, it would be invertible, and we would have $[m_k] = 0$, i.e. $m_k \in M_{k-1}$, which contradicts the way we chose it. Hence, $c_k = 0$. Continuing in like fashion, we conclude that all of the $c_i = 0$, and so the m_i are linearly independent.

I spent a long time trying to figure out how to finish the proof in an airtight manner. The normal way you would finish a proof like this is to establish the reverse inequality, $\dim_\kappa(M) \leq \operatorname{len}(M)$. However, I could see no clear way to go from an arbitrary basis $\{e_1, \ldots\}$ to a chain of submodules. E.g. for $A = M = \mathbb{C}[x]/(x^2)$, if you choose the $\kappa = \mathbb{C}$-basis $\{1, 1+x\}$, then the A-span of both basis vectors is all of M: $A \cdot 1 = A \cdot (1+x) = M$. The issue in that example is that we chose a bad basis, and we should have chosen the basis $\{1, x\}$, because this basis also generates submodules of M. The distinguishing thing about the second basis is that it comes from taking a generator for M and a generator for $\mathfrak{m}M$. So, our strategy is going to be to take generators for the submodules $\mathfrak{m}^i M$ and show that they give a chain of submodules as well as a κ-basis, and hence that $\operatorname{len}(M) = \dim_\kappa(M)$.

One more subtle thing to note: we're going to want to use Nakayama's lemma to obtain our generators, but Nakayama's lemma requires that M be finitely generated. Indeed, we must do this for the above strategy to work, as we can see from the example of $A = \mathbb{C}[x]_{(x)}$ and $M = \mathbb{C}(x)$. In this case, M is indeed both an infinite length module over A, as well as an infinite-dimensional vector space over $\kappa = \mathbb{C}$, but each submodule $(x^i)M = M$, so our strategy definitely fails in this case. However, we have already shown that $\operatorname{len}(M) \leq \dim_\kappa(M)$, so if the length is infinite then we already know that the dimension is as well, so we are done. So, we can freely assume that M has finite length, and so in particular is finitely generated.

So, suppose M has finite length. Consider the chain $M \supset \mathfrak{m}M \supset \mathfrak{m}^2 M \supset \ldots$. Since M has finite length, at some point this chain terminates $\mathfrak{m}^e M = 0$. Since M is finitely generated, by Nakayama's lemma we can find a minimal set of generators $\{x_{i,j}\}$ for $\mathfrak{m}^i M$ which descend to a basis for the κ-vector space $\mathfrak{m}^i M/\mathfrak{m}^{i+1} M$. For each $\mathfrak{m}^i M$, we have a chain $\mathfrak{m}^{i+1} M \subsetneq \mathfrak{m}^{i+1} M + Ax_{i,1} \subsetneq \mathfrak{m}^{i+1} M + Ax_{i,1} + Ax_{i,2} \subsetneq \cdots \subsetneq \mathfrak{m}^i M$. Thus, we get a chain of length equal to the total number of generators $x_{i,j}$.

For a fixed i, since the $x_{i,j}$ are κ-linearly independent in the quotient $\mathfrak{m}^i M/\mathfrak{m}^{i+1}M$, they are also κ-linearly independent in M. Furthermore, for any $y \in M$, we can write

$$y = \kappa\text{-linear combo of } x_{1,j} + y_1,$$

with $y_1 \in \mathfrak{m}M$. Continuing like this, we get that y is a κ-linear combination of the $x_{i,j}$, so the $x_{i,j}$ span M. Thus, we have found a κ-basis, and a chain of length equal to that basis. Since we already know $\operatorname{len}(M) \leq \dim_\kappa(M)$, this shows that $\operatorname{len}(M) = \dim_\kappa(M)$.

(c) For each maximal ideal $\mathfrak{m} \subset A$, let $\operatorname{len}(M_\mathfrak{m}) = n_\mathfrak{m}$. We claim $\operatorname{len}(M) \leq \sum_\mathfrak{m} n_\mathfrak{m}$. Let $M_0 \subsetneq \cdots \subsetneq M_k$ be a chain. Localizing at a maximal ideal $\mathfrak{m}$, we find there are at most $n_\mathfrak{m}$ locations where $(M_i)_\mathfrak{m} \subsetneq (M_{i+1})_{n_\mathfrak{m}}$, and the other inclusions must be equality. If k were greater than $\sum_\mathfrak{m} n_\mathfrak{m}$, then there would be some M_i so that $(M_i)_\mathfrak{m} = (M_{i+1})_\mathfrak{m}$ for all $\mathfrak{m}$. That is, $(M_{i+1}/M_i)_\mathfrak{m} = 0$ for all $\mathfrak{m}$, hence $M_{i+1}/M_i = 0$, hence $M_{i+1} = M_i$, a contradiction.

(d) An example is the $\mathbb{Z}$-module $\displaystyle\bigoplus_{p \text{ prime}} \mathbb{Z}/p\mathbb{Z}$. Let S be the set of integers not divisible by p, then localizing at S will kill every factor except $\mathbb{Z}/p\mathbb{Z}$, which is finite length as a $\mathbb{Z}[1/S]$-module. However, the whole module is clearly not finite length.

August 2024 Problem 2 Let k be a field, and let A denote the ring $k[x,y]/(xy)$. Consider the A-module M which is the quotient of the free rank 2 A-module with basis e_1 and e_2 by the submodule generated by the element $ye_1 + xe_2$.

(a) Prove that M is not a projective A-module. (Hint from the original test: find a submodule of M isomorphic to $k[x,y]/(x,y)$.) (**NB** There are other, more systematic, ways of doing this if you are comfortable with Tor.)

(b) Let $f \in A$ denote the polynomial $x + y$, and let A_f denote the localization of A at the multiplicative subset $\{1, f, f^2, \ldots\}$. Prove that $M \otimes_A A_f$ is free as an A_f-module.

Solution

(a) Consider the submodule of M generated by $ye_1 = -xe_2$. This is a cyclic submodule that is annihilated by x since $xye_1 = 0$, and is annihilated by y since $y(-xe_2) = 0$. Thus, this submodule must be isomorphic to $A/(x,y) = k[x,y]/(x,y)$. Now, in order to show that M is not projective, it suffices to show that no projective A-module can have $k[x,y]/(x,y)$ as a submodule. If P is a projective A-module, then P must be a direct summand of a free A-module, so it is enough to show that no free A-module can contain $k[x,y]/(x,y)$. If F is a free A-module, then by choosing a basis we can see that an element is annihilated by x only if it lies in yF, and is annihilated by y only if it lies in xF, and $xF \cap yF = \{0\}$, so there are no nonzero elements of F that are annihilated by both x and y. (A faster but trickier proof would be to consider the quotient P/yP over $A/(y) \cong k[x]$, which must still be a projective module. Since $k[x,y]/(x,y)$ is annihilated by y, it passes to the quotient unharmed, but then P/yP is a projective module over a domain, hence is torsion-free, while $k[x,y]/(x,y)$ is torsion as a $k[x]$-module.)

(b) We know that $M \otimes_A A_f = M_f$. Since localization is exact, we have that $M_f = ((A \oplus A)/(ye_1 + xe_2))_f = (A_f \oplus A_f)/(ye_1 + xe_2)$. We want to show this a free A_f-module, but the only way for the quotient to be free is if we quotiented out by a direct summand, so we need to show that the submodule generated by $ye_1 + xe_2$ is a direct summand of $A_f \oplus A_f$. This is very similar to row reducing a matrix, except our matrix is the column vector $(y,x)^T$ and our allowable operations are multiplying by invertible 2×2 matrices with coefficients in A_f. For instance:

$$\begin{pmatrix} 1 & 0 \\ -x & 1 \end{pmatrix} \begin{pmatrix} (x+y)^{-1} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

The existence of this row reduction implies that $ye_1 + xe_2$ generates a direct summand of $A_f \oplus A_f$, since after a change of basis it is the direct summand generated by $(1,0)^T$.

January 2024 Problem 4, notation altered Let R be a commutative ring, and let A and B be commutative R-algebras.

(a) Prove that if B is flat over R, and f is a non zero divisor in A, then $f \otimes 1$ is a non zero divisor in $A \otimes_R B$.

(b) Prove that the flatness condition above is necessary by giving an example where part (a) fails if B is not flat over R.

Solution

(a) We need to figure out a way to use the flatness of B. Flatness wins us only one thing: it tells us that if $M \hookrightarrow N$ is an injection of R-modules, then $M \otimes_R B \to N \otimes_R B$ is still an injection. This suggests we should try to phrase the condition "f is a non zero divisor" in terms of some map being injective. This we can do: f being a non zero divisor is equivalent to saying that the "multiplication by f" map $A \xrightarrow{\cdot f} A$ is injective. (Check for yourself that this is in fact a homomorphism of R-modules.) Having noticed this rephrasing we are now done: since B is flat, the map $A \otimes_R B \xrightarrow{\cdot f \otimes 1} A \otimes_R B$ is injective, which is the same as saying $f \otimes 1$ is a non zero divisor.

(b) A good example of a non-flat module is a module that has torsion. So, for example, we could take $R = A = \mathbb{Z}$, and say $B = \mathbb{Z}/2\mathbb{Z}$, and the question becomes "is there a non zero divisor in $\mathbb{Z}$ that becomes a zero divisor in $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z}$". And of course there is. For example, 2 is a non zero divisor in $\mathbb{Z}$ which becomes 0 in $\mathbb{Z}/2\mathbb{Z}$. (If you use the definition by which a zero divisor must be different from 0: first of all, cut it out, but also you could instead take say $B = \mathbb{Z}/4\mathbb{Z}$.)

August 2023 Problem 4 Two questions about homomorphisms of modules.

(a) Let R be an integral domain and M an R-module. Suppose M is free of finite rank r. Exhibit an isomorphism of R-modules ϕ from M to $\mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$ (and prove that ϕ is an isomorphism).

(b) Give an example of an integral domain R and an R-module M such that M is not isomorphic as an R-module to $\mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$.

Solution

(a) The difficulty of this problem is unpacking what it is asking for. We want to define a function $\phi\colon M \to \mathrm{Hom}_R(\mathrm{Hom}_R(M, R), R)$. For each $m \in M$, the image $\phi(m)$ will itself be a function, $\phi(m)\colon \mathrm{Hom}_R(M, R) \to R$. The inputs to the function $\phi(m)$ *will themselves be functions*, $f\colon M \to R$. So, the question becomes: how can I take an element $m \in M$ and a function $f\colon M \to R$ and produce an element $\phi(m)(f) \in R$? A hopefully reasonable thing to do would be to define $\phi(m)(f) \coloneqq f(m)$ to be the evaluation function.

With certain facts in place, you can prove this is an isomorphism in one line. Here, I'll just directly check that it is injective and surjective. To check injectivity, we want to check that if $\phi(m)(f) = 0$ for all $f\colon M \to R$, then $m = 0$. Since M is free of finite rank, we have a finite basis $m_1, \ldots, m_r$ for M, and we can construct functions $\delta_i\colon M \to R$ so that $\delta_i(m_j) = 1$ if $i = j$ and 0 otherwise. Thus, $\delta_i(m)$ is the coefficient of m_i appearing in the expansion of m in terms of the basis $\{m_i\}$. So, if we have an m such that $\phi(m)(f) = 0$ for all f, in particular $\phi(m)(\delta_i) = 0$ for all i, but then this implies $m = 0$ by the preceding observation.

To check surjectivity, notice that the functions δ_i generate $\mathrm{Hom}_R(M, R)$, because any function $M \to R$ can be specified by defining its action on a basis. In fact, the δ_i form a basis, so $\mathrm{Hom}_R(M, R)$ is also free of the same rank as M. But then given an arbitrary map $\psi\colon \mathrm{Hom}_R(M, R) \to R$, we can define $r_i = \psi(\delta_i)$. But then, $\psi = \phi(\sum r_i m_i)$, so in fact ϕ is surjective.

(b) We can take $R = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$. Then there are no homomorphisms $M \to R$, so

$$\mathrm{Hom}_{\mathbb{Z}}(\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}), \mathbb{Z}) = \mathrm{Hom}_{\mathbb{Z}}(0, \mathbb{Z}) = 0,$$

and $0 \neq \mathbb{Z}/2\mathbb{Z}$.

January 2023 Problem 3, notation altered Let $S = \mathbb{C}[x, y, z]$. In this problem, you may use the fact that the maximal ideals $\mathfrak{m} \subset S$ are of the form $\mathfrak{m} = (x - a, y - b, z - c)$ for $(a, b, c) \in \mathbb{C}^3$.

Consider the sequence

$$S \xleftarrow{g} S^3 \xleftarrow{f} S^2,$$

where the maps f and g are given by the matrices

$$g = \begin{pmatrix} x & y & z \end{pmatrix}, \qquad f = \begin{pmatrix} y & z \\ \text{-}x & 0 \\ 0 & \text{-}x \end{pmatrix}.$$

For a maximal ideal $\mathfrak{m}$, we tensor the sequence by $S/\mathfrak{m}$ and denote by $f_\mathfrak{m}$ and $g_\mathfrak{m}$ the induced maps

$$S \otimes_S (S/\mathfrak{m}) \xleftarrow{g_\mathfrak{m}} (S^3) \otimes_S (S/\mathfrak{m}) \xleftarrow{f_\mathfrak{m}} (S^2) \otimes_S (S/\mathfrak{m}).$$

(a) For which maximal ideals $\mathfrak{m}$ is $f_\mathfrak{m}$ injective?

(b) For which maximal ideals $\mathfrak{m}$ is $g_\mathfrak{m}$ surjective?

(c) For which maximal ideals $\mathfrak{m}$ is $\ker g_\mathfrak{m} = \operatorname{im} f_\mathfrak{m}$?

Solution
The important things to know/realize for all of the parts of this problem are:

- $S/\mathfrak{m} \cong \mathbb{C}$ via the isomorphism that sends $x \mapsto a$, $y \mapsto b$, and $z \mapsto c$.
- Tensor products distribute over direct sums, so $S^n \otimes_S (S/\mathfrak{m}) \cong (S/\mathfrak{m})^n \cong \mathbb{C}^n$. Under this identification, $f_\mathfrak{m}$ and $g_\mathfrak{m}$ become the maps obtained by replacing the x, y, z in the matrices above with a, b, c.

(a) The map $f_\mathfrak{m}$ is injective if and only if the columns of the matrix we obtain by replacing x, y, z with a, b, c are linearly independent, which happens if and only if $a \neq 0$.

(b) The map $g_\mathfrak{m}$ is surjective for every maximal ideal $\mathfrak{m}$ *except* $\mathfrak{m} = (x, y, z)$. In that case, g becomes the zero map, which is not surjective.

(c) Notice that for the original sequence, before we tensor, we have that $g \circ f = 0$, so $\ker g \supset \operatorname{im} f$, and this remains true after tensoring. Now, after tensoring, to check that $\ker g_\mathfrak{m} = \operatorname{im} f_\mathfrak{m}$, it is enough to check that they have the same dimension, since we already know that $\ker g_\mathfrak{m} \supset \operatorname{im} f_\mathfrak{m}$. The previous parts have shown that $\dim \ker g_\mathfrak{m} = 2$ if $\mathfrak{m} \neq (x, y, z)$, and $\dim \ker g_\mathfrak{m} = 3$ if $\mathfrak{m} = (x, y, z)$. On the other hand, we saw that $\dim \operatorname{im} f_\mathfrak{m} = 2$ if $a \neq 0$, and $\dim \operatorname{im} f_\mathfrak{m} \leq 1$ if $a = 0$. The only time the two are equal is when $a \neq 0$.

August 2022 Problem 3 Let R be a commutative ring. Suppose M and N are R-modules whose direct sum is a free module of finite rank, that is,

$$M \oplus N \simeq R^n$$

for some $n \geq 0$.

(a) Show that M is finitely generated.

(b) Show that M is finitely presented. (Recall that a module is *finitely presented* if it is isomorphic to the quotient of a finitely generated free module by a finitely generated submodule.)

(c) Besides the two properties listed above, M also has another well-known property. What is it?

Solution

(a) Write a basis $v_1, \ldots, v_n$ for R^n. Since $M \oplus N = R^n$, for each v_i we can find unique $m_i \in M, n_i \in N$ so that $(m_i, n_i) = v_i$. On the other hand, since the v_i are a basis, for each $m \in M$, we can write $(m, 0)$ as an R-linear combination of the v_i, which in particular means that m can be written as an R-linear combination of the m_i, so the m_i generate M.

(b) By symmetry, we have that N is finitely generated, and also we have that $M = R^n/N$, so M is finitely generated.

(c) I guess this is very explicitly just a vocabulary question, not a math question. The correct answer is "M is a projective module".

January 2022 Problem 3 Let R be a commutative ring.

(a) Prove that a module M over R is projective if and only if it is a direct summand of a free module.

(b) Let M be a projective module over R and $S \subset R$ a multiplicative set. Prove that $S^{-1}M$ is a projective $R[S^{-1}]$-module.

Solution

(a) For the forward direction, assume M is projective, and let $\pi\colon F \to M$ be a surjective map where F is a free module. Such a map exists since every module is a homomorphic image of a free module. Then by definition of M being projective, there exists a map $\varphi\colon M \to F$ such that $\pi \circ \varphi = id_M$. Therefore, the short exact sequence $0 \to \ker \pi \to F \xrightarrow{\pi} M \to 0$ splits, and thus $F \cong M \oplus \ker \pi$.

For the reverse direction, assume $F \cong M \oplus N$ for some free module F. Then there exist natural maps $\alpha\colon M \to F$ and $\beta\colon F \to M$ given by inclusion and projection such that $\beta \circ \alpha = id_M$. Suppose we have a surjective map $\pi\colon L \to K$ and a map $\psi\colon M \to K$. We need to show that there exists a map $\varphi\colon M \to L$ such that the following commutes:

$$\begin{tikzcd} & M \arrow[dl, dashed, "\varphi"'] \arrow[d, "\psi"] \\ L \arrow[r, two heads, "\pi"] & K \end{tikzcd}$$

Since F is free, it is projective (this is not hard to prove using that F has a basis and so any map out of F is defined by what it does to the basis elements). Therefore, considering the map $\psi \circ \beta\colon F \to M$, we know that there exists a map $\gamma\colon F \to L$ such that $\pi \circ \gamma = \psi \circ \beta$. Then defining $\varphi = \gamma \circ \alpha\colon M \to L$, we see that $\pi \circ \varphi = \pi \circ \gamma \circ \alpha = \psi \circ \beta \circ \alpha = \psi$. Thus, M is projective.

(b) Since M is projective, by part (a), we have that $M \oplus N \cong R^d$ for some R-module N and some $d \geq 1$. Therefore, we have a split SES $0 \to N \to R^d \to M \to 0$. Since localization is exact, we get that $0 \to S^{-1}N \to S^{-1}(R^d) \to S^{-1}M \to 0$ is also a split SES. Thus, $S^{-1}M \oplus S^{-1}N \cong S^{-1}(R^d) \cong R[S^{-1}]^d$, so $S^{-1}M$ is a projective $R[S^{-1}]$-module by part (a).

August 2021 Problem 4 Decide if the following statements are true. If so, provide a proof, if not, provide a counterexample. Let M, N be finitely generated modules over a commutative ring A. Let $f \colon M \to N$ be a homomorphism of A-modules, and let $g \colon B \to A$ be a homomorphism of commutative rings.

(a) "If f is injective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(b) "If f is surjective, so is the induced morphism $M \otimes_A M \to N \otimes_A N$."

(c) "If g is injective, so is the induced morphism $M \otimes_B M \to M \otimes_A M$."

Solution

(a) We should suspect this is false because tensor products usually don't play well with injections. So we're going to need either M or N to be non-flat, but how do we figure out which one we need?

The morphism $M \otimes_A M \to N \otimes_A N$ factors as $M \otimes_A M \to M \otimes_A N \to N \otimes_A N$. If the composition were injective, then in particular the first map $\mathrm{id} \otimes f$ would also be injective. However, we know $\mathrm{id} \otimes f$ need not be injective if M is not flat, so that gives us an avenue to look for a counterexample.

Here's one possible counterexample: let $A = \mathbb{Z}$, $M = \mathbb{Z}/2\mathbb{Z}$, $N = \mathbb{Z}/4\mathbb{Z}$, and $f \colon M \to N$ given by $f(1) = 2$, which is injective. Then we have that $M \otimes_A M \cong \mathbb{Z}/2\mathbb{Z}$ where the nonzero element is $1 \otimes 1$, $N \otimes_A N \cong \mathbb{Z}/4\mathbb{Z}$, but the induced map sends $1 \otimes 1$ to $2 \otimes 2 = 4 \otimes 1 = 0 \times 1$, so the induced map is the zero map.

(b) This one is true. Looking at the factorization from part (a), both $\mathrm{id} \otimes f$ and $f \otimes \mathrm{id}$ are surjective, since tensor products preserve surjections. But then the composition of two surjections is a surjection.

(c) This one is mega false. For example, let $A = \mathbb{C}$, $B = \mathbb{R}$, and $M = \mathbb{C}$. Then $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \to \mathbb{C} \otimes_{\mathbb{C}} \mathbb{C}$ is not injective, because the one on the left has real dimension 4 and the one on the right has real dimension 2. In fact, I think quite to the contrary that $M \otimes_B M \to M \otimes_A M$ is surjective when $B \to A$ is injective, and visa versa.

January 2021 Problem 3 Two questions about projective modules.

(a) Show that a projective module over an integral domain is torsion free.

(b) Let $0 \to M_1 \to M_2 \to M_3 \to 0$ be a short exact sequence of finitely generated modules over a commutative, Noetherian ring. Prove the following OR give a counterexample:

   (i) If M_1 and M_2 are projective, then so is M_3.

   (ii) If M_2 and M_3 are projective, then so is M_1.

Solution

(a) Let M be a projective R-module, where R is an integral domain. Then M is a direct summand of a free R-module, i.e. $M \oplus N = R^d$ for some R-module N and $d \in \mathbb{N}$. Let $m \in M$ be a nonzero element, and suppose that $am = 0$ for some $a \in R$. We will show that $a = 0$. We can write $m = (r_1, \ldots, r_d)$ with $r_i \in R$. Then $am = (ar_1, \ldots, ar_d) = 0$ implies that $ar_i = 0$ in R for all i. Since R is a domain, this means that either $a = 0$ or $r_i = 0$ for all i. Since $m \neq 0$, we must have that $a = 0$. Thus, M is torsion free.

(b) (i) This is not true. For example, consider the SES of $\mathbb{Z}$-modules $0 \to \mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$, where the left map is multiplication by 2 and the right map is the natural quotient map. (Or similarly, $0 \to k[x] \xrightarrow{\cdot x} k[x] \to k \to 0$.) Then $\mathbb{Z}$ is projective, but $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$-module. One way to see this directly is that if $\pi : \mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is the map given by modding out by 2 and $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ is the identity map, then there is no map that makes the following diagram commute:

$$\begin{tikzcd} & \mathbb{Z}/2\mathbb{Z} \arrow[d] \arrow[dl, dashed] \\ \mathbb{Z}/4\mathbb{Z} \arrow[r, two heads] & \mathbb{Z}/2\mathbb{Z} \end{tikzcd}$$

This because the only possible maps are the zero map and multiplication by 2, and neither of these gives the identity map when composed with π. Another way to see that $\mathbb{Z}/2\mathbb{Z}$ is not a projective $\mathbb{Z}$-module is to use that every projective module is flat and show that it is not flat.

   (ii) This is true. If M_3 is projective, then the SES is split exact, so $M_2 \cong M_1 \oplus M_3$. Recall that being projective is equivalent to being a direct summand of a free module. If M_2 is projective, then M_1 is as well since it is a direct summand of M_2 which is a direct summand of a free module.

August 2020 Problem 4 Let R be a commutative ring.

(a) Give an example of a free R-module M and an injective homomorphism $M \to M$ which is not an isomorphism. (For this part, R can be whatever commutative ring you like.)

(b) If M_1 and M_2 are free of finite ranks r_1 and r_2, show that a surjective map $M_1 \to M_2$ exists if and only if $r_1 \geq r_2$.

(c) Assume furthermore that R is an integral domain. Show that an injective map $M_1 \to M_2$ exists if and only if $r_1 \leq r_2$.

Solution

(a) Recall that if M is finitely generated as an R-module, then any surjective endomorphism is injective. This problem shows that the converse is not true. For example, consider $\mathbb{Z}$ as a module over itself. Then the map $\mathbb{Z} \to \mathbb{Z}$ given by multiplication by 2 is injective but not surjective.

(b) We have $M_1 \cong R^{r_1}$ and $M_2 \cong R^{r_2}$. Consider a map $\varphi : R^{r_1} \to R^{r_2}$, and let $\mathfrak{m}$ be a maximal ideal in R so that $R/\mathfrak{m} \cong k$ where k is a field. It is a fact that φ is surjective iff $\varphi_{\mathfrak{m}}$ is surjective for every maximal ideal $\mathfrak{m}$, so we may assume that R is a local ring with unique maximal ideal $\mathfrak{m}$. Then consider the commutative diagram

$$\begin{tikzcd} R^{r_1} \arrow[r, "\varphi"] \arrow[d, two heads] & R^{r_2} \arrow[d, two heads] \\ k^{r_1} \arrow[r, "\overline{\varphi}"] & k^{r_2} \end{tikzcd}$$

where the vertical maps are the natural quotient maps and $\overline{\varphi}$ is induced by φ. By Nakayama's Lemma, we know that $g_1, \ldots, g_{r_2}$ are generators for M_2 iff $\overline{g_1}, \ldots, \overline{g_{r_2}}$ are generators for k^{r_2}. Then φ is surjective iff there exists $f_i \in R^{r_1}$ such that $\varphi(f_i) = g_i$ for each $i = 1, \ldots, r_2$. And this occurs iff there exists $\overline{f_i} \in k^{r_1}$ such that $\overline{\varphi}(\overline{f_i}) = \overline{g_i}$ for each $i = 1, \ldots, r_2$. Since $\overline{\varphi}$ is a map of vector spaces, this happens iff $r_1 \geq r_2$.

(c) Since R is a domain, (0) is a prime ideal, so let $K = \text{Frac}(R) = R_{(0)}$. Any map $\varphi : R^{r_1} \to R^{r_2}$ induces a map $\varphi_{(0)} : K^{r_1} \to K^{r_2}$, which is a map of vector spaces. Since localization is exact, φ is injective iff $\varphi_{(0)}$ is injective, which happens iff $r_1 \leq r_2$.

January 2020 Problem 3 Let R be a commutative ring, I an ideal in R, and S a multiplicative subset of R. There are two standard correspondences involving prime ideals:

(i) Prime ideals in R/I are in bijection with prime ideals in R which contain I.

(ii) Prime ideals in $S^{-1}R$ are in bijection with prime ideals in R which do not meet S.

(a) Prove (i) or (ii).

(b) Give an example of a ring with finitely many elements that has exactly three prime ideals.

(c) Give an example of a subring of $\mathbb{Q}$ that has exactly three prime ideals.

Solution

(a) (i) Let P be a prime ideal which contains I. Then P/I is a proper ideal of R/I. Let $x, y \in R$ such that $[x][y] \in P/I$, where $[x]$ denotes the class of x in R/I. Then $xy = p + i$ for some $p \in P$ and $i \in I$. Since $P \supset I$, we in fact have $xy \in P$, which means $x \in P$ or $y \in P$. Thus, $[x] \in P/I$ or $[y] \in P/I$.

On the other hand, let $Q \subset R/I$ be a prime ideal, and consider $P = \pi^{-1}(Q)$, where $\pi \colon R \to R/I$ is the quotient map. Then P is an ideal of R. Let $xy \in P$, then $[x][y] \in Q$, and so either $[x] \in Q$ or $[y] \in Q$. Thus, either $x \in P$ or $y \in P$.

(ii) Let P be a prime ideal which does not meet S: $P \cap S = \emptyset$. Then $S^{-1}P$ is a proper ideal of $S^{-1}R$. Suppose $\frac{x}{s_1} \cdot \frac{y}{s_2} \in S^{-1}P$, then there exist $p \in P$ and $s_3 \in S$ such that $xys_3 = ps_1s_2$. Notice here that the RHS is in P, so $xys_3 \in P$. But $s_3 \notin P$, so $xy \in P$, and thus either $x \in P$ or $y \in P$. Therefore, either $\frac{x}{s_1} \in S^{-1}P$ or $\frac{y}{s_2} \in S^{-1}P$.

On the other hand, let $Q \subset S^{-1}R$ be a prime ideal, and let $P = \ell^{-1}(Q)$, where $\ell \colon R \to S^{-1}R$ is the localization map. From here, the argument is identical to the above.

(b) $\mathbb{Z}/30\mathbb{Z}$

(c) Let $S = \{n \in \mathbb{Z} : 2 \nmid n,\ 3 \nmid n\}$. Then $S^{-1}\mathbb{Z} \subset \mathbb{Q}$ has only the prime ideals (0), (2), and (3).

August 2019 Problem 2 This problem involves ideals in a commutative ring (with unit). A proper ideal I is *primary* if whenever $ab \in I$ and $a \notin I$, then $b^n \in I$ for some $n > 0$. The radical of an ideal I, denoted $\sqrt{I}$, is the set $\sqrt{I} = \{f : f^n \in I \text{ for some } n > 0\}$.

(a) Prove that if I is primary, then $\sqrt{I}$ is prime.

(b) Is the ideal $J = (x^2, xy) \subset \mathbb{Q}[x, y]$ primary? Be sure to carefully justify your answer.

Solution

(a) Suppose $fg \in \sqrt{I}$, then by definition there exists some $n_1 > 0$ such that $(fg)^{n_1} = f^{n_1}g^{n_1} \in I$. Since I is primary, either $f^{n_1} \in I$ — in which case $f \in \sqrt{I}$ — or else there exists some n_2 such that $(g^{n_1})^{n_2} = g^{n_1 n_2} \in I$. In the latter case, $g \in \sqrt{I}$. Thus, $\sqrt{I}$ is prime.

(b) The ideal J is not primary. To see this, note that $xy \in J$, but neither $x \in J$ nor $y^n \in J$ for any $n > 0$.

January 2019 Problem 3, edited Let R be a commutative ring and $S \subseteq R$ a multiplicatively closed set.

(a) Let M be an R-module. Prove that $S^{-1}M \cong S^{-1}R \otimes_R M$.

(b) Give an example of a ring R, a multiplicatively closed subset $S \subseteq R$, and an R-module M where $S^{-1}M$ is flat (as an $S^{-1}R$-module) and nonzero, but where M is not flat. (And prove that your example has those properties!)

Solution

(a) Let's start by defining an R-module homomorphism $S^{-1}R \otimes_R M \to S^{-1}M$. We have a bilinear map $S^{-1}R \times M \to S^{-1}M$ given by $(\frac{r}{s}, m) \mapsto \frac{rm}{s}$, so the universal property of the tensor product gives us a homomorphism $\phi\colon S^{-1}R \otimes_R M \to S^{-1}M$ which satisfies $\phi(\frac{r}{s} \otimes m) = \frac{r}{s} \cdot \frac{m}{1} = \frac{rm}{s}$.

Now, showing that this ϕ is an isomorphism would be a little hard on its own, in particular checking that it is injective requires us to decide whether some arbitrary tensor in the kernel is 0 or not, which is pretty tough. Instead, let's show it's an isomorphism by constructing an inverse map. Note that $S^{-1}R \otimes_R M$ is an $S^{-1}R$-module in addition to being an R-module, since we can define $\frac{r_1}{s_1} \cdot (\frac{r_2}{s_2} \otimes m) = \frac{r_1 r_2}{s_1 s_2}$. Also, note that we have an R-module homomorphism $M \to S^{-1}R \otimes_R M$ given by $m \mapsto (\frac{1}{1} \otimes m)$. Thus, the universal property of localization gives us a map $\psi\colon S^{-1}M \to S^{-1}R \otimes_R M$ which satisfies $\psi(\frac{m}{s}) = \frac{1}{s}(\frac{1}{1} \otimes m) = (\frac{1}{s} \otimes m)$.

Now, we can check that ϕ and ψ are inverses to one another, for we have

$$\begin{aligned} \phi \circ \psi(\tfrac{m}{s}) &= \phi(\tfrac{1}{s} \otimes m) \\ &= \tfrac{m}{s} \end{aligned} \qquad\qquad \begin{aligned} \psi \circ \phi(\tfrac{r}{s} \otimes m) &= \psi(\tfrac{rm}{s}) \\ &= \tfrac{1}{s} \otimes rm = \tfrac{r}{s} \otimes m. \end{aligned}$$

Therefore, ϕ and ψ are both isomorphisms, as desired.

(b) $R = \mathbb{Z}$, $M = \mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$, $S = \{2^n\}$. Then M is not a flat R module, because M is not torsion-free. However, $M \otimes_{\mathbb{Z}} \mathbb{Z}[\frac{1}{2}] \cong \mathbb{Z}[\frac{1}{2}]$ is a free $\mathbb{Z}[\frac{1}{2}]$-module, hence is flat.

August 2018 Problem 1, edited For this problem, your answer will be graded on correctness alone, and no justification is necessary.

(a) Give an example of a ring R and a short exact sequence of R-modules $0 \to A \to B \to C \to 0$ that is not split exact.

(b) Give an example of a flat $\mathbb{Z}$-module that is not free.

(c) Describe all zero-divisors and all units in $\mathbb{Q}[x]/(x^2 - 1)$.

Solution

(a) The usual $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ will do.

(b) $\mathbb{Q}$

(c) Any polynomial divisible by either $x-1$ or $x+1$ descends to a zero-divisor. (You may want to exclude the polynomials divisible by both, depending on whether or not your definition allows 0 to be a zero-divisor.) Any polynomial that does not descend to a zero-divisor descends to a unit.

January 2018 Problem 1 For this problem, your answer will be graded on correctness alone, and no justification is necessary.

(a) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization $R_f = 0$.

(b) Give an example of a commutative ring R and a nonzero element $f \in R$ where the localization map $R \to R_f$ is neither injective nor surjective.

(c) Give an example of a *local* ring R and an element $f \in R$ where $R_f \neq 0$ but R_f is no longer a local ring.

Solution

(a) Let $R = \mathbb{Z}/4\mathbb{Z}$, $f = 2$

(b) Let $R = \mathbb{C}[x,y]/(xy)$ and $f = x$. Then

$$\begin{aligned} R_f &= (\mathbb{C}[x,y]/(xy))_x \\ &= \mathbb{C}[x,y]_x/(xy)_x \\ &= \mathbb{C}[x,x^{-1},y]/(xy) \\ &= \mathbb{C}[x,x^{-1},y]/(y) \\ &= \mathbb{C}[x,x^{-1}]. \end{aligned}$$

Here, we can see this is not injective since $y \in R$ maps to 0, but it is also not surjective because nothing maps to x^{-1}.

(c) Let $R = \mathbb{C}[x,y]_{(x,y)}$ and $f = y$. Remember the notation, R is of the form $\mathbb{C}[x,y]_P$ for a prime ideal P. Let's look at the structure of the prime ideals in R, also known as its **Spectrum**, pictured in Figure 1. The ring R is local, and has unique maximal ideal $\mathfrak{m} = (x,y)$, but it has many other prime ideals contained in $\mathfrak{m}$, such as the ideals (x), (y), $(x+y)$, $(2x-3y)$, and so on. When we localize at $f = y$, all the primes which contain y will become the unit ideal, but any prime which does not contain y will remain prime in the localization. Thus, the spectrum of R_f is pictured in Figure 2. We can see that the primes like (x) or $(x+y)$ become maximal in the localization, so that R_f is no longer a local ring.

January 2018 Problem 2 Recall that that a (left) zero-divisor in a ring R is an element a such that $ab = 0$ for some nonzero $b \in R$. Consider the rings

$$R_1 = \mathbb{C}[x]/(x^3) \qquad \text{and} \qquad R_2 = M_n(\mathbb{C}) \qquad (n \times n \text{ matrices over } \mathbb{C}, \text{ where } n > 1).$$

(a) Give an example of a nonzero zero-divisor in R_1.

(b) Give an example of a nonzero left zero-divisor in the ring R_2.

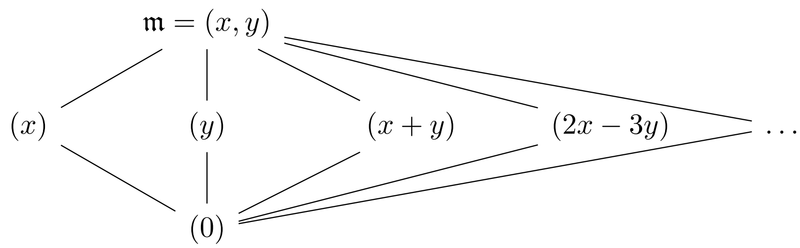

Figure 1: The spectrum of R from January 2018 Problem 1(c)

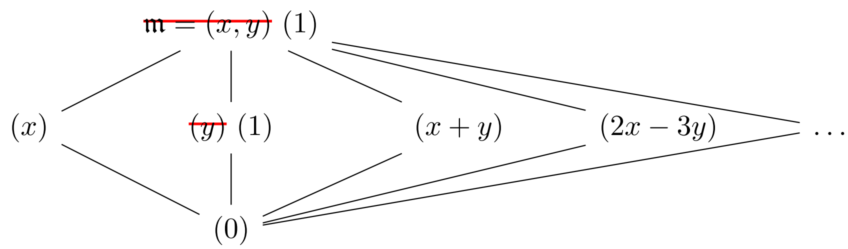

Figure 2: The spectrum of R_f from January 2018 Problem 1(c)

(c) Show that the set of zero-divisors in R_1 form an ideal, but the set of left zero-divisors of R_2 is not a left ideal.

(d) Let S be a commutative ring. Prove that if the set of zero-divisors is an ideal I, then $I \subset S$ is a prime ideal. (**NB** you also have to assume S isn't the zero ring.)

Solution

(a) The element x is a nonzero zero-divisor.

(b) In $M_2(\mathbb{C})$, the matrix $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is a nonzero left zero-divisor since $A^2 = 0$. For general n, the matrix A consisting of all 0s and a 1 in the top right corner is a left zero-divisor since $A^2 = 0$.

(c) Notice that every element of the ideal (x) is a zero-divisor. On the other hand, notice that the prime ideals of R are prime ideals of $\mathbb{C}[x]$ which contain (x^3). There is only one such prime ideal, namely (x), so R is a local ring with maximal ideal (x). Thus, any element not in (x) is invertible, and so isn't a zero-divisor. Thus, every zero-divisor lives in (x).

On the other hand, the left zero divisors of $M_2(\mathbb{C})$ do not form a left ideal because they are not closed under addition. For example, both A above and A^T are left zero-divisors, but $A + A^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is not. Similarly, for any $M_n(\mathbb{C})$, if A is as above and B is the matrix with 1s in the first subdiagonal and 0s elsewhere, then A and B are both left zero-divisors, but $A + B$ is full rank.

(d) The set of zero-divisors is always a proper subset of S, because 1 is never a zero-divisor. Suppose ab is a zero-divisor, but a is not a zero-divisor. Then there exists some nonzero c such that $(ab)c = 0$. Since a is not a zero-divisor, and since $a(bc) = 0$, it must be that $bc = 0$. But then b is a zero-divisor. Thus, if the set of zero-divisors is an ideal, it is a prime ideal.

August 2017 Problem 1, edited For this problem, your answer will be graded on correctness alone, and no justification is necessary.

Let $M = \mathbb{C}[x]/(x^2 + x)$ and $N = \mathbb{C}[x]/(x - 1)$. Compute the dimension of the following tensor products as vector spaces over $\mathbb{C}$:

(a) $M \otimes_{\mathbb{C}[x]} N$

(b) $M \otimes_{\mathbb{C}} N$

(c) $M \otimes_{\mathbb{C}[x^2]} N$.

Solution

(a) Recall that $R/I \otimes_R R/J \cong R/(I + J)$, so $M \otimes_{\mathbb{C}[x]} N \cong \mathbb{C}[x]/(x - 1, x^2 + x) = 0$. Thus, it is 0-dimensional over $\mathbb{C}$.

(b) We have $M \otimes_{\mathbb{C}} N \cong M \otimes_{\mathbb{C}} \mathbb{C} \cong M$, which is 2-dimensional over $\mathbb{C}$ with basis $1, x$.

(c) The only possible basis elements are $1 \otimes 1$ and $x \otimes 1$. Since we can multiply across the tensor by x^2, we have $x \otimes 1 = -x^2 \otimes 1 = -(1 \otimes x^2) = -(1 \otimes 1)$. So this is 1-dimensional over $\mathbb{C}$ with basis $1 \otimes 1$.

Here is a more sophisticated way of doing this problem, which is not easier but which generalizes better to other situations involving tensor products. The ring $\mathbb{C}[x^2]$ is isomorphic to $\mathbb{C}[y]$. Under this identification, the inclusion $\mathbb{C}[x^2] \hookrightarrow \mathbb{C}[x]$ is the same as the composition of the inclusion $\mathbb{C}[y] \hookrightarrow \mathbb{C}[y][t]$ and the quotient $\mathbb{C}[y][t]/(t^2 - y)$. That is to say, we have a commuting square
$$\begin{tikzcd} \mathbb{C}[y] \arrow[r, hook] \arrow[d, "\sim"] & \frac{\mathbb{C}[y][t]}{(t^2-y)} \arrow[d, "\sim"] \\ \mathbb{C}[x^2] \arrow[r, hook] & \mathbb{C}[x] \end{tikzcd}$$
where the downward arrows are obtained by sending $y \mapsto x^2$, $t \mapsto x$.

Now, we can rewrite the modules M and N as $\mathbb{C}[y]$-modules:

$$M = \frac{\mathbb{C}[x]}{(x^2 + x)} \cong \frac{\mathbb{C}[y][t]}{(y + t, t^2 - y)} \cong \mathbb{C}[y], \qquad N = \frac{\mathbb{C}[x]}{(x - 1)} \cong \frac{\mathbb{C}[y][t]}{(t - 1, t^2 - y)} \cong \frac{\mathbb{C}[y]}{(1 - y)}.$$

Thus, we get that $M \otimes_{\mathbb{C}[x^2]} N \cong M \otimes_{\mathbb{C}[y]} N \cong \mathbb{C}[y] \otimes_{\mathbb{C}[y]} \mathbb{C}[y]/(1 - y) \cong \mathbb{C}[y]/(1 - y)$, which is a 1-dimensional vector space.

August 2017 Problem 5, modified Let $B = \mathbb{Q}[x,y]/(xy)$.

(a) Consider the map $\mathbb{Q}[t] \to B$ sending t to x. This map makes B into a $\mathbb{Q}[t]$-module. Show that B is *not* a finite $\mathbb{Q}[t]$-module under this map.

(b) Consider the map $\mathbb{Q}[t] \to B$ sending t to $x + y$. Show that this map makes B into a finite $\mathbb{Q}[t]$-module.

(c) Consider B as a $\mathbb{Q}[t]$-module under the map from part (a). Is B a flat $\mathbb{Q}[t]$-module? What about when considered as a $\mathbb{Q}[t]$-module under the map from part (b).

Solution

(a) Suppose B were finitely generated, with generating set $\{f_1, \ldots, f_n\}$. Then each of the f_i has finite degree in the variable y, so there is a maximum degree, say y^k. But then the y-degree of a $\mathbb{Q}[t]$-linear combination $a_1 f_1 + \cdots + a_n f_n$ is at most k, so there is no way to write y^{k+1} (or any higher power) in this way. Thus, B cannot have a finite generating set.

(b) We will show that under this map, B has generating set $\{1, x\}$. All we need to show is that any power x^k or y^k can be written as a $\mathbb{Q}[t]$-linear combination of these two elements. We have x already in our generating set, and for $k > 1$ we have $x^k = (x+y)x^{k-1} = tx^{k-1}$. On the other hand, we have $y^k = (x+y)^k - x^k = t^k - x^k$.

(c) The ring $\mathbb{Q}[t]$ is a PID. Recall that a module over a PID is flat iff it is torsion-free. In part (a), B has torsion because $ty = xy = 0$. However, in part (b) B is torsion-free, hence flat.

January 2017 Problem 3 Let R be a commutative ring with unity. Show that a polynomial

$$f(t) = c_n t^n + c_{n-1} t^{n-1} + \cdots + c_0 \in R[t]$$

is nilpotent if and only if all of its coefficients $c_0, \ldots, c_n \in R$ are nilpotent.

Solution

We begin by showing that a sum of two nilpotent elements is nilpotent. Suppose a, b are nilpotent, with least exponents n, m respectively. Then $(a+b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{n+m-k}$. For each term of this sum, either $k \geq n$, in which case $a^k = 0$, or $k < n$ and $n + m - k > m$, in which case $b^{n+m-k} = 0$. Thus, $(a+b)^{n+m} = 0$, so $a + b$ is nilpotent.

The above immediately shows that if all the coefficients are nilpotent, then in fact $f(t)$ is nilpotent.

Now, suppose $f(t)$ is nilpotent, so that $f^k = 0$ for some $k > 0$. Then looking at the highest degree coefficient, we see that $c_n^k = 0$, so c_n is nilpotent. Thus, also $c_n t^n$ is nilpotent, and so $f(t) - c_n t^n$ is a sum of two nilpotent elements, hence is nilpotent. Repeating the argument now gives that all the coefficients are nilpotent.

August 2016 Problem 4 Let R be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(y, x)$.

(a) Show that R is generated as a $\mathbb{C}$-algebra by $x + y$ and xy.

(b) Show that the map $\mathbb{C}[u, v] \to R$ sending u to $x + y$ and v to xy is an isomorphism.

(c) Let S be the subring of $\mathbb{C}[x, y]$ consisting of the polynomials $P(x, y)$ such that $P(x, y) = P(-x, -y)$. Can S be generated as a $\mathbb{C}$-algebra by two polynomials? Either give two generators, or prove that S is not generated by 2 polynomials.

Solution

(a) Let $P(x, y) \in R$. By induction on the degree of P, we may assume that P is homogeneous of some degree n and that all lower degree polynomials in R are generated by $x + y$ and xy over $\mathbb{C}$. Because $P(x, y) = P(y, x)$, we must have that the number of x factors is the same as the number of y factors. So, if x^t divides P for some $t > 0$, then y^t divides P, but then $(xy)^t$ divides P, and we are done by the induction hypothesis. Therefore, we can assume that $t = 0$, i.e. no power of x or y divides P. In this case, we must have $P = x^n + y^n + xyQ(x, y)$ for some symmetric $Q(x, y)$ of lesser degree. But then $P - (x + y)^n$ is divisible by xy.

(b) Part (a) gives that this map is surjective. So we just need to show injectivity. Suppose $Q(u, v) \in \mathbb{C}[u, v]$ is in the kernel, i.e. $Q(x + y, xy) = 0$, and assume that $Q(u, v) \neq 0$. Observe that taking $y = 0$ gives $Q(x, 0) = 0$, which implies $Q(u, 0) = 0$, so every term of Q has a factor of v. Similarly, taking $x = 0$ gives that $Q(y, 0) = 0$ and by symmetry in R, we have $Q(0, y) = 0$, so every term of Q has a factor of u as well. Therefore, by unique factorization, we have $Q(u, v) = uvQ'(u, v)$. Applying the map gives $(x + y)(xy)Q'(x + y, xy) = 0$, which implies that $Q'(x + y, xy) = 0$ since $(x + y)xy \neq 0$ and R is a domain. This means that $Q'(u, v)$ is an element of the kernel of strictly smaller degree than $Q(u, v)$. But then we could just repeat the argument with $Q'(u, v)$ to produce infinitely many elements of the kernel of smaller and smaller degree, which is impossible. Thus, $Q(u, v) = 0$.

(c) We first show that $S = \mathbb{C}[x^2, xy, y^2]$. The backwards containment is clear. For the forward containment, note that if $P(x, y) \in S$, then $P(x, x) = P(-x, -x)$ is a polynomial is x where all terms must have even degree. Therefore, every term of $P(x, y)$ must have even total degree. So every monomial in $P(x, y)$ is of the form $x^i y^j$ with $i + j$ even and WLOG $i \geq j$. Then $x^i y^j = (xy)^j x^{i-j} \in \mathbb{C}[x^2, xy, y^2]$ since $i - j$ is also even. Then $S = \mathbb{C}[x^2, xy, y^2]$ cannot be generated as a $\mathbb{C}$-algebra by two polynomials because it is 3-dimensional as an algebra over $\mathbb{C}$.

August 2014 Problem 4 Let R be a commutative ring and M an R-module. Recall that a prime ideal P is an associated prime of M if there exists some $m \in M$ such that P consists of those f such that $fm = 0$; i.e. P is the annihilator of some $m \in M$.

(a) Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of R-modules, and let P be an associated prime of M. Prove that P is an associated prime of either M' or M''.

(b) Is the converse true? That is, if P is an associated prime of either M' or M'', must it be the case that P is an associated prime of M?

Solution

(a) Suppose P is not an associated prime of M', and write $P = \operatorname{ann}(m)$ for some $m \in M$. We want to show that $P = \operatorname{ann}(m'')$ for some $m'' \in M''$. Let ϕ be the map $M \to M''$ in the SES. Then a natural choice is $m'' = \phi(m)$. We will show that this does indeed work. Let $f \in P$ so that $fm = 0$. Then $f\phi(m) = \phi(fm) = 0$, so $f \in \operatorname{ann}(\phi(m))$. Thus, $P \subseteq \operatorname{ann}(\phi(m))$. Conversely, if $f \in \operatorname{ann}(\phi(m))$, then $\phi(fm) = f\phi(m) = 0$, so $fm \in \ker(\phi) = \operatorname{im}(\varphi)$ by the exactness of the sequence where $\varphi : M' \to M$. Suppose towards a contradiction that $f \notin P$. Then $fm = \varphi(m') \neq 0$, so $m' \neq 0$ since φ is injective. We will show that this gives $P = \operatorname{ann}(m')$, contradicting our assumption. For any $g \in P$, we have $\varphi(gm') = g\varphi(m') = g(fm) = f(gm) = 0$, so $gm' = 0$ by injectivity, which gives $g \in \operatorname{ann}(m')$. On the other hand, given $h \in \operatorname{ann}(m')$, we have $0 = \varphi(hm') = h\varphi(m') = hfm$, so $hf \in P = \operatorname{ann}(m)$. Since $f \notin P$ and P is prime, we must have $h \in P$. This gives our contradiction; thus, $f \in P$ and $P = \operatorname{ann}(\phi(m))$.

(b) No, take for example the SES of $\mathbb{Z}$-modules $0 \to \mathbb{Z} \to \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \to 0$ where the first map is multiplication by 2. Then (2) is the annihilator of 1 in $\mathbb{Z}/2\mathbb{Z}$, but $\mathbb{Z}$ is torsion-free, so its only associated prime is (0).

August 2013 Problem ? Let I and J be ideals in a commutative ring R.

(a) Show that if $I + J = R$, then $I \cap J = IJ$.

(b) Suppose that I and J are ideals in $\mathbb{C}[x]$, and suppose that $I + J = (x)$. Show that $(I \cap J)/IJ$ is 1-dimensional as a complex vector space, and moreover that it is isomorphic to $\mathbb{C}[x]/(x)$ as a $\mathbb{C}[x]$-module.

(c) On the other hand, for general commutative rings R, once $R/(I+J)$ is not trivial, the difference between $I \cap J$ and IJ can be large even if $R/(I \cap J)$ is small. Demonstrate this by showing that, if $R = \mathbb{C}[x, y]$, then there exists ideals I and J in R such that $I + J = (x, y)$ and the dimension of $(I \cap J)/IJ$ as a $\mathbb{C}$-vector space is at least 100.

Solution

(a) It is always the case that $IJ \subseteq I \cap J$, so we need to show the reverse containment. Since $I + J = R$, there exist $i \in I$ and $j \in J$ such that $i + j = 1$. Let $a \in I \cap J$, then we have $a = a \cdot 1 = a(i + j) = ai + aj \in IJ$ since $a \in I$ and $a \in J$.

(b) Since $\mathbb{C}[x]$ is a PID, we have that $I = (f)$ and $J = (g)$ for some $f, g \in \mathbb{C}[x]$. Since $I + J = (x)$, we can find $a, b \in \mathbb{C}[x]$ such that $af + bg = x$. Because x is a prime element in a UFD, this equation implies that x is the only possible common factor of f and g. But f and g must not be relatively prime, since then $I + J = \mathbb{C}[x]$. Thus, we can write $f = xf'$ and $g = xg'$ where $f', g' \in \mathbb{C}[x]$ are relatively prime. Now we will show that $I \cap J = (xf'g')$. It is clear that $xf'g' \in I \cap J$, so suppose $h \in I \cap J$. Then $h = cxf' = dxg'$ for some $c, d \in \mathbb{C}[x]$. Since h is divisible by both xf' and g', which are relatively prime, h is also divisible by $xf'g'$. Then $IJ = (fg) = (x^2f'g')$, so we have that

$$(I \cap J)/IJ = (xf'g')/(x^2f'g').$$

Consider the surjective $\mathbb{C}[x]$-module map $\varphi : (xf'g') \to \mathbb{C}[x]/(x)$ given by $xf'g' \mapsto 1$. Then $\ker\varphi = (x^2f'g')$, so we have $(I + J)/IJ = (xf'g')/(x^2f'g') \cong \mathbb{C}[x]/(x)$.

(c) Let $I = (x, y)$ and $J = (x^{99}, yx^{98}, \ldots, y^{98}x, y^{99})$. Then $J \subseteq I$ so $I + J = I = (x, y)$, and $IJ = (x^{100}, yx^{99}, \ldots, y^{99}x, y^{100})$. This gives

$$(I \cap J)/IJ = J/IJ = (x^{99}, yx^{98}, \ldots, y^{98}x, y^{99})/(x^{100}, yx^{99}, \ldots, y^{99}x, y^{100})$$

which has basis $x^{99}, yx^{98}, \ldots, y^{98}x, y^{99}$ as a vector space over $\mathbb{C}$, and thus has dimension 100 as a $\mathbb{C}$ vector space.

Noncommutative rings + applications to linear algebra

August 2025 Problem 5 Fix a prime p, and let R be a finite ring of size $|R| = p^2$. Here, we assume that rings are associative and have identity, but are not necessarily commutative.

(a) Prove that R is in fact commutative.

(b) List all possibilities for R, up to isomorphism. You need to show that the rings on your list are non-isomorphic, and that there are no other possibilities.

Solution

Compare this problem to August 2018 Problem 5.

(a) We have a ring homomorphism $\mathbb{Z} \to R$ which sends 1 to 1, and the kernel of this homomorphism must be either (p) or (p^2). If the kernel is (p^2), then this map is a surjection and $R \cong \mathbb{Z}/p^2\mathbb{Z}$ is commutative. If the kernel is (p), then the image of this homomorphism is a copy of $\mathbb{F}_p$ lying inside R. Let $x \in R$ be any element not in this $\mathbb{F}_p$, then the p^2 distinct elements $a + bx$ for $a, b \in \mathbb{F}_p$ must in fact be all of R. But all these elements commute with one another, so R is commutative.

(b) The possibilities are $R \cong \mathbb{Z}/p^2\mathbb{Z}$, $R \cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$, $R \cong \mathbb{F}_p[x]/(x^2)$, and $R \cong \mathbb{F}_{p^2}$. We have that:

- $\mathbb{Z}/p^2\mathbb{Z}$ has cyclic additive group, while the other rings do not;

- $\mathbb{F}_{p^2}$ is a field while the other rings are not;
- $\mathbb{F}_p[x]/(x^2)$ has nilpotent elements, while the other rings do not.

These facts are enough to conclude that the four rings are non-isomorphic.

To prove that this list is exhaustive, let R be a ring of order p^2 other than $\mathbb{Z}/p^2\mathbb{Z}$. Then the characteristic of R is p, so we have a homomorphism $\mathbb{F}_p[x] \to R$ sending 1 to 1 and x to any $x \in R \setminus \mathbb{F}_p$, and by part (a) this homomorphism is surjective, so $R \cong \mathbb{F}_p[x]/\ker$. Since $\mathbb{F}_p[x]$ is a PID, the kernel of this homomorphism is of the form $(f(x))$ for some polynomial f, and $\deg f = \dim_{\mathbb{F}_p}(R) = 2$. If f is irreducible, then $R \cong \mathbb{F}_{p^2}$. If $f = f_1 f_2$ is reducible but not square, then by the Chinese Remainder Theorem $R \cong \mathbb{F}_p[x]/(f_1) \times \mathbb{F}_p[x]/(f_2) \cong \mathbb{F}_p \times \mathbb{F}_p$. If $f = g^2$ for a linear polynomial g, then after a change of variables we get $R \cong \mathbb{F}_p[x]/(x^2)$. Thus, our original list was exhaustive.

January 2024 Problem 5 Let $\mathbb{C}$ denote the field of complex numbers. Pick a nonzero $q \in \mathbb{C}$ such that $q^n \neq 1$ for all positive integers n. Let A denote the $\mathbb{C}$-algebra with generators x, y subject to the relation $yx = qxy$.

(a) Let V denote a nonzero A-module on which x and y are injective. Show that V has infinite dimension. (**NB** The module V is a $\mathbb{C}$-vector space, and the problem is asking you to show that V has infinite dimension as a $\mathbb{C}$-vector space.)

(b) Give an example of a nonzero A-module on which x and y are injective.

(c) Show that the following is a basis for the $\mathbb{C}$-vector space A:

$$x^i y^j \qquad\qquad i, j \geq 0.$$

Solution

I like this problem because part (a) is a rephrasing of a problem I like from the 2015 qual.

(a) I think it might be tricky to look at this problem and get going down the right track. A few key facts that would help to point you in the right direction are: an A-module is the same thing as a vector space V together with two linear operators x, y which satisfy the given commutation relation; any linear operator on a finite-dimensional complex vector space has an eigenvalue; and commuting linear operators have a tendency to interact well with eigenvalues.

Suppose that V has finite dimension D, and that x is injective. Since V is a finite-dimensional complex vector space, y has an eigenvector, so say v is an eigenvector of eigenvalue λ. Then $yx(v) = qxy(v) = q\lambda xv$, and since x is injective, $xv \neq 0$, so xv is an eigenvector of y of eigenvalue $q\lambda$. Continuing similarly, we get that each of the vectors $x^i v$ is an eigenvector of y of eigenvalue $q^i\lambda$, and by the assumption that $q^i \neq 1$, these eigenvectors are all linearly independent. However, this contradicts the assumption that V is finite-dimensional.

(b) A is a module over itself on which x and y act injectively. For suppose it didn't, then we would have some nonzero $f \in A$ so that say $x \cdot f = 0$. Using the commutation relation, we can write all the terms of f as $c_{ij}x^iy^j$, but then $x \cdot f = \sum c_{ij}x^{i+1}y^j$, which is not zero because by part (c), the x^iy^j are a vector space basis for A.

(c) As above, we can certainly write any element of A as a linear combination of such elements, because whenever we have a y that comes before an x we can use $yx = qxy$ to swap them. I'm not sure how careful the problem wants you to be to show that these are linearly independent, but to me it seems enough to say that the only relation we have allows us to swap x and y, and so there's no way for us to convert some linear combination $\sum c_{ij}x^iy^j$ into 0.

August 2022 Problem 5, edited Let V be a vector space over $\mathbb{C}$ of dimension $n \geq 2$. Let

$$A\colon V \to V$$

denote a $\mathbb{C}$-linear map with n mutually distinct eigenvalues. Prove that V contains one-dimensional subspaces $V_1, \ldots, V_n$ such that

(i) we have
$$V = \sum_{i=1}^{n} V_i$$
and if $i \neq j$, $V_i \cap V_j = \{0\}$ (**NB** these two conditions are equivalent in this case);

(ii) $AV_i \subseteq V_i + V_{i+1}$ for $1 \leq i \leq n-1$ and $AV_n \subseteq V_n + V_1$; and

(iii) V_i is not an eigenspace of A for $1 \leq i \leq n$.

Solution

Ivan's homely solution: Let's begin by translating what the problem asks into things we can actually work with. First, we note that since A has n distinct eigenvalues, we can choose a basis $e_1, \ldots, e_n$ for V so that the matrix of A is diagonal with distinct diagonal entries. So, from now we can just pretend we started with a diagonal matrix with distinct entries.

Decomposing V into 1-dimensional subspaces V_i is almost the same as choosing a set of n vectors v_i, except that we can scale the vectors arbitrarily without changing the 1-dimensional subspaces chosen. Condition (i) above tells us that these vectors should form a basis for V. Condition (ii) says that the matrix of A with respect to this basis looks like (in the case $n = 3$)

$$\begin{pmatrix} a & 0 & f \\ b & c & 0 \\ 0 & d & e \end{pmatrix}.$$

(In general, A will have this shape of having possibly nonzero entries only on the main diagonal, the first subdiagonal, and one entry in the top right corner. To myself, I call this shape "a staircase under a star".) Condition (iii) says that the off-diagonal entries of A are not allowed to be 0 (or else v_i is an eigenvector for A for that i).

I found it useful to do one more bit of translation. Since we only care about the v_i up to scaling, and since we know the subdiagonal entries are not zero, we can rescale the v_i so that the subdiagonal entries are equal to 1 (in the $n = 3$ case, $b = d = 1$). We cannot also make the "star" be 1 in this way, because scaling all the v_i's by the same amount doesn't change the matrix of A, so rescaling only gives us $n-1$ degrees of freedom in this regard. After all this translation, our original problem is equivalent to: "given a diagonal matrix with distinct entries A, does there exist $d_1, \dots, d_n, s$ so that A is similar to the matrix

$$\begin{pmatrix} d_1 & 0 & \dots & 0 & s \\ 1 & d_2 & \ddots & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 & d_n \end{pmatrix} ?\text{"}$$

Now, I played around with examples for a long time. Eventually, I came around to thinking about the characteristic polynomial of such a matrix. In general, two matrices with the same characteristic polynomial need not be similar, but in our case we have been told that A has distinct eigenvalues. So, a matrix with the same characteristic polynomial as A has the same (distinct) eigenvalues as A, and is therefore diagonalizable to the same diagonal matrix as A, and therefore is similar to A. So, in this case, it is enough to construct a "staircase under a star" matrix that has the same characteristic polynomial as A. But this is easy, since a staircase under a star matrix has characteristic polynomial $s + \prod_{i=1}^{n}(x - d_i)$. Let $a(x)$ be the characteristic polynomial of A, then for *any* $s \neq 0$ let $d_1, \dots, d_n$ be the n complex roots of $a(x) - s$ (which must exist by the fundamental theorem of algebra), so $a(x) - s = \prod_{i=1}^{n}(x - d_i)$. Then the staircase under a star matrix for that s and those d_i has the same characteristic polynomial as A, and so must be similar to A by our reasoning above. (Indeed, we usually get $n!$ different matrices for a fixed s, by permuting the d_i, and we get a different set of matrices by changing s, so the representation is highly non-unique.)

Josh's fast solution: Consider V to be a $\mathbb{C}[x]$-module, where x acts by multiplication by A. Let $p(x)$ be the minimal polynomial of A. By the classification of finitely generated modules over a PID, V is isomorphic to a module of form $\mathbb{C}[x]/f_1 \oplus \cdots \oplus \mathbb{C}[x]/f_r$, where the f_i are polynomials, $f_1 \mid f_2 \mid \cdots \mid f_r$, and $f_r = p(x)$. In this case, since the eigenvalues of A are distinct, $\deg p(x) = n$, so $\mathbb{C}[x]/p(x)$ has dimension n as a complex vector space, so in fact $V \cong \mathbb{C}[x]/p(x)$.

Under this isomorphism, there is some vector $v \in V$ which corresponds to the element $1 \in \mathbb{C}[x]/p(x)$. (Such a vector is called a "cyclic vector" for A, because $v, Av, \dots A^{n-1}v$ form a basis for V.) Define $v_1 = v$, and $v_{i+1} = (A - d_i)v_i$ for $1 \leq i < n-1$, for constants d_i yet to be determined. We want to have $Av_n = d_n v_n + v_1$, and we observe that by unravelling the construction, we get $(A - d_n)v_n = g(x)v_1$ where $g(x) = \prod_{i=1}^{n}(x - d_i)$. So, choose $g(x) = p(x) + 1$, and set the d_i to be the roots of g, and we are done.

(This solution highlights nicely that the hypothesis that A have distinct eigenvalues is weaker than what we need: all we actually need is that the minimal polynomial of A has degree n, in which case it will be equal to the characteristic polynomial. A matrix A whose minimal polynomial is equal to its characteristic polynomial is called a "regular matrix".)

(Notice also that the form of the matrix we're trying to produce is very similar to a Jordan form matrix with a single Jordan block. I think if I were familiar enough with the theory of Jordan normal form, from the perspective of thinking about $\mathbb{C}[x]$-modules, this solution might have been easier to think of.)

August 2021 Problem 5 Let A be an algebra over $\mathbb{C}$ with two generators e and f and defining relations $e^2 = e, f^2 = f$.

(a) Let B be the ring of 2×2 matrices over the polynomial ring $\mathbb{C}[t]$. Consider B as an (infinite-dimensional) algebra over $\mathbb{C}$. Show that there exists a unique $\mathbb{C}$-algebra homomorphism $\phi\colon A \to B$ such that $\phi(e) = \begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix}$ and $\phi(f) = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

(b) Prove that the homomorphism ϕ from part (a) is injective.

(c) Consider the algebra A as a vector space over $\mathbb{C}$. Write down a basis for A and prove that your answer is correct.

Solution

(a) We need to show that the map ϕ given is well-defined. So, we just need to show that it respects the relations in A, i.e. that $\phi(e^2 - e) = \phi(f^2 - f) = 0$. It is straightforword to compute that $\phi(e^2) = \phi(e)^2 = \phi(e)$ and similarly $\phi(f^2) = \phi(f)$, so this is a well-defined $\mathbb{C}$-algebra homomorphism $A \to B$. It is unique because it is defined on the generators of A.

(b) Any element in A can be written as a $\mathbb{C}$-linear combination of $\{1, e, f, ef, fe, efe, fef, ...\}$. So, let's think about the image of these elements under ϕ. We find that $\phi(efe) = t\phi(e)$, $\phi(fef) = t\phi(f)$, $\phi(efef) = t^2\phi(ef)$, $\phi(fefe) = t^2\phi(fe)$,$\ldots$, $\phi(efef\cdots e) = t^n\phi(e)$, $\phi(fefe\cdots f) = t^n\phi(f)$, $\phi(efef\cdots f) = t^n\phi(ef)$, and $\phi(fefe\cdots e) = t^n\phi(fe)$ for some n. Therefore, $\phi(a)$ for any $a \in A$ can be written as

$$\phi(a) = \begin{pmatrix} a_1 & 0 \\ 0 & a_1 \end{pmatrix} + p_e(t)\begin{pmatrix} 1 & t \\ 0 & 0 \end{pmatrix} + p_f(t)\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + p_{ef}(t)\begin{pmatrix} t & t \\ 0 & 0 \end{pmatrix} + p_{fe}(t)\begin{pmatrix} 0 & 0 \\ 1 & t \end{pmatrix}$$

where the $p_i(t) \in \mathbb{C}[t]$ are determined by the coefficients of the elements in a. Therefore, if $\phi(a) = 0$, we have

$$\begin{pmatrix} a_1 + p_e(t) + tp_{ef}(t) & t(p_e(t) + p_{ef}(t)) \\ p_f(t) + p_{fe}(t) & a_1 + p_f(t) + tp_{fe}(t) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Solving this system gives that $p_e(t) = p_f(t) = -p_{ef}(t) = -p_{fe}(t)$, which then gives that $a_1 + (1-t)p_e(t) = 0$. Considering the degrees of the polynomials, we see this is only possible if $p_e(t) = 0$, and then $a_1 = 0$. Thus, $a = 0$, and ϕ is injective.

(c) A basis for A as a $\mathbb{C}$-vector space is given by $\{1, e, f, ef, fe, efe, fef, \cdots\}$, so this is an infinite-dimensional $\mathbb{C}$-vector space.

January 2021 Problem 2 Let k be a field, and let $R = M_n(k)$, the non-commutative ring of $n \times n$ matrices over k.

(a) Give examples of a simple left R-module M and a simple right R-module N.

(b) Can you find another simple left R-module M' that is not isomorphic to M? (Either find one or explain why there is none such.)

(c) Compute $\dim_k(N \otimes_R M)$.

Solution

(a) To do this problem, you have to remember (or make an educated guess about) what a simple module is. A (left/right/two-sided)-module is *simple* if it is nonzero, and contains no proper, nonzero (left/right/two-sided)-submodules. Luckily, the ring R has a module that we already know and love: the vector space k^n. Indeed, let v be a nonzero vector, and $v' \in k^n$ an arbitrary vector, then there exists a linear transformation that takes v to v' (and say, takes some complementary subspace of $\langle v \rangle$ to 0). Thus, $Rv = k^n$, so any nontrivial submodule of k^n must be all of k^n, so k^n is a simple left module.

For a simple right module, we could take k^n viewed as row vectors rather than column vectors with matrix multiplication on the right, or equivalently we could take k^n viewed as column vectors with $v \cdot A = A^T v$, or equivalently we could take the dual vector space $\mathrm{Hom}_k(k^n, k)$ with $\phi \cdot A = \phi \circ A$. Confirm to yourself that these are the same.

(b) (Previously, there was a subtle error in my argument here. Thanks to Gabriela Brown for bringing it to my attention.) If I were guessing the answer this problem based only on the phrasing, I would guess that the answer is "none exists". Let us show that is the case.

Suppose we have a simple left R-module M'. We will show that M' is isomorphic to k^n. Since M' is simple, it is generated by any nonzero $m' \in M'$. So, we can define an R-module homomorphism $\phi\colon M' \to M$ by specifying where to send just m' and then extending R-linearly. The only thing we need to do to make sure the resulting map is a homomorphism is to make sure that whenever $Am' = 0$, we also have $A\phi(m') = 0$. So, let $I = \{A \in R : Am' = 0\}$ be the annihilator of m', which is a left ideal of R. Any matrix in I must have nontrivial kernel, because an invertible matrix cannot kill a nonzero element. Indeed, the matrices in I must all have a nontrivial mutual kernel, because if they didn't then a linear combination of them would have trivial kernel, which we just said cannot happen. Thus, there is a nonzero vector $v \in k^n$ that is in the kernel of every matrix in I. We define our map ϕ by sending $m' \mapsto v$.

The image of this map is a nontrivial submodule of k^n, hence is all of k^n since k^n is simple, and so this map is onto. Notice now that its kernel is a submodule of M', hence is either 0 or all of M'. Since this map is not the zero map, the kernel cannot be all of M', so is 0, and so this is an isomorphism between M' and k^n. (There's actually some cool afterlife to this fact: look up "Morita equivalence".)

(c) (Previously I had an incorrect solution here. Thanks to Abbi Jones, Hannah Ashbach, and Bella Finkel for pointing out my error.) Let's view M as k^n viewed as column vectors and matrix multiplication on the left, and N as k^n viewed as row vectors and matrix multiplication on the right. Let e_1 be the first standard basis (column) vector for M. Then, given any other vector $v \in M$, we can choose the matrix A which has v as its first column and all other entries 0, and this matrix satisfies $Ae_1 = v$. We can therefore rewrite any simple tensor $u \otimes v = u \otimes Ae_1 = uA \otimes e_1$. Now, since $u \in k^n$, we should view u as a row vector, and by our choice of A, $uA = (\lambda, 0, \ldots, 0)$ for some $\lambda \in k$. Thus, every vector in $N \otimes_R M$ is equivalent to a vector of the form $(\lambda, 0, \ldots, 0) \otimes e_1$, and projection onto the first factor gives an isomorphism between $N \otimes_R M$ and $\langle(\lambda, 0, \ldots, 0)\rangle$, so $\dim_k N \otimes_R M = 1$.

Here's another, more abstract way of putting it. Any simple left (right) module is isomorphic to $R/\mathfrak{m}$ for some left (right) maximal ideal $\mathfrak{m}$. I know of some maximal ideals of $M_n(\mathbb{C})$: there's the maximal left ideal I consisting of matrices where the first column contains all 0s, and the maximal right ideal J consisting of matrices where the first row contains all 0s. So, I can write $M = R/I$, $N = R/J$. Then, we have a theorem that tells us that for any right module N and left ideal I, $N \otimes_R R/I \cong N/NI$. Thus, we get that $N \otimes_R M \cong N/NI = R/(I + J)$ where by $I + J$ I mean the additive subgroup of R consisting of matrices that can be written as a sum of matrices from I and matrices from J. The subgroup $I + J$ is just the vector subspace of R consisting of matrices with a 0 in the $(1, 1)$ position, and when we mod out by this subgroup we get a 1-dimensional vector space (parametrized by the possible values in the $(1, 1)$ position of a matrix).

August 2020 Problem 5, edited Let $U\colon \mathbb{Z} \to \mathbb{Z}$ be the function defined by $U(n) = -n$, and let $T\colon \mathbb{Z} \to \mathbb{Z}$ be the function defined by $T(n) = n+1$. Denote by M the space of all functions from $\mathbb{Z}$ to $\mathbb{C}$. Then T and U can be thought of as elements of $\mathrm{End}_{\mathbb{C}}(M)$ (the algebra of $\mathbb{C}$-linear maps from M to M) by the operation of composition of functions. (For instance, if m is the function $m(n) = 3n$, then mU is the function sending n to $3(-n) = -3n$, mT is the function sending n to $3(n+1) = 3n+3$, and, taking V to be $2TU + 1$, mV is the function sending n to $2 \cdot 3(-n + 1) + 3n = -3n + 6$.

(**NB** there are a couple confusing things here. One is that the function composition mU and mT occurs *inside* the parentheses, i.e. $mU(n) = m(U(n))$, but the scalar multiplication occurs *outside* the parentheses, $m(2U)(n) = 2m(U(n))$. This is because the scalar multiplication is scalar multiplication in the complex vector space M, and the scalar multiplication there is scaling the function, not the input. Indeed, I can scale by any complex number, but the input cannot be scaled by an arbitrary complex number and remain an integer. The second thing is that the element written above as 1 in $2TU + 1$ is really the identity map on M, so $m(2TU + 1)(n) = 2mTU(n) + m(n)$.)

Let R be the subalgebra of $\mathrm{End}_{\mathbb{C}}(M)$ generated by U and T; that is, it is a complex vector space spanned by all linear maps obtainable by repeatedly composing copies of T and copies of U.

(a) Show that R is not commutative.

(b) M is an infinite-dimensional complex vector space which carries the structure of a right R-module by function composition. The space P_0 of constant functions is a 1-dimensional submodule of M. Give an example of a 1-dimensional submodule of M containing non-constant functions, and show that it is the only one. (**NB** This is *not* asking for a submodule that is generated by just one $f \in M$, it is asking for a submodule that is *also* a 1-dimensional complex vector space.)

(c) There is a ring homomorphism $\phi \colon R \to \mathbb{C}$, called *augmentation*, defined by $\phi(T) = \phi(U) = 1$. Let I be the kernel of ϕ, which is a two-sided ideal. Prove that $M \otimes_R R/I = 0$.

Solution

(a) Let $\iota \colon \mathbb{Z} \to \mathbb{C}$ be the inclusion $\iota(n) = n$. Then $\iota TU(n) = -n+1$, while $\iota UT(n) = -n-1$. Thus, $TU \neq UT$. (Why am I mentioning the function ι at all? For technical reasons: it's hypothetically possible that the functions TU and UT are different *as functions* $\mathbb{Z} \to \mathbb{Z}$, but nevertheless act identically *as linear maps* $M \to M$. This would be the case if, e.g., we restricted to the subspace of functions $\mathbb{Z} \to \mathbb{C}$ satisfying $f(n+2) = f(n)$.)

(b) To specify such a submodule, it is enough to demonstrate a function $f \colon \mathbb{Z} \to \mathbb{C}$ so that fT and fU are both scalar multiples of f (i.e. f is an eigenfunction of both T and U). If f is nonzero and $fT = \lambda f$, this says that $f(n+1) = \lambda f(n)$, and so $\lambda \neq 0$ and $f(n) = \lambda^n f(0)$ for all $n \in \mathbb{Z}$. On the other hand, $fU = \mu f$ means that $f(-n) = \mu f(n)$, so $\lambda^{-n} = \mu \lambda^n$, so λ satisfies $\mu x^{2n} - 1$ for all values of n. The only way that can be true is if $\mu = 1$ and $\lambda = \pm 1$, and if $\lambda = 1$, then f is a constant function. So, we must have that $f(n) = (-1)^n f(0)$. The set of such functions f forms a 1-dimensional $\mathbb{C}$-vector space (parametrized by $f(0)$), and the argument above shows that it is an R-submodule of M.

(Alternative argument: such a submodule must be a module over the abelianization of R, $R/(TU - UT)$, and a function on which TU and UT act identically must satisfy $f(n+2) = f(n)$, as I noted in the parenthetical comment in part (a). Similar to above, the assumption that $fT = \lambda f$ implies that $\lambda^2 - 1 = 0$, and so if f is nonconstant, then $\lambda = -1$.)

(c) By a standard result, $M \otimes_R R/I \cong M/MI$, where MI is the right R-submodule of M generated by elements of the form mi for some $m \in M$, $i \in I$. So, our goal is to show that $MI = M$. To do this, we need to know some specific elements of I. Looking at ϕ, I can spot some elements in the kernel: for example $T-1$ and $U-1$. (It is true that these elements actually generate I, but we won't need that fact.) Notice in particular what $T-1$ does to an element $f \in M$: $f(T-1)(n) = f(n+1) - f(n)$ takes the difference between successive terms of f. If we start with some targe $m \in M$, it seems reasonable to say that there *exists* some $f \in M$ where $m(n) = f(n+1) - f(n) = f(T-1) \in MI$. Indeed, define f inductively by

$$f(n) = \begin{cases} 0 & \text{if } n = 0 \\ f(n-1) + m(n) & \text{if } n > 0 \\ f(n+1) - m(n) & \text{if } n < 0. \end{cases}$$

Then $f(n+1) - f(n) = m(n)$, and so $MI = M$.

January 2020 Problem 1 On this problem, no justification is required; only the answer will be graded. Let S_3 be the symmetric group on 3 letters, and let R be the group ring $\mathbb{Z}[S_3]$.

(a) Write down a nonzero element of R which is a zero-divisor.

(b) Write down an element in the center of R which is not in $\mathbb{Z}$.

(c) Write down an R-module which is not a free module.

Solution

(a) Let e be the identity in S_3 and let $\sigma = (12)$. Then $e + \sigma$ is a nonzero element of R that is a zero-divisor since $(e+\sigma)(e-\sigma) = 0$.

(b) Let $\tau = (123)$, and consider the element $x = e + \tau + \tau^2$. The element x clearly commutes with τ, so it is enough to check that x commutes with σ. We have

$$\begin{aligned}\sigma x &= \sigma + \sigma\tau + \sigma\tau^2 \\ &= \sigma + \tau^{-1}\sigma + \tau^{-2}\sigma \\ &= \sigma + \tau^2\sigma + \tau\sigma \\ &= x\sigma.\end{aligned}$$

(c) Consider $\mathbb{Z}^3$ as an R-module, where S_3 acts by permuting the entries. So for example, $(2\sigma + \tau)(a, b, c) = (2b+c, 3a, 2c+b)$. For any free R-module M of finite rank we have $\mathrm{rank}_{\mathbb{Z}\text{-mod}}(M) = 6\ \mathrm{rank}_{R\text{-mod}}(M)$, since R has rank 6 as a free $\mathbb{Z}$-module. However, $\mathbb{Z}^3$ has rank 3 as a $\mathbb{Z}$-module, which is not divisible by 6, so it cannot be a free R-module.

January 2018 Problem 3 Consider the field with 11 elements $F = \mathbb{F}_{11}$. Let G denote the cyclic group of order 11 (written multiplicatively), with generator $r \in G$. Denote by FG the group algebra of G (also sometimes denoted $F[G]$). We consider r as an element of FG, and let $T\colon FG \to FG$ be the F-linear map given by $T(x) = rx$ for all $x \in FG$. Find the Jordan canonical form of T.

Solution

We'll start by determining the minimal and characteristic polynomials of T. On the one hand, we know that $T^{11} = Id$ so the minimal polynomial $f(y)$ divides $y^{11} - 1$. On the other hand, if the degree of f is less than 11, then $f(T) : FG \to FG$ takes 1 to $f(r) \neq 0$ so $f(T) \neq 0$. This tells us that $y^{11} - 1$ is the minimal polynomial, and since it has degree 11 it is also the characteristic polynomial of T. In characteristic 11, $y^{11} - 1 = (y-1)^{11}$ so 1 is the only eigenvalue of T and the size of the largest Jordan block with eigenvalue 1 is 11, so the Jordan canonical form of T consists of a single 11×11 Jordan block with eigenvalue 1.

January 2017 Problem 5 Consider the ring $R = \mathbb{C}[x]$.

(a) Describe all simple R-modules.

(b) Give an example of an R-module that is indecomposable, but not simple. (Recall that a module is *indecomposable* if it cannot be written as a direct sum of non-trivial submodules.)

(c) Consider R-modules $M = R/(x^3 + x^2)$, and $N = R/(x^3)$, and take their tensor product over R: $M \otimes_R N$. It is an R-module, and in particular a vector space over $\mathbb{C}$. What is its dimension over $\mathbb{C}$?

(d) Let M be any R-module such that $\dim_{\mathbb{C}}(M) < \infty$, and let $N = R/(x^3)$ as before. Show that

$$\dim_{\mathbb{C}}(M \otimes_R N) = \dim_{\mathbb{C}} \operatorname{Hom}_R(N, M).$$

Solution

(a) Recall that a simple R-module is a non-zero R-module M whose only R-submodules are the 0 module and M itself. In this case, we know that a $\mathbb{C}[x]$-module is the same thing as a complex vector space V together with a choice of a linear transformation $x \colon V \to V$. Suppose V is simple, then, since V is a complex vector space, we know that the linear transformation x has an eigenvector v of some eigenvalue λ. Then the subspace generated by v is stable under multiplication by x, and of course is stable under addition and scalar multiplication. So, $\langle v \rangle$ is a non-zero submodule of V, and so must in fact equal all of V. This shows that any simple R-module must be a 1-dimensional complex vector space. On the other hand, given any scalar $\lambda \in \mathbb{C}$, $\mathbb{C}$ considered as an R-module where $x \colon \mathbb{C} \to \mathbb{C}$ is multiplication by λ is a simple R-module, so we have described all the possible simple modules.

(b) As we discovered in part (a), a submodule of an R-module V is a vector subspace which is stable under multiplication by x, and such subspaces are closely related to eigenspaces of x. Indeed, if x is diagonalizable, then its eigenvectors form a basis for V, which is the same as saying that V is the direct sum of the eigenspaces of x, which is to say that V is decomposable as an R-module. So, we certainly need x not to be diagonalizable, so consider $V = \mathbb{C}^2$ considered as an R-module with x acting as

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

This module is not simple, because we already showed every simple module is 1-dimensional as a vector space. However, x has only one eigenspace, so V has only one R-submodule, and thus there's no way to write V as a direct sum of nontrivial R-submodules (we would need at least 2 submodules in order to do that). Thus, V is indecomposable.

(c) Using standard properties of tensor products, we have

$$\begin{aligned}\mathbb{C}[x]/(x^3+x^2)\otimes_{\mathbb{C}[x]}\mathbb{C}[x]/(x^3) &= \mathbb{C}[x]/(x^3, x^3+x^2)\\ &= \mathbb{C}[x]/(x^2, x^3)\\ &= \mathbb{C}[x]/(x^2).\end{aligned}$$

Thus, $\dim_{\mathbb{C}}(M\otimes_R N) = \dim_{\mathbb{C}} \mathbb{C}[x]/(x^2) = 2$.

(d) Here is the longer, but perhaps more "elementary" way to approach this one. Since $\dim_{\mathbb{C}}(M) < \infty$, in particular M must be finitely generated as an R-module. Since R is a PID, by the classification theorem, we must have that $M \cong \bigoplus_{i=1}^n (R/(f_i(x)))^{\oplus e_i}$ for some n and polynomials $f_i(x)$. Then, as in part (c), we have

$$\begin{aligned}M\otimes_R N &\cong \left(\bigoplus_{i=1}^n \left(\frac{R}{(f_i(x))}\right)^{\oplus e_i}\right)\otimes_R N\\ &\cong \bigoplus_{i=1}^n \left(\frac{R}{(f_i(x))}\right)^{\oplus e_i}\otimes_R \frac{R}{(x^3)}\\ &\cong (R/(x))^{\oplus e_1}\oplus (R/(x^2))^{\oplus e_2}\oplus (R/(x^3))^{\oplus e_3}.\end{aligned}$$

Now, all you need to check really is that $\operatorname{Hom}_R(N, R/(x^i))$ is an i-dimensional vector space for $i = 1, 2, 3$, and that the dimensions add correctly over these direct sums, and then you will conclude that $\dim_{\mathbb{C}}(M\otimes N) = e_1 + 2e_2 + 3e_3 = \dim_{\mathbb{C}} \operatorname{Hom}_R(N, M)$.

Alternatively, we know that $M\otimes_R N = M\otimes_R (R/(x^3))$, and by a standard result, this is isomorphic to $M/(x^3)M$. On the other hand, to specify a function from $N = R/(x^3)$ to M, we just need to say where $1 \in R/(x^3)$ will map to, and the rest of the map will be determined by R-linearity. Furthermore, since $1 \in R/(x^3)$ satisfies $x^3 \cdot 1 = 0$, the image of 1 in M must be an element m satisfying $x^3 \cdot m = 0$. That is, $\operatorname{Hom}_R(N, M) \cong M[x^3]$, the x^3-torsion submodule of M. Notice that we have an exact sequence (but not a short exact sequence) of R-modules

$$0 \to M[x^3] \to M \xrightarrow{\cdot x^3} M \to M/(x^3)M \to 0.$$

In the case where $\dim_{\mathbb{C}} M < \infty$, this is also an exact sequence of finite-dimensional $\mathbb{C}$-vector spaces. Since the sequence is exact, its Euler characteristic must be 0, so we have

$$\dim_{\mathbb{C}} M[x^3] - \dim_{\mathbb{C}} M + \dim_{\mathbb{C}} M - \dim_{\mathbb{C}} M/(x^3)M = 0.$$

Thus, $\dim_{\mathbb{C}} M[x^3] = \dim_{\mathbb{C}} M/(x^3)M$ as desired.

(Notice also that the module N is another example of an indecomposable R-module which is not simple.)

6 Other problems I like

From a Purdue qual Let $M_2(\mathbb{C})$ be the space of 2×2 matrices over $\mathbb{C}$. The space $M_2(\mathbb{C})$ is a complex vector space of dimension 4. There is a linear map $T \colon M_2(\mathbb{C}) \to M_2(\mathbb{C})$ that sends A to A^T.

(a) Compute the eigenvalues of T.

(b) For each eigenvalue of T, compute a basis for the associated eigenspace.

(c) Is the map T diagonalizable?

Solution

(a) Since $(A^T)^T = A$, the map T satisfies $T^2 = I$, but $T \neq I$, so the minimal polynomial of T is $x^2 - 1$, and thus the eigenvalues are ± 1.

(b) The $\lambda = 1$ eigenspace consists of symmetric matrices. We may take a basis for this eigenspace consisting of the matrices $S_{i,j}$ which have a 1 in the (i, j) and (j, i) entries (if distinct) and 0s elsewhere. The $\lambda =$ -1 eigenspace consists of skew-symmetric matrices. We may take a basis for this eigenspace consisting of the matrices $A_{i,j}$, $i < j$, which have a 1 in the (i, j) entry, a -1 in the (j, i) entry, and 0s elsewhere.

(c) Yes, T is diagonalizable. We can see this both because its minimal polynomial has no repeated roots, and also because we just found an eigenbasis.

From a Purdue qual Let $f \in \mathbb{Q}[x]$ be a degree 5 irreducible polynomial, let $K/\mathbb{Q}$ be its splitting field, and let $\alpha, \beta, \gamma, \delta, \epsilon \in K$ be the roots of f. Suppose that $\mathrm{Gal}(K/\mathbb{Q}) = S_5$, the symmetric group on 5 elements.

(a) Find the Galois group of $K/\mathbb{Q}(\alpha, \beta)$.

(b) Find the Galois group of $K/\mathbb{Q}(\alpha + \beta, \alpha\beta)$.

(c) The field $\mathbb{Q}(\alpha, \beta)$ contains the field $\mathbb{Q}(\alpha + \beta, \alpha\beta)$. Find the degree $[\mathbb{Q}(\alpha, \beta) : \mathbb{Q}(\alpha + \beta, \alpha\beta)]$.

Solution

(a) The Galois group of $K/\mathbb{Q}(\alpha, \beta)$ will consist of all those elements of S_5 which fix α and β, which is a subgroup isomorphic to S_3.

(b) The Galois group of $K/\mathbb{Q}(\alpha + \beta, \alpha\beta)$ will consist of all those elements of S_5 which fix $\alpha + \beta$ and $\alpha\beta$. Such a permutation can swap α and β, and can freely permute the remaining roots, so this subgroup is isomorphic to $S_2 \times S_3$.

(c) We have that

$$[\mathbb{Q}(\alpha,\beta):\mathbb{Q}(\alpha+\beta,\alpha\beta)][K:\mathbb{Q}(\alpha,\beta)] = [K:\mathbb{Q}(\alpha+\beta,\alpha\beta)]$$
$$[\mathbb{Q}(\alpha,\beta):\mathbb{Q}(\alpha+\beta,\alpha\beta)]|\mathrm{Gal}(K/\mathbb{Q}(\alpha,\beta))| = |\mathrm{Gal}(K/\mathbb{Q}(\alpha+\beta,\alpha\beta))|$$
$$[\mathbb{Q}(\alpha,\beta):\mathbb{Q}(\alpha+\beta,\alpha\beta)]\cdot 6 = 12.$$

Hence, $[\mathbb{Q}(\alpha,\beta):\mathbb{Q}(\alpha+\beta,\alpha\beta)] = 2$.

From a Michigan qual Let A be an integral domain, and let M be an A-module. We say that M is *torsion-free* if whenever $a\cdot m = 0$ either $a = 0$ or $m = 0$.

(a) Let A be a principal ideal domain. Suppose that M and N are torsion-free A-modules. Prove that $M\otimes_A N$ is torsion-free. (You may assume that M and N are finitely generated, but this is not necessary!)

(b) Let A be the ring $\mathbb{C}[x,y]$, and let M be the ideal $(x,y)\subset A$. Show that $M\otimes_A M$ is **not** torsion-free.

Solution

(a) Low-ish-tech way: suppose M and N are finitely generated. Then, by the classification of finitely-generated modules over a PID, the assumption that M and N are torsion free implies that they are free, say of ranks r_1, r_2 respectively. Then, since tensor products distribute over direct sums, $M\otimes_A N$ is free of rank r_1r_2. (I thought about trying to work hands-on with the tensors, and I think it's real bad.)

Higher-tech way: it is a fact that a torsion-free module over a PID is flat. The fact that a module B is torsion-free can be rephrased as saying that, for any $a\in A\backslash\{0\}$, the multiplication by a map $B\xrightarrow{\cdot a} B$ is injective. Since M is torsion-free, we have this condition for M. Since N is flat, we have that $M\otimes_A N\xrightarrow{(\cdot a)\otimes\mathrm{id}} M\otimes_A N$ is injective. But this is exactly the multiplication by a map on $M\otimes_A N$ by bilinearity, so $M\otimes_A N$ is torsion-free.

(b) Consider the elements $m_1 = x\otimes y$ and $m_2 = y\otimes x$ in $M\otimes_A M$. These seem like they would be the same element, and indeed if we were talking about $A\otimes_A A$ they would be, as they would both be equivalent to $xy\otimes 1$. However, in $M\otimes_A M$, we can't write them as $xy\otimes 1$ since $1\notin M$. Also, neither m_1 nor m_2 is equal to 0, because we have a map $M\otimes_A M\to M$ sending $f\otimes g$ to fg, and under this map they both map to xy. However, notice that $x(x\otimes y) = x\otimes xy = xy\otimes x = x(y\otimes x)$, so $x\otimes y - y\otimes x$ is x-torsion (and also y-torsion for the same reason).

So, we have found a torsion element if, as suggested above, we can show that $x\otimes y\neq y\otimes x$. To do so formally, consider the map $M\to M/M^2$. This induces a map

$$M\otimes_A M\to (M/M^2)\otimes_A (M/M^2)$$

sending $f \otimes g$ to the reduction in each component. Notice that x and y act as the zero map on M/M^2, so M/M^2 is really an $A/M \cong \mathbb{C}$-vector space with basis vectors x, y. Thus, $(M/M^2) \otimes_A (M/M^2)$ is a 4-dimensional $\mathbb{C}$-vector space, with basis $x \otimes x$, $x \otimes y$, $y \otimes x$, $y \otimes y$. Thus, $x \otimes y, y \otimes x \in M \otimes_A M$ get sent to linearly independent vectors under this map, and therefore cannot be equal in $M \otimes_A M$.

The high-tech way: we have an exact sequence of A-modules $0 \to M \to A \to A/M \to 0$. Tensoring this sequence with M, we get a long exact sequence of Tor modules, which begins $0 \leftarrow M/M^2 \leftarrow M \leftarrow M \otimes_A M \leftarrow \mathrm{Tor}_1(A/M, M) \leftarrow 0 \ldots$. Luckily, it is easy to write down a free resolution of A/M to compute that $\mathrm{Tor}_1(A/M, M) \cong A/M$. Thus, A/M sits as a submodule of $M \otimes_A M$, and A/M is clearly torsion.